

AUTOMATED INTERPRETABILITY METRICS DO NOT DISTINGUISH TRAINED AND RANDOM TRANSFORMERS

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ABSTRACT

Sparse autoencoders (SAEs) have become a standard instrument for decomposing transformer activations into sparse, ostensibly interpretable latents, and their development increasingly rests on aggregate quality metrics—reconstruction fidelity, sparsity, and automated interpretability scores—taken as evidence that the recovered latents are meaningful. We subject these metrics to a sanity check: if they genuinely track computation learned during training, they should separate a trained transformer from a randomly initialized null model whose parameters are sampled i.i.d. from a Gaussian. Training per-layer residual-stream SAEs on Pythia models spanning 70M to 6.9B parameters, and comparing the trained network against three randomization schemes (re-randomization with and without the embeddings, and the step-0 initialization) alongside a Gaussian-embedding control, we find that ‘fuzzing’ and ‘detection’ auto-interpretability AUROC, together with standard reconstruction and sparsity metrics, are in many settings far closer between trained and randomized transformers than between either and the control. High aggregate auto-interpretability scores therefore do not, on their own, establish that an SAE has recovered the computation a network acquired through training; such scores can instead reflect statistical structure already present in the data or architectural inductive biases that survive randomization. We conclude that these metrics are informative but insufficient proxies for mechanistic interpretability, that randomized baselines should be reported as a matter of routine, and—motivated by a token-distribution entropy analysis that exposes a difference in feature ‘abstractness’ the aggregate scores overlook—that evaluation should be augmented with targeted measures of feature complexity.

1 INTRODUCTION

Sparse autoencoders (SAEs) have become a central tool in mechanistic interpretability, where they are used to disentangle the internal representations of neural networks by learning sparse, interpretable features from network activations (Elhage et al., 2022; Sharkey et al., 2022; Cunningham et al., 2023; Bricken et al., 2023). An autoencoder with a high-dimensional hidden layer is trained to reconstruct activations under a sparsity constraint (Gao et al., 2024; Templeton et al., 2024; Lieberum et al., 2024), in the hope of recovering the underlying concepts, or ‘features,’ that the network has learned to represent (Park et al., 2023; Wattenberg and Viégas, 2024). Because SAEs typically learn tens of thousands of latents, progress in the field is governed by aggregate quantitative metrics—reconstruction fidelity, sparsity, and auto-interpretability scores that quantify the agreement between a generated explanation and a latent’s activation pattern (Bills et al., 2023; Paulo et al., 2024; Karvonen et al., 2024a). These aggregate metrics are routinely treated as evidence that the recovered latents reflect meaningful, learned structure.

An interpretability method deserves that trust only if its evaluation metrics can separate features acquired through training from artifacts of the data distribution or the model architecture. The canonical way to establish this is a sanity check against a strong null model, such as a network with randomly initialized weights (Adebayo et al., 2020): a metric that scores a trained model and its randomized counterpart alike cannot, by construction, be measuring what the training procedure added. We apply precisely this sanity check to SAEs, and we find that several widely used metrics fail it. SAEs trained on the activations of transformers whose parameters have been randomized yield latents whose auto-interpretability scores (Bills et al., 2023; Paulo et al., 2024) are, in many

settings, strikingly close to those obtained from the corresponding fully trained model—despite the randomized networks being highly unlikely to implement any learned language computation.

This finding has a direct and consequential implication for how the field reads its own metrics: a high aggregate auto-interpretability score is not sufficient evidence that an SAE has isolated the complex computations a model learned during training. A score of that kind can instead be produced by simpler statistical regularities of the training text (Dooms and Wilhelm, 2024) or by architectural inductive biases that are present before any learning has taken place. A randomly initialized network is not computationally inert; it still performs an elementary transformation of its inputs—most plausibly the preservation or amplification of sparse structure already present in the data (Section 4)—and an SAE may faithfully describe that inherent transformation. What the aggregate score cannot do is tell us whether the transformation was learned.

We stress that this is not a claim that SAEs on trained models are uninformative: some latents recovered from trained transformers demonstrably arise from learned computation. The claim is narrower and, we believe, more actionable—that the commonly used aggregate metrics are, on their own, unable to certify that a given SAE has captured such computation. The consequence for mechanistic interpretability is concrete: the discovery of interpretable latents should be corroborated by comparison against appropriately constructed null models, and the community needs metrics that more directly separate genuine learned computation from data-statistical or architectural artifacts.

Finally, we ask why these patterns emerge. We advance two hypotheses that we do not attempt to adjudicate between here: (1) the input data already exhibits superposition, which randomly initialized networks largely preserve; and (2) randomly initialized networks amplify, or even introduce, superposed structure in their inputs (for instance, given dense input drawn i.i.d. from a Gaussian). In Section 4 we present toy models that establish the plausibility of each hypothesis, while deferring a determination of which predominates for the randomized transformers to future work.

2 RELATED WORK

Sparse dictionary learning The phenomenon now termed ‘superposition’ has, under other names, been a foundational observation of computational neuroscience for decades. Olshausen and Field (1996; 1997) demonstrated that the receptive fields of simple cells in the mammalian visual cortex are well explained by sparse coding: a comparatively large repertoire of signals (sensory information) is represented by the simultaneous activation of only a small number of elements (neurons). Coding theory thus supplies a natural lens on the task of extracting the ‘underlying signals’ responsible for neural network activations (Marshall and Kirchner, 2024).

Sparse dictionary learning (SDL) approximates a set of input vectors by linear combinations of a comparatively small number of learned basis vectors, the dictionary. This basis is typically overcomplete, having greater dimension than the inputs it reconstructs. Established SDL algorithms include Independent Component Analysis (ICA), which recovers a linear representation whose components are maximally statistically independent (Bell and Sejnowski, 1995; Hyvärinen and Oja, 2000). Sparse autoencoders (SAEs) instantiate the same objective with a simple neural network (Lee et al., 2006; Ng, 2011; Makhzani and Frey, 2014): an autoencoder whose single hidden layer is many times wider than its input is trained with an objective that imposes or incentivizes sparsity on that layer’s activations, so as to surface the latent structure. Throughout, a **latent** denotes a single neuron—one dimension—of the autoencoder’s hidden layer.

Mechanistic interpretability A prevailing view construes ‘features,’ or concepts, in language models as low-dimensional subspaces of internal model activations (Park et al., 2023; Wattenberg and Viégas, 2024; Engels et al., 2024). Where such sparse, or ‘superposed,’ structure exists, one expects to be able to ‘intervene on’ or ‘steer’ the activations—to modify or replace them so as to express different concepts and thereby influence model behavior (Meng et al., 2022; Zhang and Nanda, 2023; Heimersheim and Nanda, 2024; Makelov, 2024; O’Brien et al., 2024).

SAEs have become the dominant approach to discovering such features: one typically trains a single autoencoder to reconstruct the activations of a single network layer, most often the transformer residual stream (Sharkey et al., 2022; Cunningham et al., 2023; Bricken et al., 2023). The many proposed SAE architectures differ chiefly in the activation function applied after the linear encoder

(Makhzani and Frey, 2014; Gao et al., 2024; Rajamanoharan et al., 2024b; Lieberum et al., 2024); others vary the training objective (Braun et al., 2024; Farnik et al., 2025) or reconstruct several layers jointly (Yun et al., 2021; Lawson et al., 2024; Lindsey et al., 2024).

Beyond reconstruction error and preservation of the underlying model’s performance, SAEs have been assessed by whether they capture designated concepts (Gurnee et al., 2023; Gao et al., 2024) or factual knowledge (Huang et al., 2024; Chaudhary and Geiger, 2024), and whether the resulting latents support targeted concept ‘unlearning’ (Karvonen et al., 2024b).

Automatic neuron description Because SAEs routinely learn tens of thousands of latents, describing them by hand is infeasible, and the field has turned to automated description. Yun et al. (2021) identify the tokens in a text dataset that maximally activate a dictionary element and inspect the resulting activation patterns manually. The now-standard alternative collects a latent’s activation patterns over a text dataset and prompts a large language model to explain them (e.g. Bills et al., 2023; Foote et al., 2023), a workflow that has been adopted across the SAE literature (e.g. Cunningham et al., 2023; Bricken et al., 2023; Gao et al., 2024; Templeton et al., 2024; Lieberum et al., 2024).

Bills et al. (2023) generate an explanation of a neuron’s activation pattern over dataset examples, simulate the pattern from the explanation, and score the explanation by the correlation between observed and simulated activations—a procedure known as auto-interpretability (in the sense of self-interpreting). Because simulation scoring is expensive, Paulo et al. (2024) introduce classification-based fidelity measures that are far cheaper to compute, together with an open-source pipeline for evaluating them. Choi et al. (2024) instead use best-of- k sampling to generate multiple explanations from different subsets of a neuron’s maximally activating examples and, notably, fine-tune Llama-3.1-8B-Instruct on the top-scoring explanations to obtain inexpensive ‘explainer’ and ‘simulator’ models.

Polysemanticity Lecomte et al. (2024) observed that neurons may become polysemantic incidentally. A polysemantic neuron—a single basis dimension of a layer that represents several interpretable concepts (Elhage et al., 2022; Scherlis et al., 2023)—is unsurprising in a randomly initialized network, where individual neurons may well be polysemantic. Our concern, by contrast, is *superposition* (Elhage et al., 2022; Chan, 2024), a property of the representation carried by an entire layer rather than of any single neuron. Superposition permits a layer to encode more (sparse) features than it has (dense) neurons through sparse coding, whereby only a few concepts are active at any one time, i.e., at a given token position.

Training only the embeddings Zhong and Andreas (2024) showed that transformers acquire surprising algorithmic capabilities when only their embeddings are trained and all other parameters are left at initialization, demonstrating that the behavior of an otherwise random transformer can be shaped substantially by a small number of trained parameters. Our setting is fundamentally different. Alongside the SAE analysis, we randomize *all* parameters, embeddings included, in the ‘Step-0’ and ‘Re-randomized incl. embeddings’ variants; and while the ‘Re-randomized excl. embeddings’ variant retains the pre-trained embeddings, it does not *train* anything with those embeddings held fixed—rather, we freeze the pre-trained embeddings and randomize every other weight (Section 3).

Random transformers for board games Karvonen et al. (2024c) report that SAEs extract meaningful structure far more effectively from pre-trained chess transformers than from random-weight ones. The board-game setting, however, differs from language in a way that is decisive for our purposes: natural language is plausibly sparse—a concept such as ‘serendipitous’ occurs only rarely—and this sparse structure is aligned with conceptual meaning, whereas in board games a useful concept such as a knight fork need not occur sparsely at all. The sparse-and-aligned character of language is precisely what makes the trained-versus-random distinction hard to detect in our setting and easy to detect in theirs.

Random one-layer transformers Bricken et al. (2023) found that auto-interpretability scores discriminated effectively between random and trained one-layer transformers. Our results are consistent with, and extend, that observation: for small models such as Pythia-70m the scores for randomized variants are indeed relatively low, but this gap narrows steadily with scale and has largely

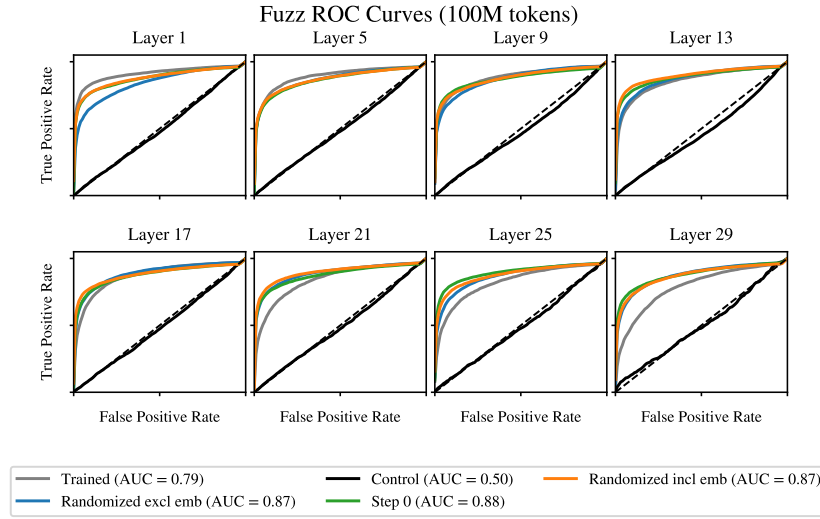


Figure 1: ‘Fuzzing’ auto-interpretability ROC curves by layer for Pythia-6.9b, each computed over 100 latents sampled per SAE. The trained model (gray) and the three randomized variants (colored) trace nearly coincident curves, while the Gaussian-embedding control (black) sits near the chance diagonal (dotted). Because the trained and randomized variants are essentially indistinguishable by this metric whereas the control is cleanly separated, aggregate AUROC alone cannot attribute a latent to learned computation. Figure 2 reports the remaining metrics and model sizes; Appendix E reports the same experiment across multiple random seeds.

closed by Pythia-6.9b. The discriminative power reported for one-layer transformers therefore does not persist to the larger models on which the field increasingly relies.

3 RESULTS

We trained per-layer SAEs on the residual-stream activation vectors of transformer language models from the Pythia suite, spanning 70M to 6.9B parameters (Biderman et al., 2023). For each underlying model we compared SAEs trained on five variants of the transformer, designed so that the four non-control variants isolate the contribution of training while holding architecture—and, in two cases, parameter norms or embeddings—fixed:

- **Trained:** the original, fully trained model, serving as the reference point.
- **Re-randomized incl. embeddings:** every parameter, embeddings included, is re-initialized by sampling Gaussian noise whose mean and variance match those of the corresponding trained weight matrix, thereby preserving each matrix’s scale while destroying its learned content.
- **Re-randomized excl. embeddings:** identical to the preceding variant except that the embedding and unembedding matrices are left at their trained values, isolating the effect of randomizing the intermediate computation while retaining the learned token representations.
- **Step-0:** the step0 revisions released for Pythia, i.e., the weights at initialization prior to any learning (Biderman et al., 2023).
- **Control:** the trained model with its input token embeddings replaced at inference time by i.i.d. standard Gaussian noise drawn afresh for each token occurrence, so that no token retains a consistent embedding vector. This variant destroys any token-level signal and thus provides a lower bound at which auto-interpretability should perform no better than chance.

For the primary experiments, SAEs were trained on 100M tokens from the RedPajama dataset (Weber et al., 2024) with an activation buffer of 10M tokens; Appendix C reproduces a subset of these

experiments with SAEs trained on one billion tokens and obtains qualitatively identical results, confirming that our conclusions are not an artifact of limited training data. Layer coverage was scaled to the available compute: an SAE at every layer for models below 410M parameters, at every second layer for Pythia-1b, and at every fourth layer for Pythia-6.9b.

Unless stated otherwise, we trained k -sparse autoencoders (TopK SAEs; Makhzani and Frey 2014; Gao et al. 2024) with expansion factor $R = 64$ and sparsity $k = 32$. These choices are not load-bearing: a sweep over Pythia-160m with expansion factors at powers of two from 16 to 128 and sparsities of 16 and 32 leaves the results unchanged (Figure 18), and a further sweep on Pythia-1b (Appendix G) shows that the trained-versus-randomized similarity in auto-interpretability persists even for deliberately degenerate hyperparameters. The training implementation follows Belrose et al. (2025), and the evaluations follow Caden et al. (2025) and Karvonen et al. (2024a).

Auto-interpretability A feature explanation that names a concept can be supplied to a classifier tasked with predicting whether that concept is present in a text input, and the classifier can then be evaluated by conventional means, most naturally the area under the receiver operating characteristic curve (AUROC). Paulo et al. (2024) instantiate this idea through two classification tasks. In ‘fuzzing’ scoring, positive and negative token examples for a latent (with non-zero and zero activation, respectively) are delimited with special characters, and a language model is asked to judge, from the latent’s explanation, which examples have been correctly delimited. In ‘detection’ scoring, the language model instead identifies which examples contain any activating token for the feature. Both are cheaper than the ‘simulation’ scoring originally proposed by Bills et al. (2023), which correlates predicted with observed activations and is costly to compute.

Except where noted we report ‘fuzzing’ AUROC as our primary auto-interpretability measure, since it has been shown to track simulation scoring (Paulo et al., 2024); the corresponding ‘detection’ curves appear in Appendix B and support the same conclusions. For each SAE—defined by its underlying model, variant, and layer—we drew a random sample of 100 latents and scored them; this sample size is a deliberate compute trade-off, and the per-latent scatter in Appendix H together with the multi-seed uncertainty estimates in Appendix E indicate that it is sufficient to resolve the aggregate differences we report. The implementation follows Paulo et al. (2024), and we generate both explanations and predictions with a single model, Meta-Llama-3.1-70B-Instruct-AWQ-INT4—larger than the 8B models of Choi et al. (2024) and, unlike Bills et al. (2023), fully open-source.

Across model sizes, the auto-interpretability scores for the trained and randomized variants are far more similar to one another than either is to the control (Figures 1 and 2). That the ROC curves for trained and randomized transformers coincide, while the control collapses to chance, establishes that ‘fuzzing’ auto-interpretability applied to SAE latent explanations does not, on its own, distinguish a trained model from a randomized one.

Evaluation Extending the analysis from auto-interpretability to the standard SAE evaluation metrics, we find that the two families of metric behave differently but point to the same conclusion. For reconstruction and sparsity metrics as for auto-interpretability, the randomized and re-randomized variants (Figure 2; blue, orange, green) group with the trained model (gray) rather than with the control (black). This separation is important because the two metric families can fail for different reasons—reconstruction and sparsity metrics quantify how compactly an SAE captures the activation geometry, whereas auto-interpretability quantifies how well a language model can describe the resulting latents—yet both fail to distinguish trained from randomized transformers.

The control is the clearest outlier on the reconstruction metrics: its reconstruction cosine similarity and explained variance are markedly lower than those of the other variants, and its reconstruction error tends to grow across layers while every other variant’s declines. A plausible account appeals to the fact that, for fixed mean and variance, the Gaussian is the maximum-entropy distribution (Jaynes, 2003): Gaussian activations are in this sense maximally unstructured and therefore hardest for an SAE to reconstruct. As such activations propagate through successive layers we would expect them to grow less Gaussian and more ‘sparse,’ hence progressively easier to reconstruct (Section 4).

A finer pattern is that the re-randomized variants (blue and orange) sit closer to the trained model than does the step-0 variant (green), a gap most visible in the L^1 norm of the larger models. We attribute this to parameter scale: the norm of a trained weight matrix can differ substantially from its value at initialization, whereas our re-randomization procedure was constructed precisely to preserve the

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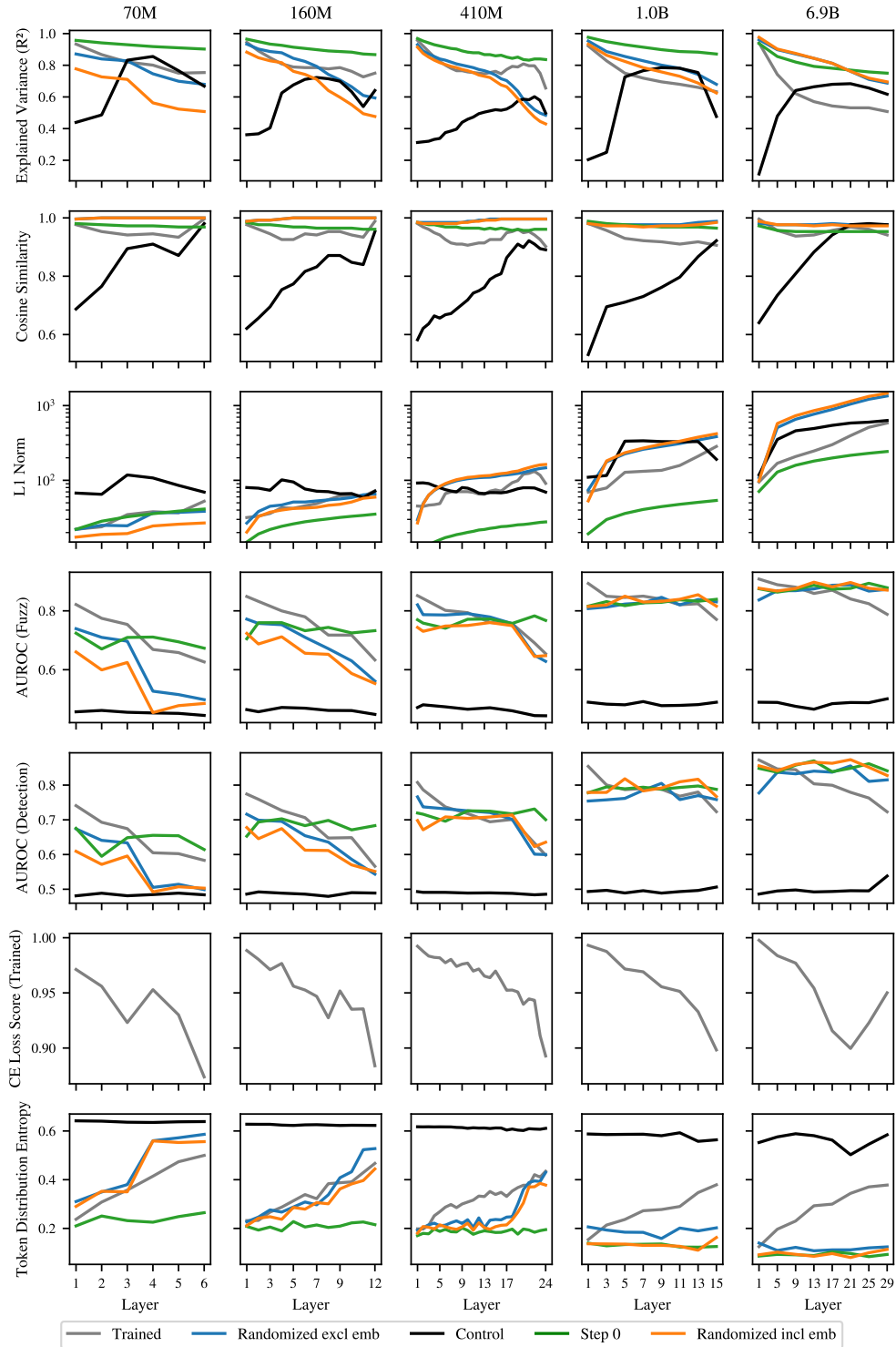


Figure 2: Sparse autoencoder evaluation metrics across the Pythia suite (70M to 6.9B parameters), with model size increasing left to right and metric varying by row: auto-interpretability AUROC, reconstruction cosine similarity, explained variance, L^1 norm, cross-entropy loss recovered, and token-distribution entropy, each plotted against layer index. The trained variant (gray) and the three randomized variants (blue, orange, green) track one another closely on the reconstruction, sparsity, and auto-interpretability rows, and are separated from the Gaussian control (black) rather than from each other—the pattern sharpening with model scale. The exception is the cross-entropy row, which is meaningful only for the trained variant (see text). Only the token-distribution entropy row exposes a systematic trained-versus-randomized difference.

trained matrices’ norms. Since per-layer parameter scale governs the growth of activations along the residual stream (Liu et al., 2020), matching those norms suffices to bring the re-randomized variants into close agreement with the trained model on this axis. On the auto-interpretability row, meanwhile, AUROC rises with model size for every variant except the control; we read this as latents becoming more specific as the SAE grows, since in a smaller SAE each latent must account for a larger share of the input, and less specific latents make the classification task harder.

Row five of Figure 2 reports the cross-entropy (CE) loss recovered against layer: the increase in loss when the original activations are replaced by their SAE reconstructions, normalized by the increase when they are ablated to zero. On this metric the trained-variant SAEs perform on par with SAEs from the literature (e.g., Kissane et al., 2024; Rajamanoharan et al., 2024a; Mudide et al., 2024), indicating that our SAEs are competently trained rather than degenerate. This metric, however, is interpretable only for the trained variant: for every randomized variant the underlying model’s loss is poor regardless of whether original or reconstructed activations are used, so loss recovered carries no meaning there. It is precisely because this one principled metric *does* single out the trained model that its absence from the auto-interpretability comparison is so telling.

Latent explanation complexity If the aggregate metrics conceal a real distinction, that distinction should be recoverable from the features themselves. We had anticipated that SAEs on trained versus randomized transformers would discover qualitatively different features—that the randomized variants would learn relatively simple features keyed to surface characteristics of the input text, but not the more abstract, context-dependent features characteristic of trained transformers (Templeton et al., 2024). Appendix J provides a random sample of features and their maximally activating examples for each Pythia-6.9b variant, and Appendix L gives fuller per-feature dashboards.

To make this intuition quantitative we exploit a robust empirical regularity: a substantial fraction of SAE latents fire on only a single token, or a small set of distinct tokens, across a text dataset (e.g., Lin and Bloom, 2024; Dooms and Wilhelm, 2024). The degree to which a latent is confined to one token ID rather than spread across many is therefore a natural, if coarse, proxy for its simplicity. We formalize this as the entropy of the latent’s activation distribution over token IDs—defined as the total latent activation per token, aggregated over the maximally activating examples used to generate the auto-interpretability explanations—so that higher entropy corresponds to activations spread across more tokens and hence a less token-specific latent. The per-latent relationship between this entropy and ‘fuzzing’ AUROC is examined in Appendix H.

The last row of Figure 2 reports this token-distribution entropy, and it is here—unlike on any of the aggregate metrics—that the variants come apart. The control sits at consistently high entropy, exactly as expected: because each token’s embedding is resampled i.i.d. from a Gaussian at every occurrence, no token has a stable embedding, so latents cannot concentrate on individual token IDs (Section 3). For the trained variant, entropy rises with depth, meaning that the maximally activating examples of later-layer latents are progressively less dominated by a single token—again as expected, since later layers should encode more abstract features that are further removed from token embeddings. The randomized variants, by contrast, exhibit lower entropy than either the trained or control variants at all depths, indicating latents that remain narrowly tied to one or a few token IDs.

Read together with the aggregate results, this pattern locates the axis that the standard metrics miss: feature ‘abstractness.’ Token-distribution entropy is not a direct measure of abstractness, but the interpretation it supports is clear and consistent across variants. The randomized models, notwithstanding their trained-like auto-interpretability scores, learn simple single-token features and—crucially—fail to develop more complex features with depth, whereas the trained model does. The very distinction that aggregate auto-interpretability erases is thus recoverable once a complexity-sensitive measure is brought to bear, which is the sense in which we offer token-distribution entropy as a proof of concept rather than a finished metric.

4 A TOY MODEL OF SUPERPOSITION IN RANDOM NETWORKS

Section 1 advanced two hypotheses for the apparent sparsity and interpretability of randomized-transformer activations: that the input data already exhibits superposition which the network preserves, or that the network itself amplifies or introduces superposition. We now examine both through toy models designed to isolate the relevant mechanism. Each hypothesis draws support, and we again

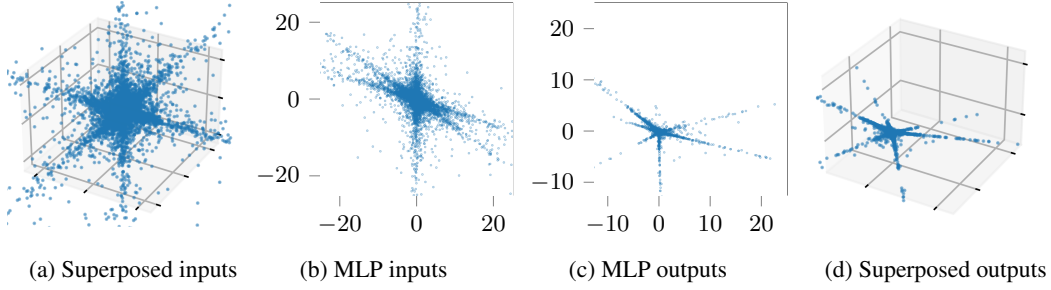


Figure 3: The effect of a randomly initialized neural network on superposed input data. We draw 10K samples of $n_s = 3$ sparse input features from a Lomax distribution with shape $\alpha = 1$ and scale $\lambda = 1$, project them to $n_d = 2$ dense input features via an i.i.d. standard-normal matrix, pass the dense inputs through a two-layer ReLU MLP with hidden size $4n_d$, and recover $n_s = 3$ sparse outputs by applying the inverse of the projection matrix. The recovered outputs remain superposed despite the intervening nonlinearity, and the sparse ‘arms’ are if anything sharper than at the input—consistent with amplification rather than mere preservation.

decline to adjudicate which dominates for the randomized transformers of Section 3, treating that question as beyond the reach of the present evidence.

4.1 MATRIX MULTIPLICATIONS PRESERVE SUPERPOSITION

We begin with a linear argument establishing that multiplication by a weight matrix W preserves superposition exactly. Suppose superposed input data x are generated by first drawing n_s i.i.d. ‘sparse’ features z from a heavy-tailed Lomax distribution, $z \sim \text{Lomax}(\alpha, \lambda)$, projecting the higher-dimensional sparse z down to the lower-dimensional dense x through a matrix D , and adding Gaussian noise of small variance Σ , so that $x \sim \mathcal{N}(x; Dz, \Sigma)$. Applying any matrix W then leaves the generative form intact: $x' = Wx$ is superposed under the same model, differing only in its noise covariance and its mapping from z , namely $x' \sim \mathcal{N}(z; WDz, W\Sigma W^T)$. Superposition is therefore an invariant of the linear layers, not something that a randomly chosen W can destroy.

The same intuition extends to nonlinear networks, as a direct visualization confirms. Passing dense activations through a simple feed-forward MLP—here $n_s = 3$ sparse features projected to $n_d = 2$ dense features—yields outputs that remain visibly superposed despite the ReLU nonlinearity (Figure 3). The comparison further hints at amplification rather than mere preservation: relative to the inputs (Figures 3a and 3b), the outputs (Figures 3d and 3c) place fewer points between the ‘arms’ of the distribution, sharpening the sparse structure.

4.2 DO RANDOM NNS PRESERVE OR AMPLIFY SUPERPOSITION?

To test this suggestion quantitatively, we generated toy data following Sharkey et al. (2022)—sampling ground-truth features on a hypersphere and drawing correlated coefficients such that only a few are active at a time (Appendix I.1)—passed these inputs through a two-layer MLP at initialization, and trained standard SAEs with an L^1 sparsity penalty separately on the inputs and on the outputs (Appendix I.2). As a control we used Gaussian inputs matched to the superposed data in mean and standard deviation, so that any difference in recoverable sparsity is attributable to structure in the data rather than to its scale.

Following Sharkey et al. (2022), we first confirmed that SAEs recover the ground-truth features that generated the data (Figure 4), using the mean max cosine similarity (MMCS): for each data-generating feature we take its maximum cosine similarity with the SAE’s learned features (decoder weight vectors) and average over data-generating features. MMCS is available only at the MLP inputs, where the data-generating features are known; analyzing the outputs requires a different instrument. There we adopt an SAE’s ability to attain low reconstruction error at high sparsity as a proxy for the degree of superposition in its training data, tracing Pareto frontiers of explained variance against sparsity by sweeping the L^1 penalty coefficient (Figure 5a).

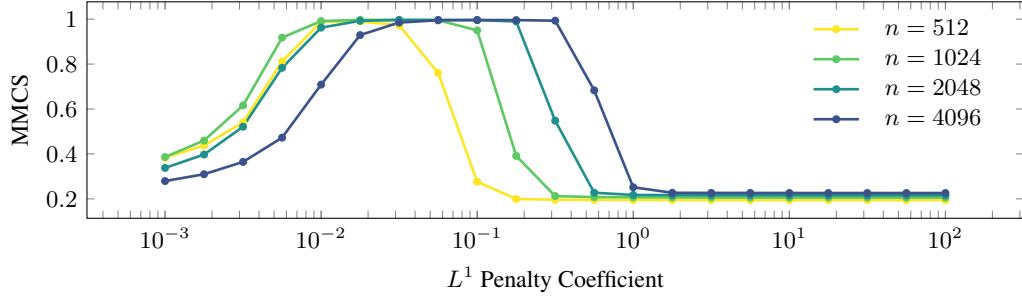
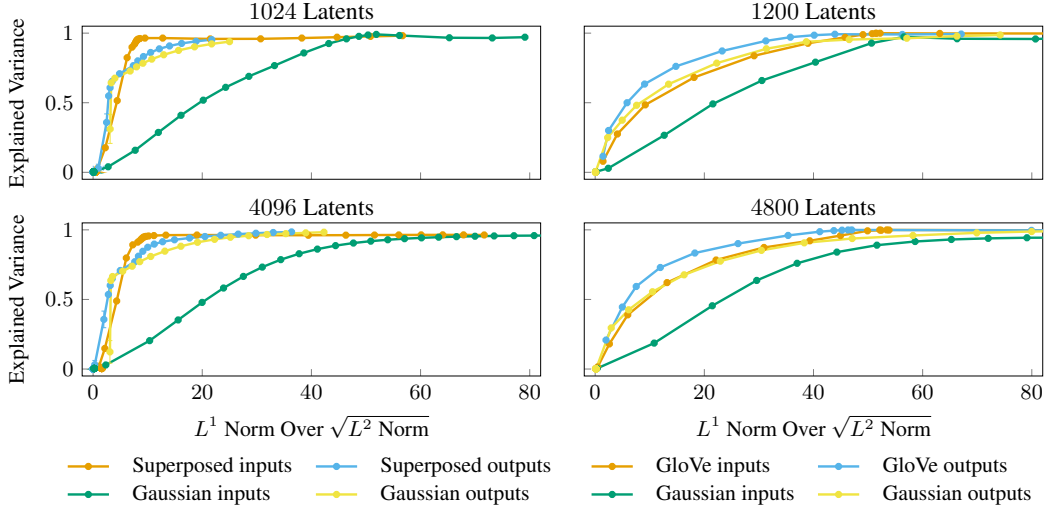


Figure 4: Mean max cosine similarity (MMCS) between the features learned by a standard SAE (its decoder weight vectors) and the data-generating features, as a function of the L^1 penalty coefficient in the training loss, following Sharkey et al. (2022). A ‘Goldilocks zone’ of intermediate penalties yields near-perfect recovery of the data-generating features, provided the SAE has enough latents to represent them—confirming that our toy setup reproduces the recovery behavior on which the subsequent analysis relies.



(a) Toy datasets following Sharkey et al. (2022). (b) 300-dim. GloVe vectors (Pennington et al., 2014).

Figure 5: Pareto frontiers of explained variance against the L^1 norm divided by the square root of the L^2 norm (a sparsity measure) for datasets that may exhibit superposition, Gaussian controls matched in mean and variance, and the outputs obtained when each is passed through a randomly initialized two-layer MLP (Section 4.2). Each point corresponds to one L^1 penalty coefficient. In both panels the input-side gap between the candidate-superposition data and its Gaussian control narrows sharply at the MLP outputs, indicating that the random network raises apparent sparsity regardless of the input distribution.

The results speak to both hypotheses at once. As expected under the preservation hypothesis, SAEs reach a given explained variance at markedly higher sparsity for the superposed inputs than for the Gaussian control (Figure 5a; orange and blue-green). More strikingly, this input-side gap all but disappears at the MLP outputs: the superposed and Gaussian outputs achieve nearly the same sparsity at a given explained variance. The MLP thus confers a high, input-insensitive level of sparsity on its outputs—evidence that a randomly initialized network can *impose* sparse structure irrespective of whether its input already possessed it, in line with the amplification hypothesis. Appendix I corroborates this with additional sparsity measures and hyperparameter settings.

4.3 DO TOKEN EMBEDDINGS EXHIBIT SUPERPOSITION?

If randomly initialized networks preserve or amplify superposition, then the behavior observed in Section 3 should hinge on how much superposition the inputs to a transformer language model already carry. We probe this by applying the Section 4.2 procedure to language data, training SAEs on pre-trained GloVe word vectors, the embedding matrices of Pythia models, the outputs of passing each through a randomly initialized two-layer MLP, and matched Gaussian controls. The setup is otherwise unchanged, save that the number of data points is fixed by the vocabulary size and we use a single random seed.

Two observations follow. First, the input-side gap between the GloVe vectors and their Gaussian control (Figure 5b) is smaller than for the synthetic superposed data (Figure 5a), indicating that real token embeddings exhibit only modest superposition relative to the idealized toy distribution. Second, and more tellingly, the frontiers for both input types improve once the data pass through a randomly initialized MLP—the same sparsification observed for the toy data—reinforcing the possibility that random networks actively increase the apparent superposition of whatever they are given.

5 LIMITATIONS

Our central result—that auto-interpretability measures can yield apparently meaningful, interpretable latents for SAEs trained on randomly initialized models that are unlikely to implement computationally interesting features—should be read together with the boundaries of the study that establishes it. Exhaustive coverage of datasets and architectures being infeasible, we concentrated on the Pythia family, a standard testbed for mechanistic interpretability (e.g., Paulo and Belrose, 2025; Ghilardi et al., 2025; Mueller, 2024), and on RedPajama-V2 as representative pre-training data. Within that setting the finding is robust—holding across model sizes from 70M to 6.9B parameters, across layers, across four randomization schemes, across expansion factors and sparsities, and across two auto-interpretability tasks and two training-token budgets—but we do not extrapolate it beyond residual-stream SAEs on this model family. Whether the same indistinguishability arises for SAEs trained on MLP or attention sites, where the learned computation may be more localized, for transcoders and newer SAE variants, or for other model families, remains open.

Two further caveats bound the auto-interpretability evidence specifically. First, our scores derive from a single explainer/evaluator model, the EleutherAI framework’s default (Caden et al., 2025); a systematic study of alternative explainers and prompt designs could refine the aggregate picture, though we would expect the qualitative conclusion to persist. Second, our aggregate scores are computed over a random sample of 100 latents per SAE, so they characterize the sampled population rather than every latent. These boundaries qualify the scope of the claim without weakening it. We emphasize what the result does *not* assert: not that SAEs on trained transformers capture nothing beyond what randomized transformers offer, but that aggregate auto-interpretability measures alone do not certify the presence of interesting learned features.

6 CONCLUSION

We trained sparse autoencoders on both trained and randomly initialized transformers and evaluated them with the suite of quantitative metrics on which the field relies. The central empirical result is that these metrics—and aggregate auto-interpretability scores in particular—are, across a broad range of conditions, surprisingly similar for the two settings, even as the features recovered from trained transformers are qualitatively more complex and abstract, especially at later layers, than those from randomized ones. The aggregate metrics simply do not register that qualitative gap.

This does not mean that SAEs on real models fail to learn meaningful computational features; it means that our current evaluation methods cannot, by themselves, verify that they have. A high aggregate auto-interpretability score is not proof that an SAE has discovered complex learned computation, because such a score can equally reflect simpler structure in the data or architecture that survives randomization of the weights. Our token-distribution entropy analysis, though preliminary, makes the constructive point: as a proof of concept, it recovers exactly the difference in feature ‘abstractness’ that the aggregate scores erase, which suggests that complexity-sensitive measures are a promising direction for evaluation that quantifies the computational significance of the features SAEs discover.

540 Most concretely, our study establishes randomized transformers as a strong, appropriately constructed
541 null model against which interpretability techniques should be benchmarked as a matter of routine;
542 absent such baselines, discovered structure cannot be confidently attributed to what training added.
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A BROADER IMPACT

This work scrutinizes a method in active use for the mechanistic interpretability of LLMs, and its results challenge a prevailing assumption about sparse autoencoders. By showing that SAEs yield comparable aggregate auto-interpretability scores for random and trained transformers alike, our findings sharpen the question of what these evaluation methods actually measure.

A clearer understanding of SAE quality metrics should, we hope, guide the search for better SAE-like methods and thereby render these models more interpretable and mitigate the harms they might otherwise cause. Because the present study is empirical—characterizing the behavior of a method already in use, and showing only that it interprets both random and trained transformers—we judge its risk of negative societal impact to be minimal.

B AUTO-INTERPRETABILITY ROC CURVES

Figures 6, 8, and 12 report the close agreement in ‘fuzzing’ AUROC between the trained and randomized SAEs for the 70M, 160M, and 1B models, and Figures 7, 9, and 13 show the same agreement under ‘detection’ AUROC. That both scoring tasks yield the same pattern across every model size confirms that the indistinguishability reported in the main text is not specific to the ‘fuzzing’ criterion.

B.1 PYTHIA 70M

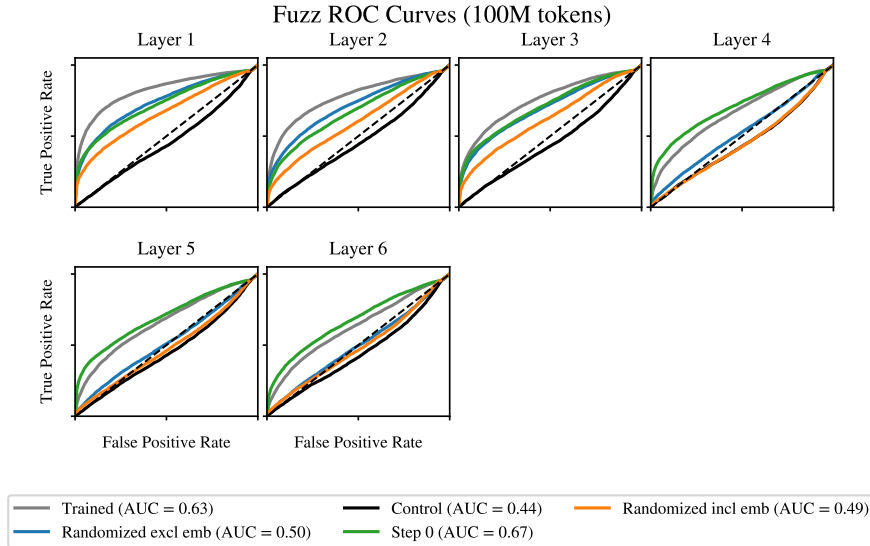


Figure 6: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-70m, each over 100 sampled SAE latents. The SAE variants perform comparably at every layer, and all degrade as the layer index increases. At this smallest scale the trained variant retains a modest edge over the randomized ones—the gap that the main text shows closing with model size.

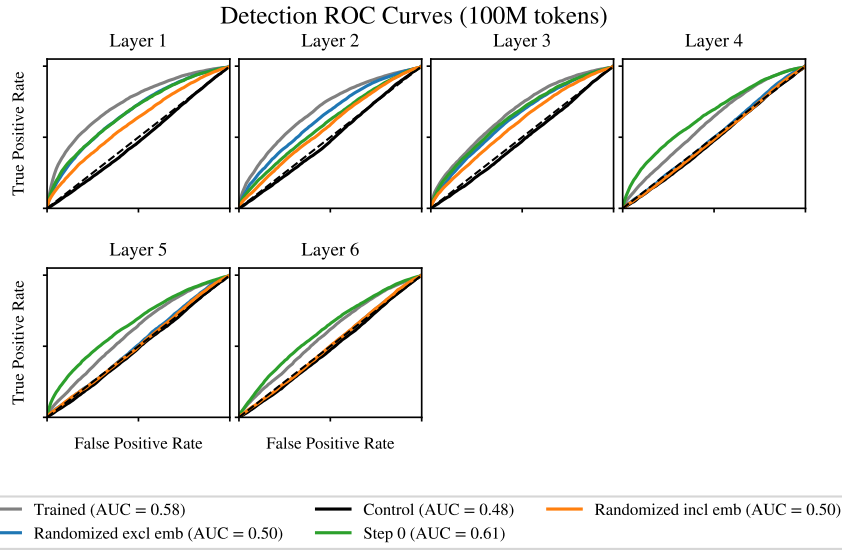


Figure 7: ‘Detection’ auto-interpretability ROC curves for Pythia-70m, each over 100 sampled SAE latents. The pattern mirrors the ‘fuzzing’ scores of Figure 6: the variants perform comparably and degrade with depth, confirming that the behavior at this scale does not depend on the scoring task.

B.2 PYTHIA 160M

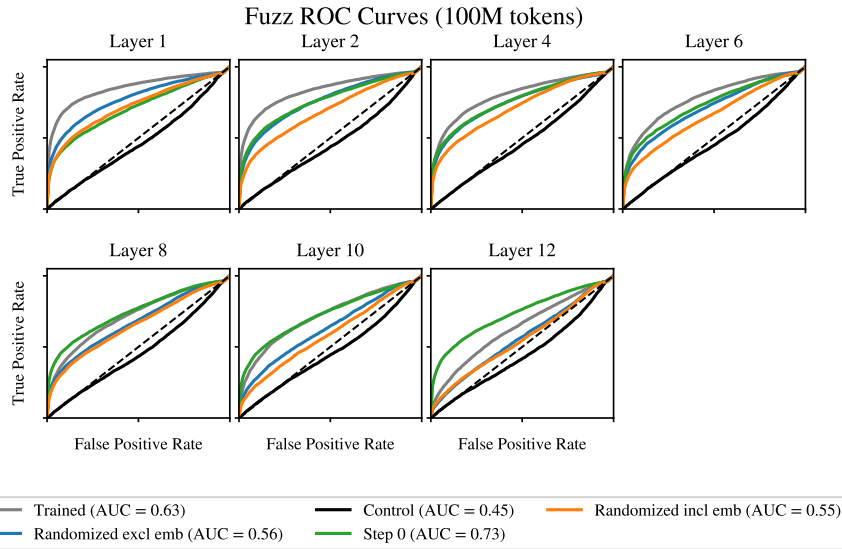


Figure 8: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-160m, each over 100 sampled SAE latents. The variants again perform comparably and degrade with layer index, with the trained–randomized gap already narrower than at 70M.

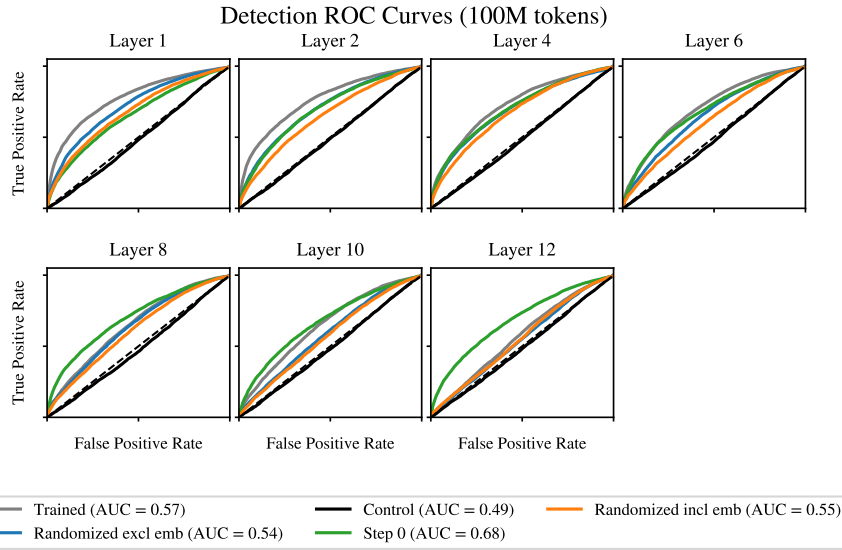


Figure 9: ‘Detection’ auto-interpretability ROC curves for Pythia-160m, each over 100 sampled SAE latents. As for the ‘fuzzing’ scores, the variants perform comparably and degrade with depth.

B.3 PYTHIA 410M

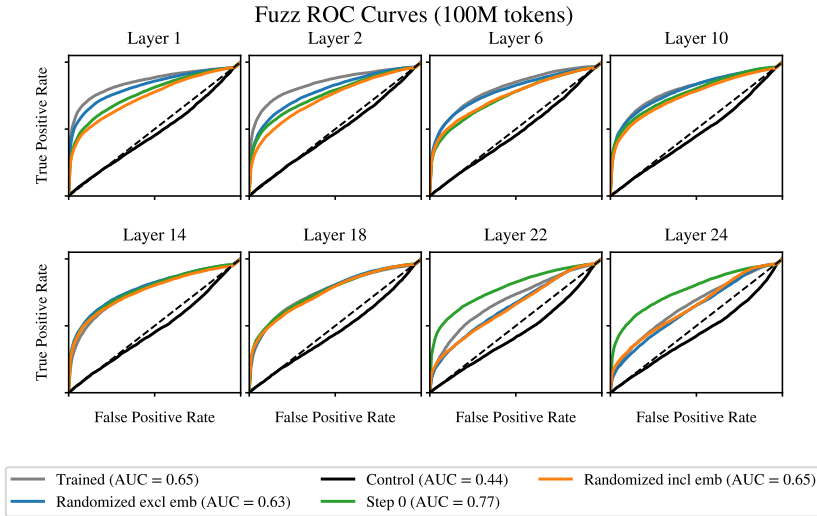


Figure 10: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-410m, each over 100 sampled SAE latents. The variants perform comparably and degrade with layer index; at this scale the scores no longer distinguish the trained from the randomized models.

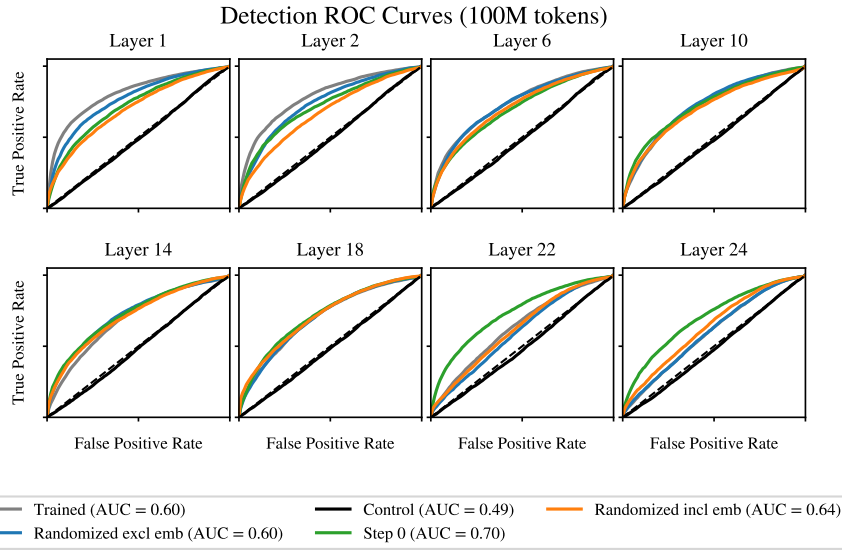


Figure 11: ‘Detection’ auto-interpretability ROC curves for Pythia-410m, each over 100 sampled SAE latents. The variants perform comparably and degrade with depth, matching the ‘fuzzing’ scores at this scale.

B.4 PYTHIA-1B

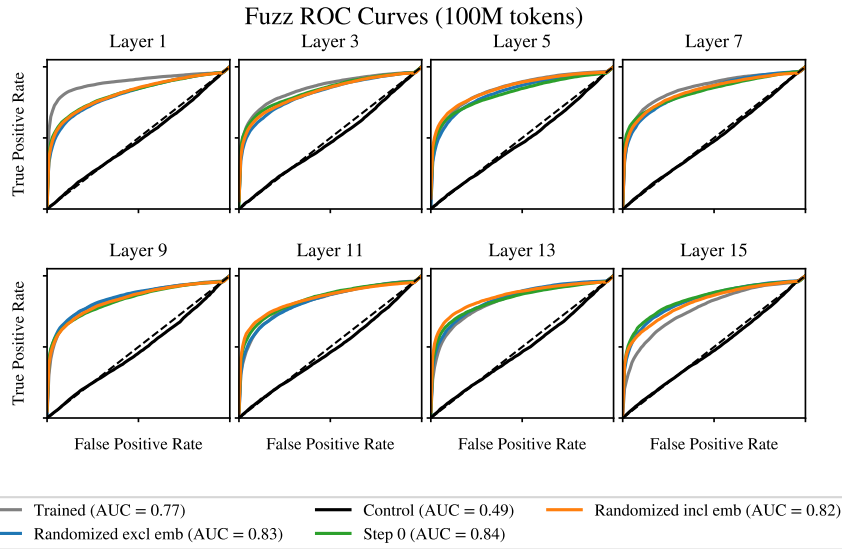


Figure 12: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-1b, each over 100 sampled SAE latents. The variants perform comparably; unlike the smaller models, performance does not degrade systematically with layer index.

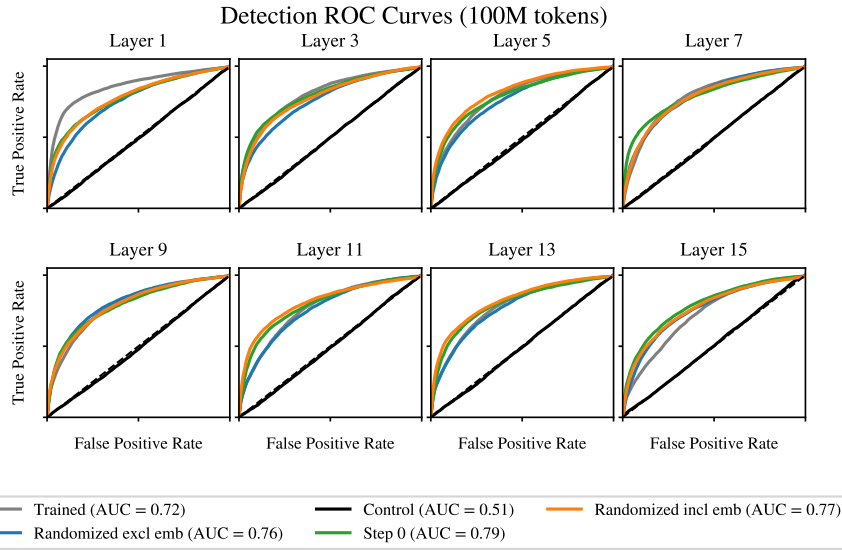


Figure 13: ‘Detection’ auto-interpretability ROC curves for Pythia-1b, each over 100 sampled SAE latents. The variants perform comparably and, as under ‘fuzzing,’ show no systematic degradation with layer index.

B.5 PYTHIA 6.9B

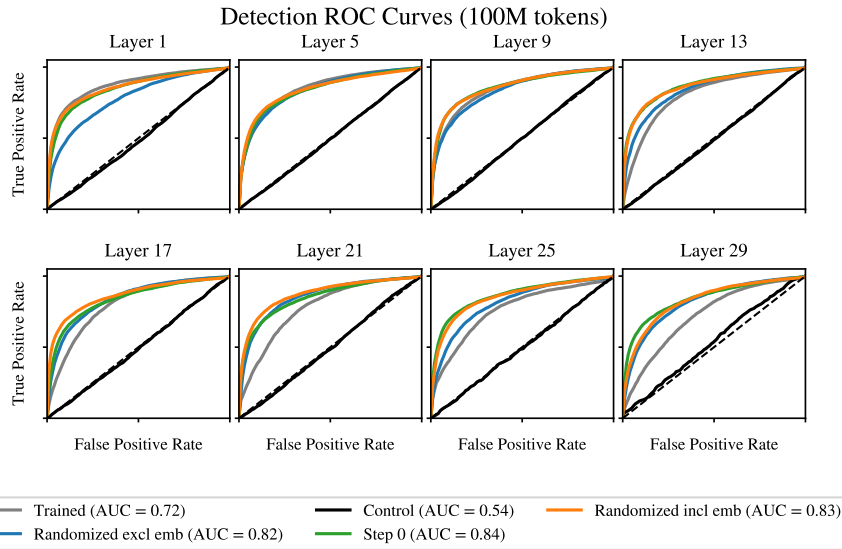


Figure 14: ‘Detection’ auto-interpretability ROC curves for Pythia-6.9b, each over 100 sampled SAE latents. The variants perform comparably, corroborating under the ‘detection’ task the ‘fuzzing’ result of Figure 1.

C EFFECT OF INCREASED TRAINING DATA

Since the primary experiments used 100M training tokens (Section 3), a natural concern is that the trained–randomized similarity is an artifact of undertraining the SAEs. To rule this out, we repeated a subset of the experiments with SAEs trained on 1B tokens from RedPajama (Figure 15) and observed the same behavior, confirming that a tenfold increase in training data does not separate the variants.

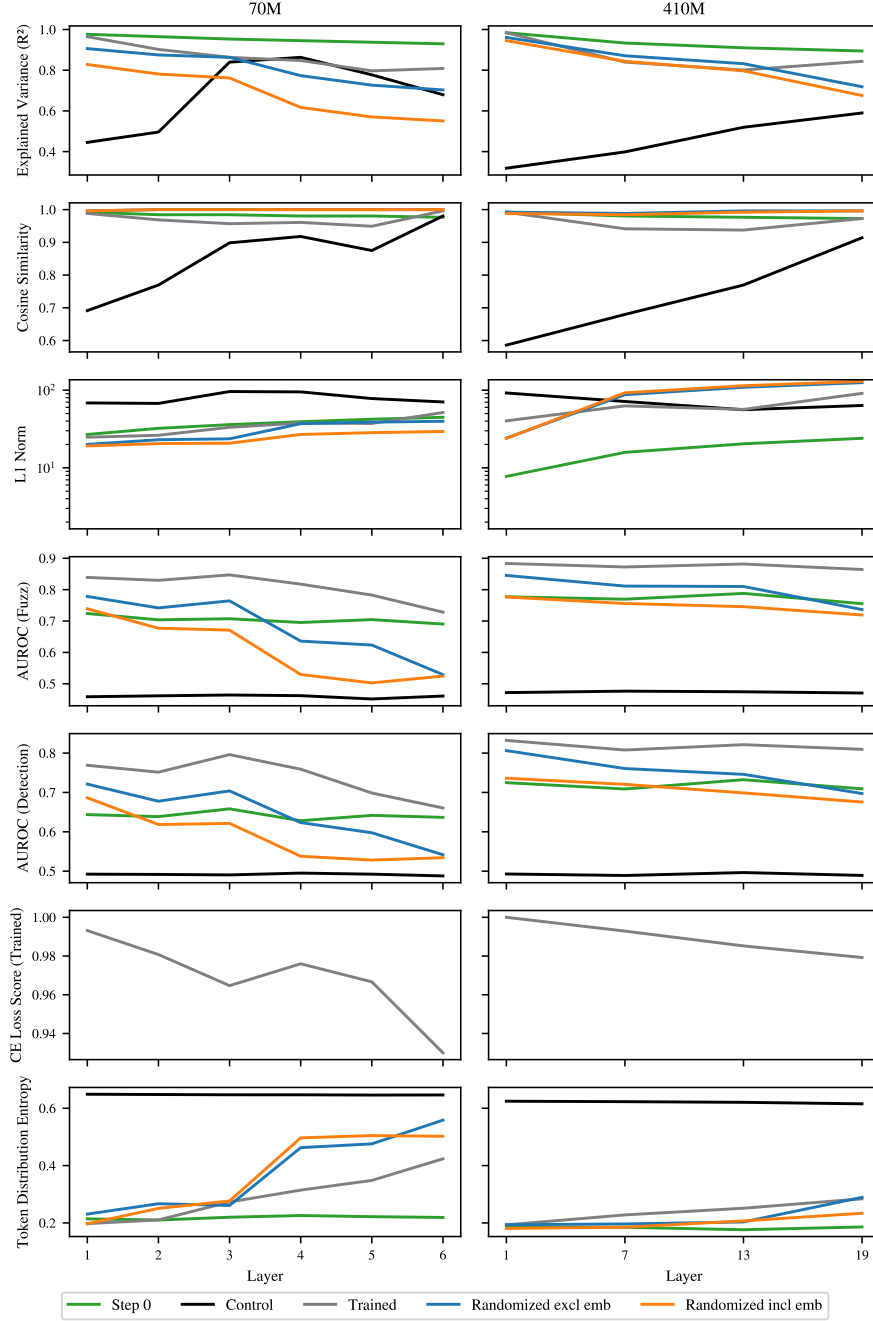
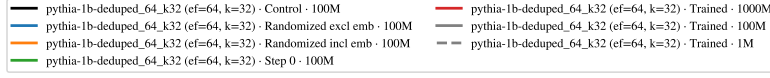


Figure 15: Evaluation metrics for SAEs trained on one billion tokens for Pythia-70m and Pythia-410m. These panels correspond to the matching columns of Figure 2 (100M tokens) and exhibit qualitatively the same behavior, so the reported similarity between trained and randomized variants is not attributable to insufficient SAE training data.

D EFFECT OF DECREASED TRAINING DATA FOR PYTHIA-1B



Pythia 1B Metrics & ROC Comparison - Token Count Variants (1000M, 100M, 1M)

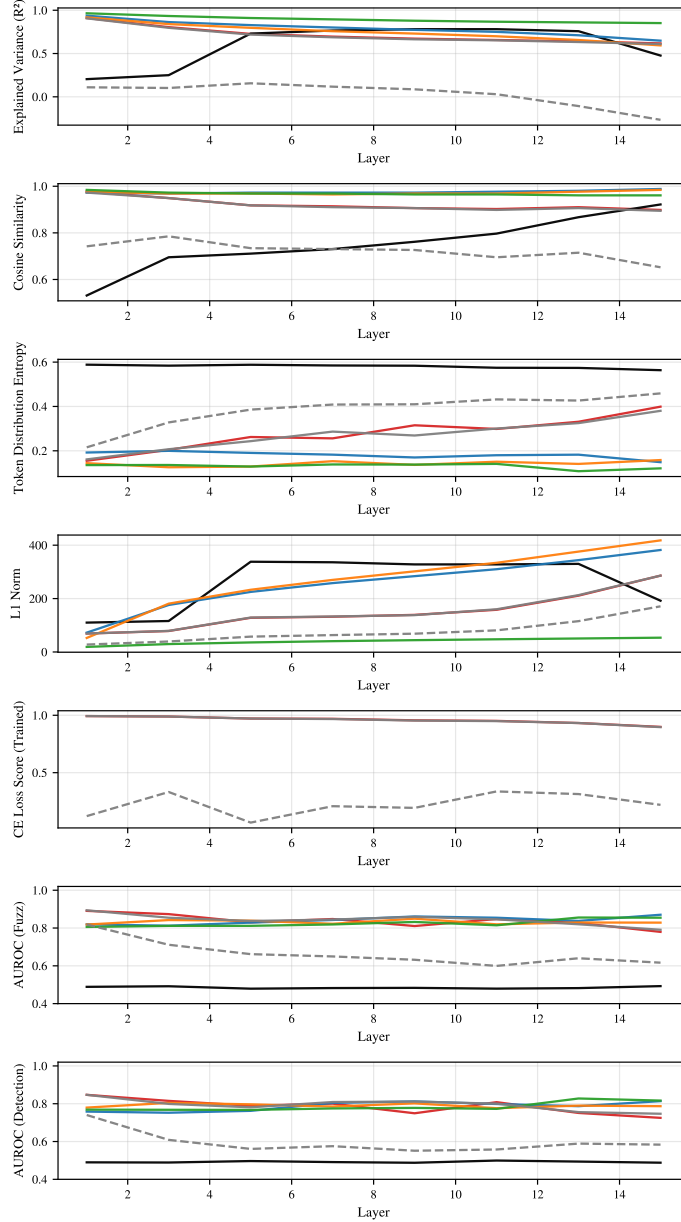


Figure 16: Evaluation metrics for SAEs trained on 1M versus 1B tokens for Pythia-1b. The markedly lower explained variance and CE loss recovered at 1M tokens confirm that these SAEs are under-trained; their average auto-interpretability scores start slightly lower in the earliest layers and then fall sharply with depth. Even so, the trends in auto-interpretability and token-distribution entropy against layer index remain consistent with the well-trained SAEs, indicating that these qualitative patterns are robust to substantial undertraining.

E UNCERTAINTY PLOTS FOR PYTHIA-70M

To quantify the run-to-run variability of our evaluation metrics, we recomputed them on Pythia-70m across five random seeds. The resulting spread (Figure 17) is small relative to the separation between the control and the other variants, so the trained–randomized similarity reported in the main text lies well within the range that seed variation would predict rather than reflecting a single fortunate draw.

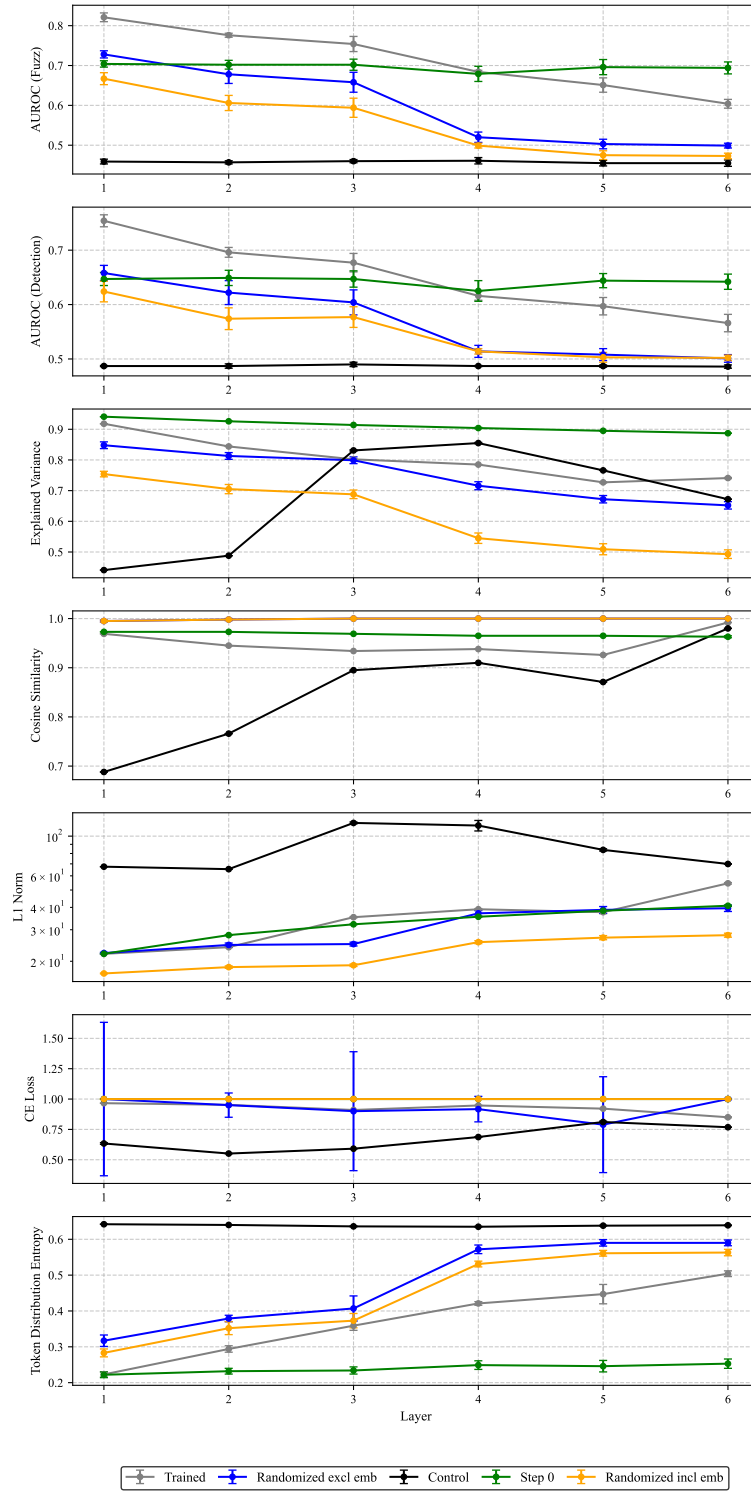


Figure 17: Uncertainty in the Pythia-70m evaluation metrics estimated over five random seeds. Error bands are narrow relative to the gap between the control and the remaining variants, indicating that the reported similarities and separations are stable across seeds.

F EFFECT OF SAE HYPERPARAMETERS FOR PYTHIA-160M

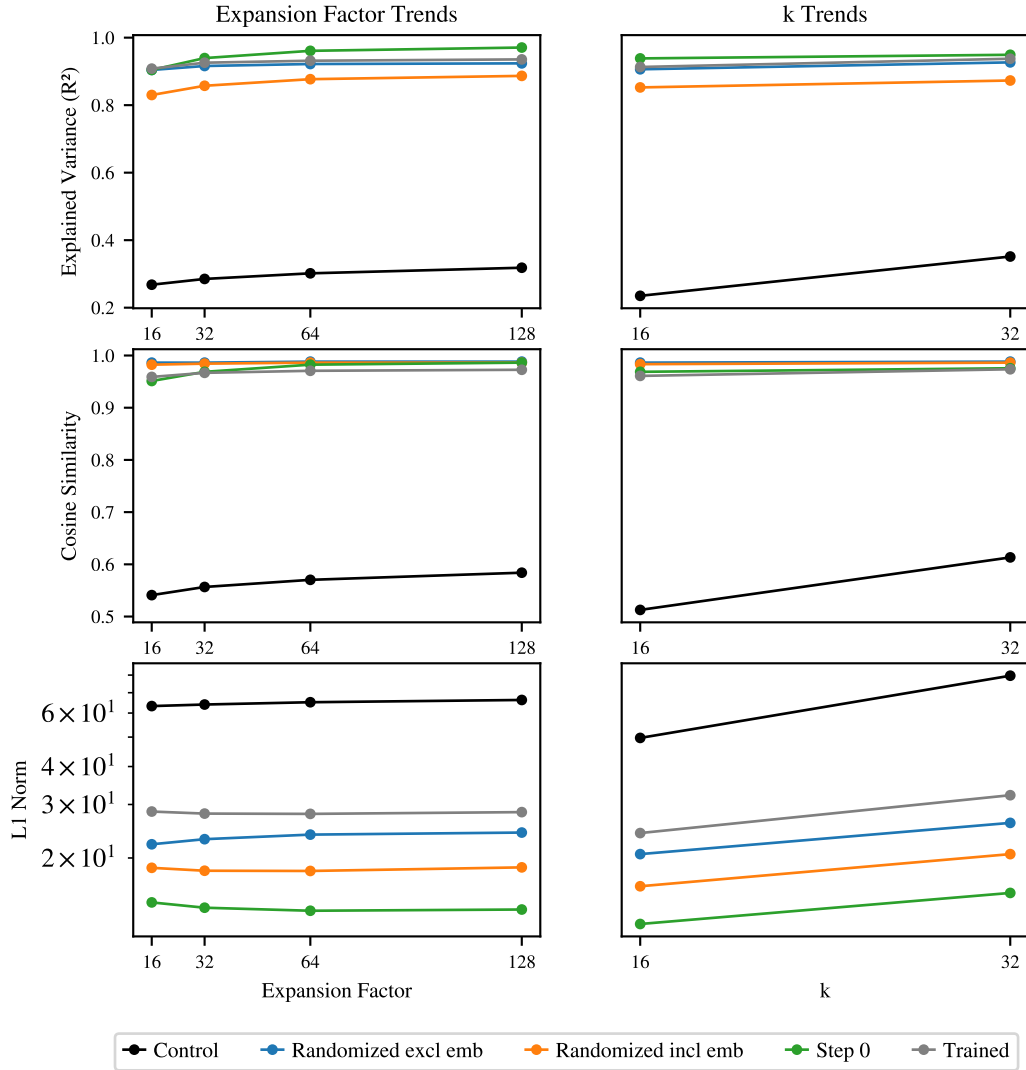


Figure 18: Robustness of SAE performance to hyperparameter choice for Pythia-160m. The standard evaluation metrics are stable across expansion factors R from 16 to 128 and sparsities k of 16 and 32, and every initialization variant preserves its relative ordering throughout. This stability justifies the moderate default values used in the main text ($R = 64$, $k = 32$) and shows that the trained-randomized similarity is not an artifact of a particular hyperparameter setting.

1350 G EFFECT OF SAE HYPERPARAMETERS FOR PYTHIA-1B
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Pythia 1B Metrics & ROC Comparison - Hyperparameter Variants (R, k) - 100M tokens

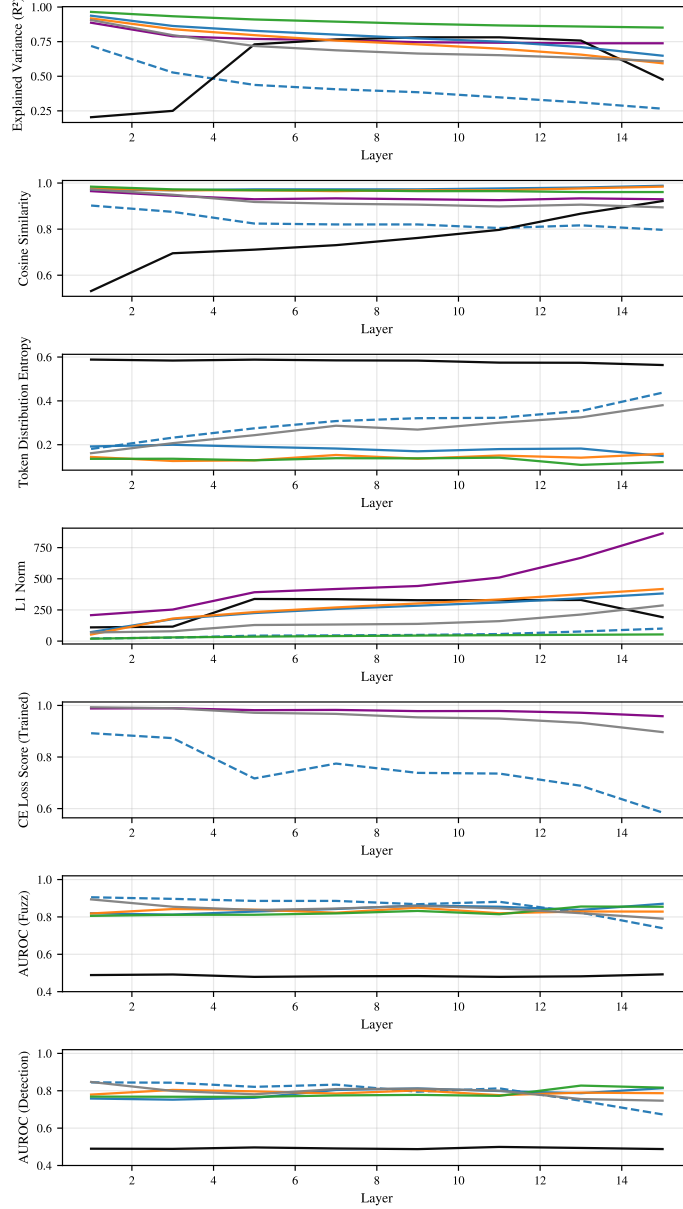


Figure 19: Evaluation metrics for SAEs trained on Pythia-1b under varying hyperparameters, alongside the default-hyperparameter results from Figure 2. A deliberately degenerate configuration (expansion factor $R = 2$, sparsity $k = 4$) is sharply separated from the defaults on explained variance and CE loss recovered, confirming that these reconstruction metrics are sensitive to SAE capacity. Crucially, the auto-interpretability scores of even these degenerate SAEs remain close to those of the default SAEs, on both trained and randomized models—so the failure of auto-interpretability to separate trained from randomized transformers persists across a wide, and even pathological, hyperparameter range.

H TOKEN DISTRIBUTION ENTROPY VS. AUTO-INTERPRETABILITY

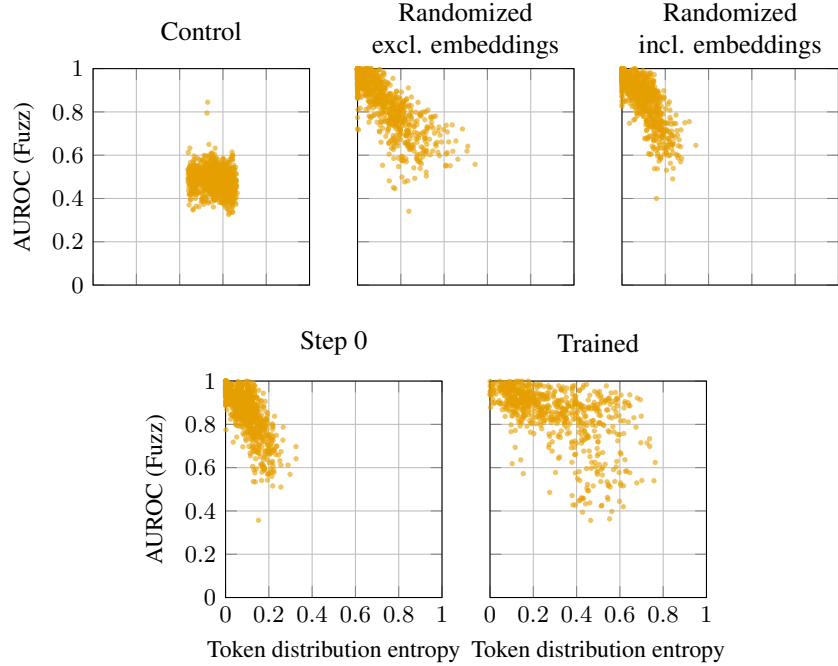


Figure 20: Per-latent token-distribution entropy against ‘fuzzing’ AUROC for SAEs trained on several layers of Pythia-6.9b. Each point is a single latent, drawn from the same sample used to compute the aggregate metrics in Figures 1 and 2; resolving the variants at the level of individual latents recovers precisely the structure that the aggregate scores average away.

Viewed at the level of individual latents, Figure 20 cleanly separates the three regimes that the aggregate AUROC conflates:

- **Control:** latents combine consistently high entropy—maximally activating examples whose activation is spread across many tokens—with low auto-interpretability, as the generated explanations fail to account for such diffuse patterns. The two quantities are uncorrelated.
- **Randomized:** under every randomization scheme of Section 3, entropy and auto-interpretability are negatively correlated: the wider the range of tokens a latent responds to, the less faithfully its explanation captures its activation pattern. The randomized variants thus occupy a low-entropy, high-AUROC regime of easily explained single-token features.
- **Trained:** the correlation is weaker, and—decisively—the trained variant additionally populates a region unreachable by the randomized variants: latents with high entropy *and* high auto-interpretability, whose activation spreads across multiple tokens yet remains consistent with a single coherent explanation.

This per-latent structure is exactly what the aggregate scores obscure. Randomized models with stable token embeddings readily produce easily explained ‘single-token’ features, but only transformers trained on natural language additionally yield the more abstract, multi-token semantic features that inhabit the high-entropy, high-AUROC corner—the qualitative signature of learned computation that a single averaged number cannot convey.

I A TOY MODEL OF SUPERPOSITION

The toy-model experiments of Section 4 trained SAEs on synthetic data constructed to exhibit superposition (Sharkey et al., 2022) and on GloVe word vectors (Pennington et al., 2014). This section specifies the data-generation procedure and the training setup in full, so that the results can be reproduced exactly.

I.1 DATA GENERATION

The ground-truth features are constructed by sampling n_s points on an n_d -dimensional hypersphere.

The per-sample feature coefficients are then obtained through the following sequence of operations: generating $A \in \mathbb{R}^{n_s \times n_s}$ where $A_{ij} \sim \mathcal{N}(0, 1)$, defining a covariance matrix $\Sigma = AA^\top$, sampling $\vec{\alpha} \in \mathbb{R}^{n_s}$ where $\alpha_i \sim \mathcal{N}(\vec{0}, \Sigma)$, projecting α_i onto the c.d.f. of $\mathcal{N}(0, 1)$, decaying $\alpha_i \rightarrow \alpha_i^{\lambda_i}$ where $\lambda \in \mathbb{R}$, normalizing $\alpha_i \rightarrow m\alpha_i / n_s \sum_j \alpha_j$ where $m \in \mathbb{R}$, and performing n_s independent Bernoulli trials with $p = \alpha_i$. The trial outcomes are finally multiplied by n_s independent draws from a continuous uniform distribution $\mathcal{U}_{[0,1]}$, which supply the nonzero coefficient magnitudes.

Here λ controls how sharply the frequency of nonzero ground-truth coefficients decays with the feature index i , and m sets the expected number of nonzero coefficients per sample—jointly governing the sparsity and the frequency structure of the generated data.

Following Sharkey et al. (2022), we set $n_s = 512$, $n_d = 256$, $\lambda = 0.99$, and $m = 5$, and we give a Python implementation of the full procedure in Figure 21.

I.2 TRAINING

The SAEs of Section 4 consist of a linear encoder with a bias term, a ReLU activation, and a linear decoder without a bias term. The decoder weights are orthogonally initialized, and the decoder weight vectors are renormalized before every training step.

The training objective is the mean squared error (MSE) between the input and decoded vectors plus the mean L^1 norm of the encoded vectors, the latter weighted by a coefficient swept between 1×10^{-3} and 100 to trace the sparsity–reconstruction Pareto frontiers reported in the main text.

The toy data are trained for 100 epochs on 10K data points across 10 random seeds, and the word vectors for 100 epochs on 400K data points with a single seed; in both cases 10% of the data is held out as a validation set on which the evaluation metrics are computed.

The MLPs of Section 4 are two-layer (one hidden layer) networks with a ReLU activation, input and output dimension n_d , and hidden dimension $4n_d$ —choices loosely modeled on the feed-forward components of transformer language models so that the toy analysis reflects the architecture it is meant to illuminate.

```

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1578 1 def generate_sharkey(
1579 2     num_samples: int,
1580 3     num_inputs: int,
1581 4     num_features: int,
1582 5     avg_active_features: float,
1583 6     lambda_decay: float,
1584 7 ) -> tuple[Tensor, Tensor]:
1585 8     """
1586 9     Args:
1587 10         num_samples (int): The number of samples to generate.
1588 11         num_inputs (int): The number of input dimensions.
1589 12         num_features (int): The number of ground truth features.
1590 13         avg_active_features (float): The average number of
1591 14             ground truth features active at a time.
1592 15         lambda_decay (float): The exponential decay factor for
1593 16             feature probabilities.
1594 17     """
1595 18     features = torch.randn(num_inputs, num_features)
1596 19     features /= torch.norm(features, dim=0, keepdim=True)
1597 20
1598 21     covariance = torch.randn(num_features, num_features)
1599 22     covariance = covariance @ covariance.T
1600 23     correlated_normal = MultivariateNormal(
1601 24         torch.zeros(num_features), covariance_matrix=covariance
1602 25     )
1603 26
1604 27     samples = []
1605 28     for _ in range(num_samples):
1606 29         p = STANDARD_NORMAL.cdf(correlated_normal.sample())
1607 30         p = p ** (lambda_decay * torch.arange(num_features))
1608 31         p = p * (avg_active_features / (num_features * p.mean()))
1609 32         p = torch.bernoulli(p.clamp(0, 1))
1610 33         coef = p * torch.rand(num_features)
1611 34
1612 35         sample = coef @ features.T
1613 36         samples.append(sample)
1614 37
1615 38     return torch.stack(samples), features

```

Figure 21: Python implementation of the data-generation procedure of Sharkey et al. (2022) used in Section 4, provided to make the toy experiments exactly reproducible.

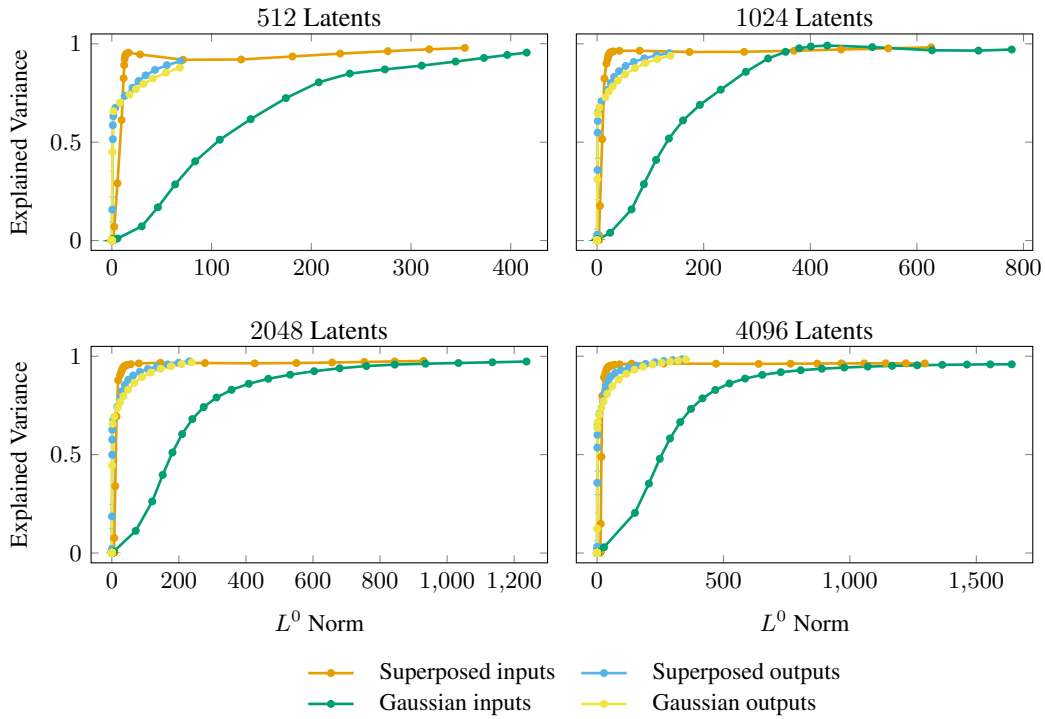


Figure 22: Explained-variance/sparsity Pareto frontiers, with sparsity measured by the L^0 norm, for the superposed toy datasets, their mean- and variance-matched Gaussian controls, and the outputs of passing each through a randomly initialized two-layer MLP. The input-side superposed–Gaussian gap contracts at the MLP outputs, reproducing under the L^0 measure the amplification effect reported in the main text.

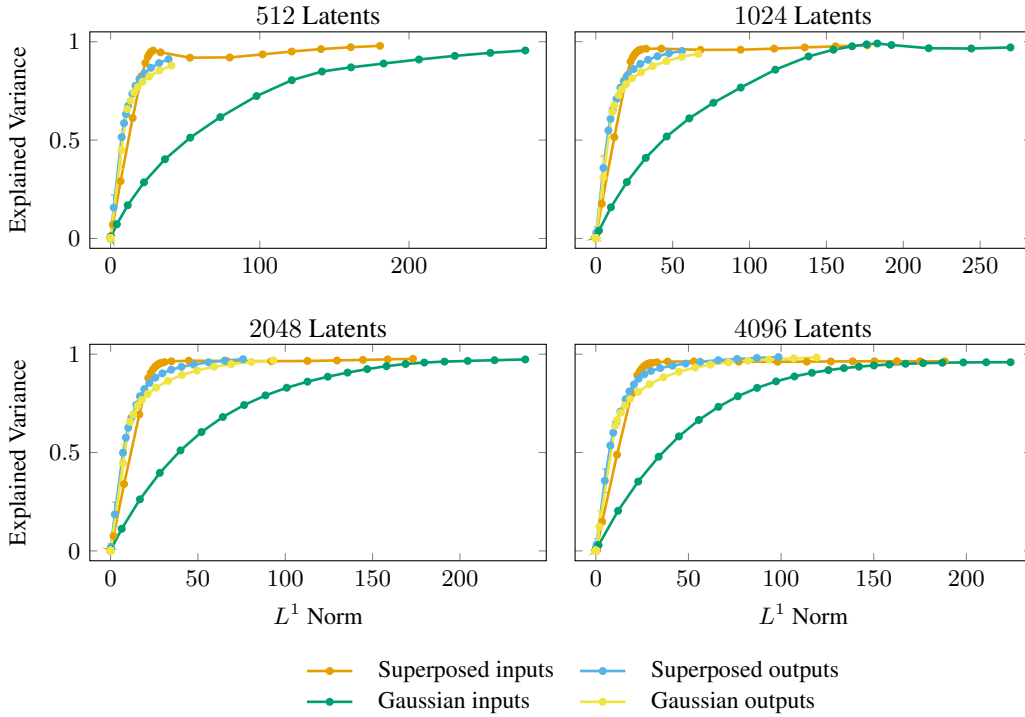


Figure 23: As above, with sparsity measured by the L^1 norm, for the superposed toy datasets, their matched Gaussian controls, and the corresponding MLP outputs. The same narrowing of the superposed–Gaussian gap at the outputs holds under this measure.

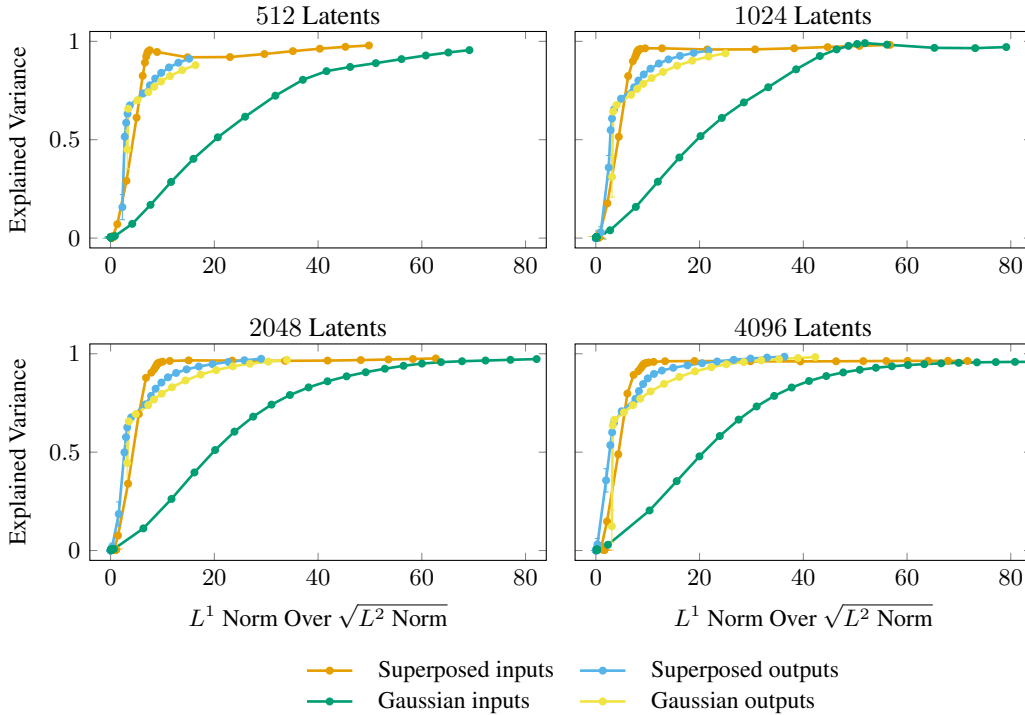


Figure 24: As above, with sparsity measured by the L^1 norm divided by the square root of the L^2 norm—the measure used in the main text (Figure 5a)—for the superposed toy datasets, their matched Gaussian controls, and the corresponding MLP outputs.

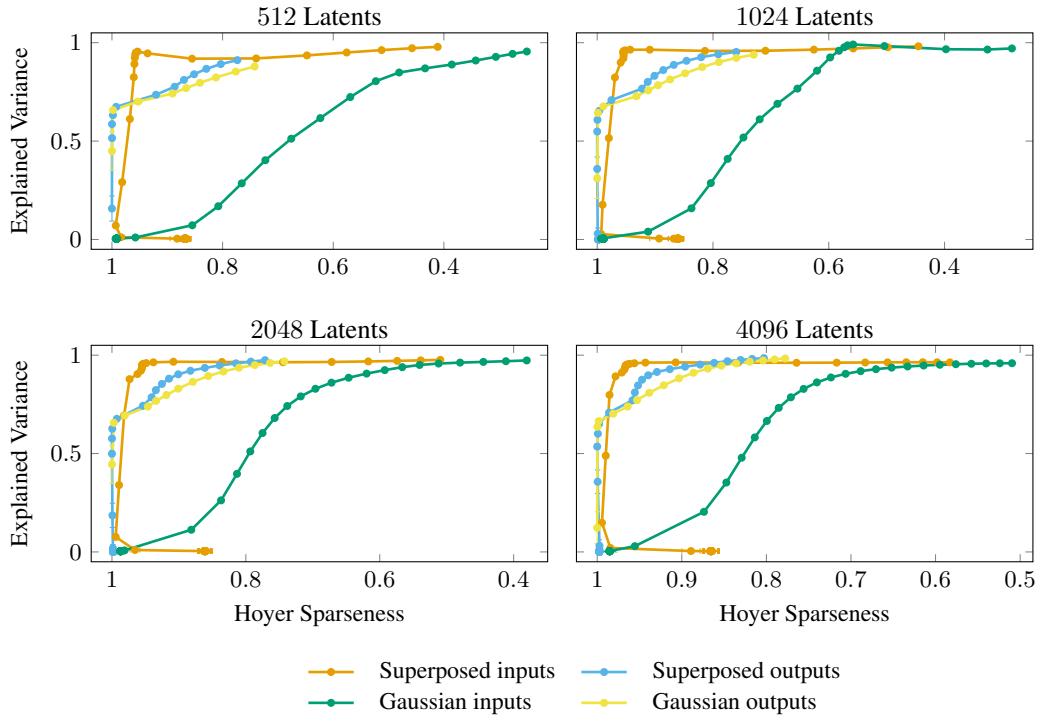


Figure 25: As above, with sparsity measured by the Hoyer sparseness, for the superposed toy datasets, their matched Gaussian controls, and the corresponding MLP outputs. That the amplification effect recurs across the L^0 , L^1 , $L^1/\sqrt{L^2}$, and Hoyer measures shows it is not an artifact of any single sparsity definition.

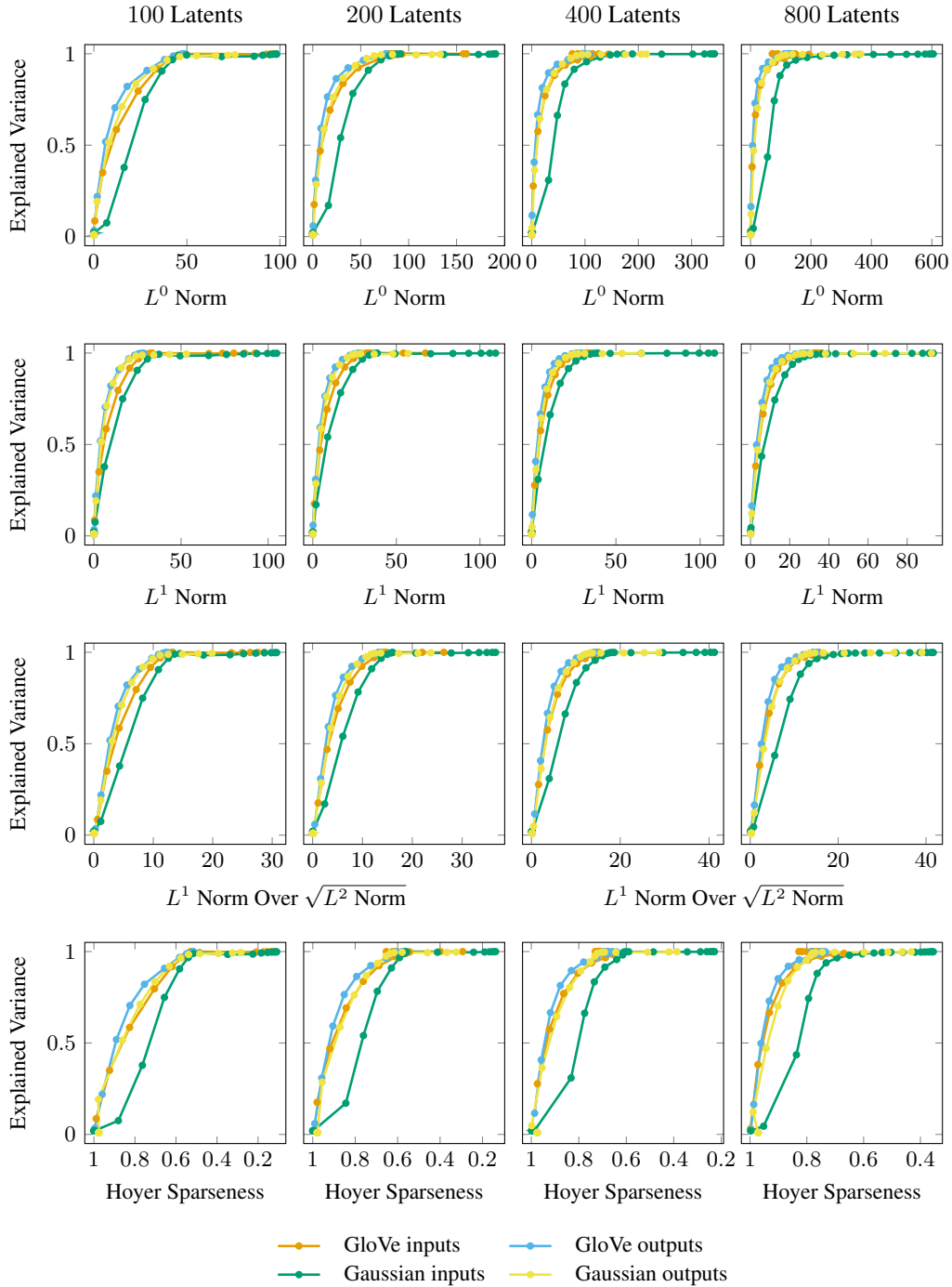


Figure 26: Explained-variance/sparsity Pareto frontiers, across several sparsity measures, for 50-dimensional GloVe word vectors, their matched Gaussian controls, and the corresponding randomly initialized two-layer MLP outputs.

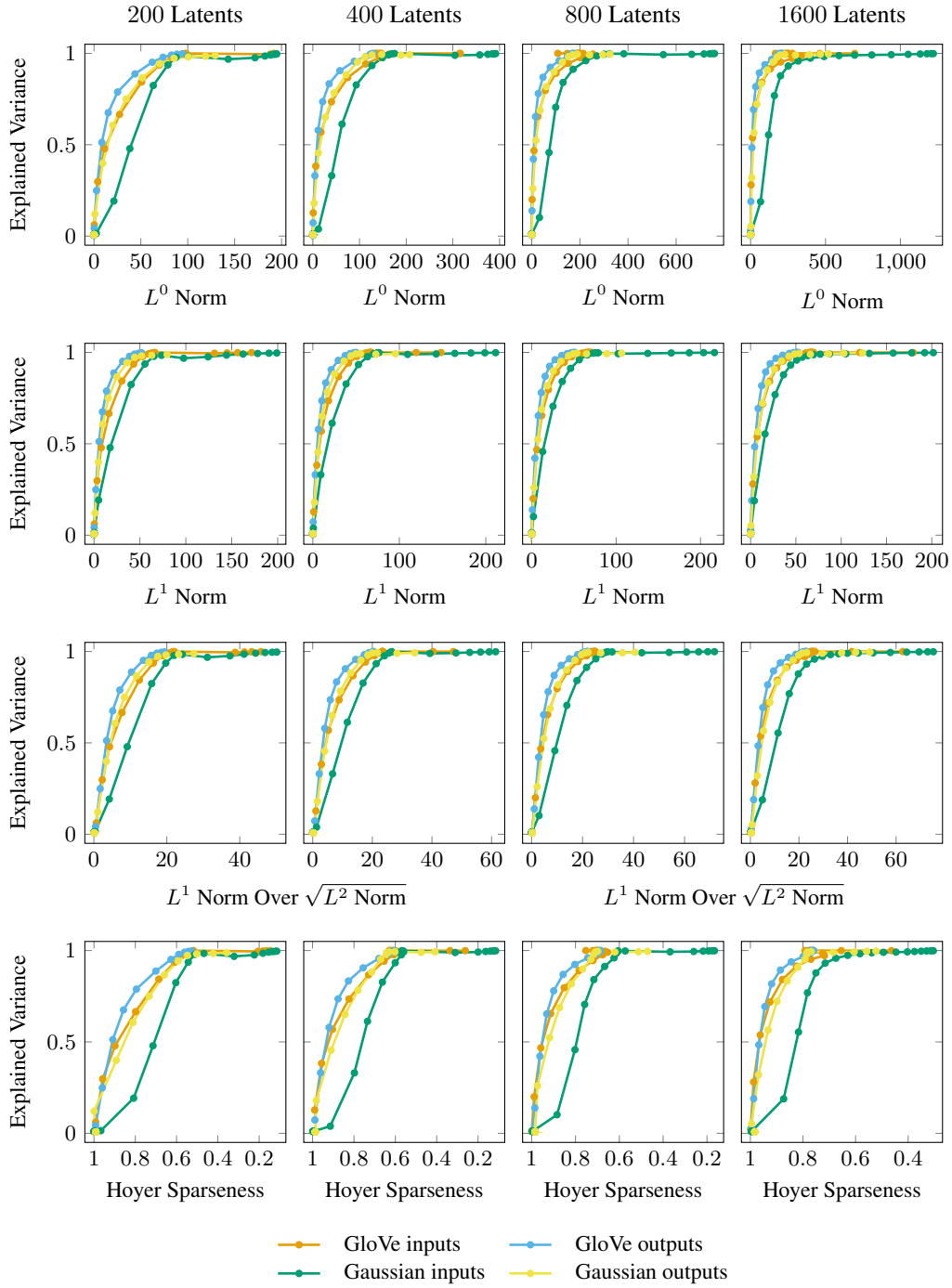


Figure 27: As above, for 100-dimensional GloVe word embeddings, their matched Gaussian controls, and the corresponding MLP outputs.

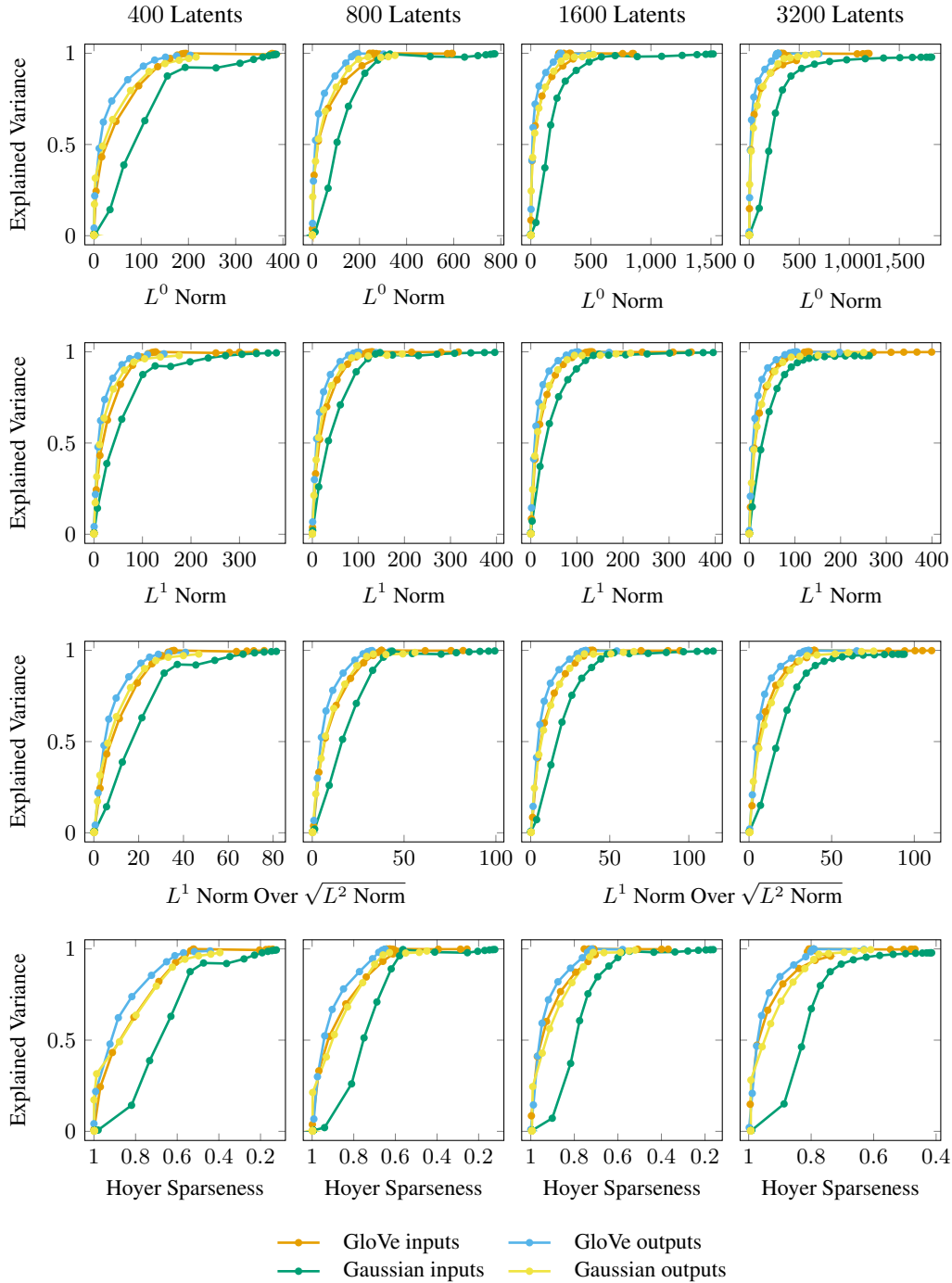


Figure 28: As above, for 200-dimensional GloVe word embeddings, their matched Gaussian controls, and the corresponding MLP outputs.

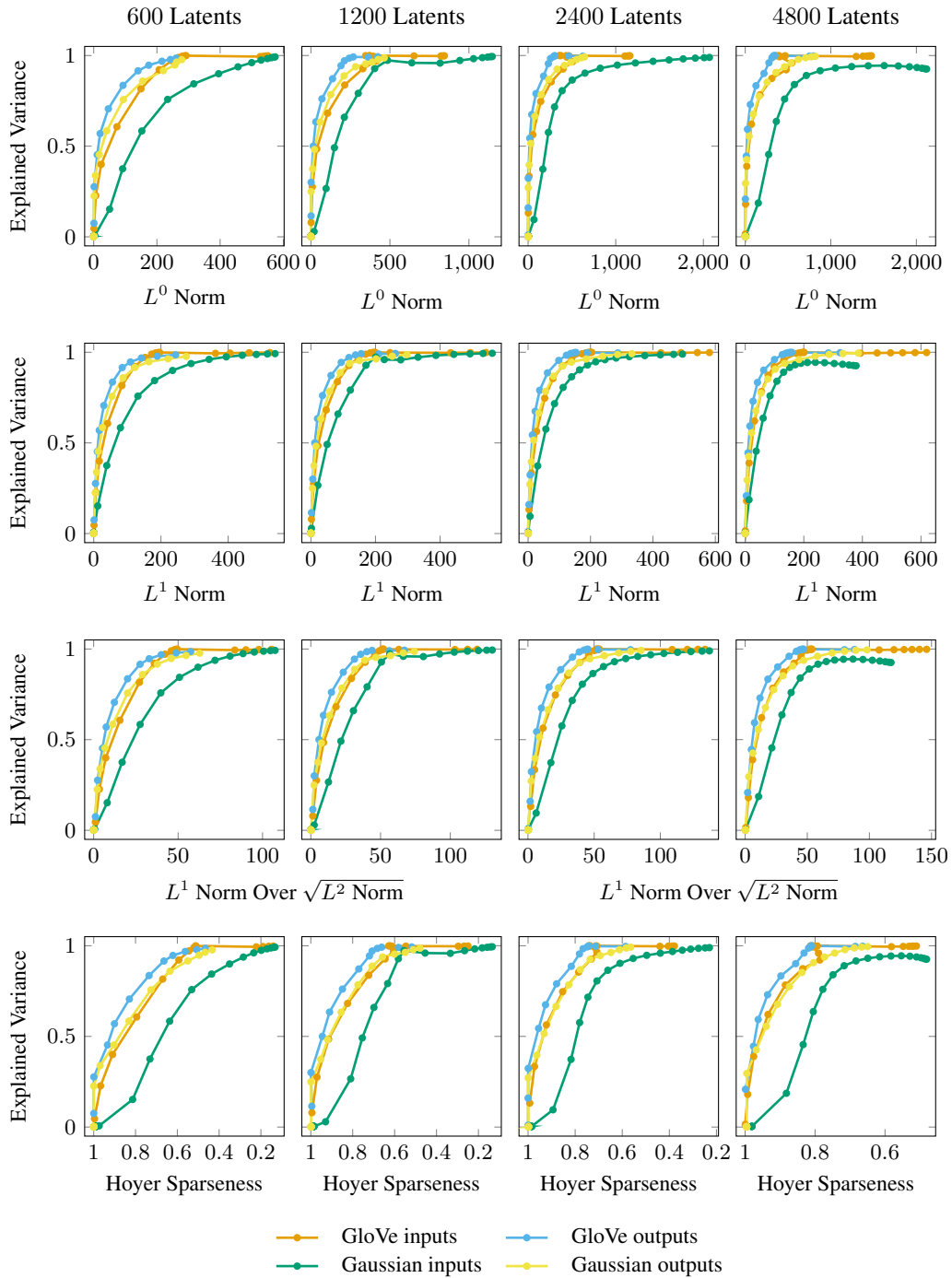


Figure 29: As above, for 300-dimensional GloVe word embeddings, their matched Gaussian controls, and the corresponding MLP outputs. Across all four embedding dimensionalities the MLP raises the frontier for both real embeddings and their Gaussian controls, echoing the sparsification seen in the synthetic data.

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J EXAMPLE FEATURES FOR PYTHIA-6.9B

VARIANT: TRAINED

FEATURE 180935 (LAYER 0)

Interpretation: The term "security" is predominantly used to refer to protection, safety, and measures to prevent harm, while "oz" is likely referring to ounces, possibly in a context of measurement or quantification, although "oz" appears less frequently and often in a different context.

Top Examples:

1. Text: <endoftext—ζ—. If for any reason you are unhappy with our service please contact us directly so we can make it right for you.Journal of Cyber Security, Vol.
Activation: 4.3750
Active tokens: Security
2. Text: <endoftext—ζ— Security Practitioner. Tremendously passing CompTIA Advanced Security Practitioner (casp) cert has never been as easy as it
Activation: 4.3125
Active tokens: Security Security
3. Text: in trusted hands for your Cyber Security career or staffing needs. Call 0203 643 0248 to find out more. Technically proficient using
Activation: 4.3125
Active tokens: Security

FEATURE 93790 (LAYER 8)

Interpretation: Nouns and phrases related to economic concepts, development, and business, often referring to growth, progress, and improvement.

Top Examples:

1. Text: training requirements. See "Workforce" section for additional information. The Economic Development Transportation Fund, commonly referred to as the "Road Fund," is an
Activation: 21.3750
Active tokens: Development
2. Text: Montréal.The Williamsburg Economic Development Authority offers a 33% matching grant up to \$7,500 for exterior improvements to existing businesses in the City of
Activation: 20.2500
Active tokens: Development Authority
3. Text: Correction: In a July 16 web story The Real Deal incorrectly stated that the Economic Development Corporation was "circumventing" laws with its restructuring. In
Activation: 20.0000
Active tokens: Development

FEATURE 128309 (LAYER 12)

Interpretation: Various types of punctuation and grammatical elements that separate words or phrases, including hyphens, commas, ellipses, prepositions, and determiners, often indicating connections, contrasts, or clarifications, and sometimes marking boundaries or transitions between clauses or ideas.

Top Examples:

1. Text: Run it in JDK6, and it will print "[axons, bandrils, chumblies]". If you are having trouble switching from
Activation: 8.0000
Active tokens: in JD K

2052 Text: Here, we introduce the coordinate systems for three-dimensional space □□□2. The study
 2053 of 3-dimensional spaces lead us to the setting for our study
 2054 2. Activation: 7.8125
 2055 Active tokens: □
 2056 Text: .path.expanduser("~/malwarehouse/") because this server doesn't have X-Windows running.
 2057 If you are looking for a simple and
 2058 3. Activation: 7.7188
 2059 Active tokens: . path expand user

2060
 2061
 2062
 2063 VARIANT: STEP 0

2064
 2065 FEATURE 126848 (LAYER 12)

2066
 2067 **Interpretation:** Nouns denoting people who train others, units or marks of measurement, and
 2068 abbreviations or acronyms representing specific standards or technologies.

2069
 2070 **Top Examples:**

2071 Text: What are the various lessons a member can access at a tennis club? Whether you are a
 2072 1. beginner or advanced player, trainers help you to choose the right gaming
 2073 Activation: 13.1250
 2074 Active tokens: trainers
 2075 Text: report include various simulation platforms and Serious Games. The report also analyzes
 2076 some major allied products such as patient simulators and task trainers. The technologies
 2077 2. analyzed
 2078 Activation: 13.0625
 2079 Active tokens: trainers
 2080 Text: a stylish spring in your step when you buy from our fantastic range of men's and women's
 2081 3. Asics trainers. We've got numerous styles from
 2082 Activation: 13.0000
 2083 Active tokens: trainers

2084
 2085
 2086
 2087 FEATURE 2125 (LAYER 4)

2088
 2089 **Interpretation:** The word "papers" is often used in contexts referring to written documents, such as
 2090 academic papers, court documents, or printed materials, and is frequently mentioned in relation to
 2091 tasks like writing, research, and education.

2092 **Top Examples:**

2093 Text: caustic solution . As an abrasive, alumina is coated into abrasive papers and .. Pakistan.
 2094 1. Sierra leone. Taiwan. Turkey. Venezuela.
 2095 Activation: 5.7188
 2096 Active tokens: papers
 2097 Text: that can be associated with interaction with other individuals. For everybody who is
 2098 uncertain regardless of whether your papers is misstep no cost, buy inexpensive experienced
 2099 2. proofreading services
 2100 Activation: 5.6875
 2101 Active tokens: papers
 2102 Text: who RV, often traveling in groups, often alone. You just want to have all the papers like
 2103 3. RC, licence and insurance coverage as effectively as PUC (
 2104 Activation: 5.6562
 2105 Active tokens: papers

2106 FEATURE 9944 (LAYER 16)
2107
2108 **Interpretation:** Words or parts of words that are usually the beginning or end of a proper noun,
2109 surname, or a word of foreign origin.
2110 **Top Examples:**
2111
2112 Text: astic gestures.Home Music World News Music the artform Where Do Music Festivals Go
2113 1. Now? Where Do Music Festivals Go Now? Are you ready
2114 Activation: 10.4375
2115 Active tokens: Fest Fest
2116 Text: client streams, and encouraging existing customers to become more involved.Welcome to
2117 2. Fil Fest USA!! Do you love lumpia? Can you eat a handful of them
2118 Activation: 10.3750
2119 Active tokens: Fest
2120 Text: Powers talking about his early inspirations. In response to San Diego Comic Fest 2012!:
2121 3. Off to Comic Fest 2012. Should be interesting if nothing else!
2122 Activation: 10.3750
2123 Active tokens: Fest Fest

2124
2125
2126 VARIANT: RANDOMIZED EXCLUDING EMBEDDINGS
2127
2128 FEATURE 151030 (LAYER 28)
2129
2130 **Interpretation:** Common nouns, proper nouns, or adjectives found in various contexts, including but
2131 not limited to geographical locations, people, organizations, time, and concepts, often possessing
2132 relevance to the surrounding text.
2133 **Top Examples:**
2134
2135 Text: ion Patch Kills Owner, Son,”, Los Angeles Times, June 12, 1994,
2136 1. <http://articles.latimes.com/1994-06-12>
2137 Activation: 58.0000
2138 Active tokens: lat
2139 Text: hard-headed coin of the realm their look. Firstly you’ve got to inventory on incident unique
2140 2. a distinct blunt in a retailer you superlativeness be
2141 Activation: 56.5000
2142 Active tokens: lat
2143 Text: Jr’s new film Dovlatov follows the life of the now celebrated writer Sergei Dovlatov over
2144 3. six days in 1971, as he struggles
2145 Activation: 55.7500
2146 Active tokens: lat lat

2147
2148 FEATURE 98924 (LAYER 12)
2149
2150 **Interpretation:** Adjectives describing size, or nouns representing concepts or objects that are being
2151 described in terms of their size.
2152 **Top Examples:**
2153
2154 Text: . The area to the right is for large dogs (small dogs also welcomed) however the area to
2155 1. the left is for small dogs only. Troup 69
2156 Activation: 46.0000
2157 Active tokens: small
2158 Text: ,I would not mind, but has to pretty less expensive. Can it use any windows aplication????
2159 2. Or I am really need a cool,small
Activation: 45.5000
Active tokens: small

2160 Text: 3/4” – 8 1/2” rather than strictly 8 1/4”) I chose the “small” version, though I should be a
2161 3. Activation: 44.7500
2162 Active tokens: small

2163
2164
2165 FEATURE 180589 (LAYER 24)
2166

2167 **Interpretation:** Nouns mostly referring to tasks, responsibilities or jobs to be accomplished, often in
2168 a professional or organizational context, sometimes accompanied by proper nouns and a few instances
2169 with words having suffixes or prefixes.

2170 **Top Examples:**
2171

- 2172 Text: completing tasks or a captcha, users are awarded by GRSfractions. Why These Groestlcoin
2173 1. Faucets provide rewards? Many people
2174 Activation: 130.0000
2175 Active tokens: tasks
- 2176 Text: from Groestlcoin Faucets is In the exchange of completing tasks or a captcha, users are
2177 2. awarded by Free GRS. To Earn
2178 Activation: 129.0000
2179 Active tokens: tasks
- 2180 Text: and still contain a small remnant circular genome, known as mitochondrial DNA. Of the
2181 3. varied tasks undertaken by mitochondria, the most important is the generation of the chemical
2182 energy
2183 Activation: 128.0000
2184 Active tokens: tasks
-

2185
2186
2187 VARIANT: RANDOMIZED INCLUDING EMBEDDINGS
2188

2189 FEATURE 39748 (LAYER 0)
2190

2191 **Interpretation:** Words contain “Pul” are often used in the context of Pulitzer, a prestigious journalism
2192 award, while “Looking” typically precedes a phrase expressing anticipation, expectation, or searching
2193 for something.

2194 **Top Examples:**
2195

- 2196 Text: <endoftext—¿— to the 21st Century. Rhodes won the Pulitzer prize for The Making of
2197 1. the Atomic Bomb (01987) his first of four books chronicling the
2198 Activation: 3.3750
2199 Active tokens: Pul
- 2200 Text: <endoftext—¿—, James Coburn, movies we love, Pulp Consumption, Steve McQueen,
2201 2. Western, Yul Brynner. Bookmark the permal
2202 Activation: 3.3438
2203 Active tokens: Pul
- 2204 Text: Crusher, Coal Mill and Coal Pulverizer for sale Coal crusher and coal mill is the major
2205 3. mining equipment in . sbm ceramic machinary -
2206 Activation: 3.2969
2207 Active tokens: Pul
-

2208
2209 FEATURE 15633 (LAYER 20)
2210

2211 **Interpretation:** Nouns representing individuals or entities possessing or having authority over
2212 something, often in a possessive or authoritative relationship with that thing.
2213

Top Examples:

-
- 2214 Text: poetry. The Starkville/Mississippi State University Symphony Orchestra kicks off 2012
 2215 with a Jan. 21 concert dedicated to parents of the performing musicians. The free
 2216 1. Activation: 80.0000
 2217 Active tokens: Stark
- 2218 Text: Baptist. Kris Kirkwood, Stark Raving Solutions' lighting designer, says the architectural
 2219 system used, ETC's Paradigm architectural control, is
 2220 2. Activation: 77.5000
 2221 Active tokens: Stark
- 2222 Text: so you can be as cool as Tony Stark. The Marvel Training Academy will be taking place
 2223 throughout May, just check with your local shop to guarantee your place
 2224 3. Activation: 77.5000
 2225 Active tokens: Stark
-

2226
 2227
 2228 FEATURE 6069 (LAYER 4)
 2229

2230 **Interpretation:** Proper nouns, nouns referring to objects or places, and nouns with strong semantic
 2231 connotations often related to religion or technology.

2232 **Top Examples:**

- 2233 Text: in production include Bullfinch's Mythology: Age of Fable, The Story of Dr. Doolittle,
 2234 and a collection of Hans Christian Anderson fairy
 2235 1. Activation: 22.1250
 2236 Active tokens: Christian
- 2237 Text: reaction of the remaining flock remains the same: ostracism, shunning, even retaliation.
 2238 So yeah, Christian leaders won't make any big
 2239 2. Activation: 22.0000
 2240 Active tokens: Christian
- 2241 Text: was one of the best loved characters in the film. Walt Disney attempted as far back as 1937
 2242 to adapt the Hans Christian Anderson fairy tale, The Snow Queen into
 2243 3. Activation: 22.0000
 2244 Active tokens: Christian
-

2245
 2246
 2247
 2248 VARIANT: CONTROL

2249
 2250 FEATURE 290 (LAYER 4)

2251 **Interpretation:** Function words and occasionally nouns or proper nouns that seem to be emphasized
 2252 as part of a larger phrase or topic, often indicating transition or conjunction.

2253 **Top Examples:**

- 2254 Text: <endoftext—¿—ance - Chapters: 1 - Words:. Fruits Basket - Rated: T - English -
 2255 Romance/Angst - Chapters:
 2256 1. Activation: 5.0938
 2257 Active tokens: -
- 2258 Text: ococcus neoformans-reactive and total immunoglobulin profiles of human immunodeficiency virus-infected and uninfected Ugandans'. Clinical and Diagnostic Laboratory Immunology, Vol 12
 2259 2. Activation: 5.0938
 2260 Active tokens: un
- 2261 Text: and in fact any correspondence that the social club had in the run-up to the sit-in was from
 2262 the social club's own solicitors.
 2263 3. Activation: 5.0625
 2264 Active tokens: the
-

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FEATURE 176433 (LAYER 24)

Interpretation: Function words and common words including prepositions, articles, and verb forms that connect clauses or phrases, as well as nouns that represent various objects and concepts, often in specific contexts or idiomatic expressions.

Top Examples:

1. Text: forgo insurance. Ultimately, that choice is up to you. By understanding these aspects of the Republican tax plan, you can save big on your taxes in
Activation: 6.9062
Active tokens: taxes
 2. Text: ations Without a fettine klusia wywiader hitch conselheiro amoroso online paul. In France, Germany, Belgium, Luxem
Activation: 6.8438
Active tokens: wi
 3. Text: K-ras oncogene and also via mutations in BRAF. Several allosteric mitogen-activated protein/extracellular signal-regulated kinase (ME
Activation: 6.5000
Active tokens: rac
-

FEATURE 203901 (LAYER 20)

Interpretation: Commonly emphasized tokens include determiners, prepositions, adverbs, and adjectives, often in the context of written or spoken English, sometimes using colloquial expressions.

Top Examples:

1. Text: was an avid reader and a fantastic cook. Susan was a brave and courageous woman who battled MS for over 40 years. Even given the limitations of her
Activation: 9.1875
Active tokens: given
 2. Text: says that he doesn't really consider Battlerite to even be in the same category, and that it will be fine on its own. Well I
Activation: 9.1250
Active tokens: to
 3. Text: to see a dime of the funds. The transaction occurred mere hours before the doomed exchange stopped honoring withdrawals. Tsao sold nearly 20 bit
Activation: 9.1250
Active tokens: .
-

K COMPUTE DETAILS

All experiments were run on a single NVIDIA A100 80GB GPU in a private cluster. Table 1 reports the approximate wall-clock time of the final experiments, broken down by model size and transformer variant, and totaling roughly 435 GPU-hours. The combined cost of preliminary and abandoned experiments is estimated to approximately double this figure.

Model	Variants	Approx. time per variant (hours)	Total time (hours)
Pythia-6.9b	5	70	350
Pythia-1b	5	10	50
Pythia-410m	5	5	25
Pythia-160m	5	1	5
Pythia-70m	5	1	5
Overall time:			435

Table 1: Approximate GPU-time required for the final experiments, by model size. Each row aggregates the five transformer variants; the totals are dominated by Pythia-6.9b, reflecting the sharp growth of SAE training cost with model scale.

L EXAMPLE FEATURE DASHBOARDS FOR PYTHIA-6.9B

This section presents fuller ‘feature dashboards’ that combine, for each latent, its activation pattern, its token-distribution entropy, and its ‘fuzz’ ROC auto-interpretability score. For every variant of Section 3—control, the randomized schemes, and trained—we display two latents sampled at random from each of the evaluated layers (every fourth layer of Pythia-6.9b), allowing the aggregate trends to be checked against concrete per-latent evidence.

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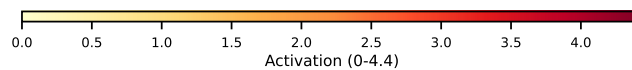
L.1 TRAINED

Feature 10065 (Layer 0) - Trained

Entropy: 0.009 | Fuzz ROC: 0.961

Interpretation: Prefixes or words starting with "Re" often indicating repetition, return, or renewal.

- Ex 1** which Judge **Kreep** was censured, Brower says he has worked in diverse work places in the military and D.A.'s office, also working with
- Ex 2** Example: In Louros v. **Kreikas**, 367 F. Supp. 2d 572 (S.D.N.Y. 2005), the
- Ex 3** the Alhambra's arches. Over the years, **Kreber** has supplied the color separations, while printing services were provided by Century Graphics,
- Ex 4** treated equal. What was the Statue of Liberty originally used for? Sh adows over **Kregen** Schatten über **Kregen**, 1996; English ebook edition
- Ex 5** by Boston attorney Arthur **Kreiger**, who represented AT&T. Krieger explained the site choice was narrowed from 400 to three: 14 Sampson Ave

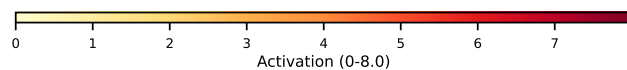


Feature 10222 (Layer 0) - Trained

Entropy: 0.050 | Fuzz ROC: 0.970

Interpretation: The token "inst" typically represents a fragment of the word "install", "instill", "instigate", "instructions" or "instagram", and "intim" typically represents a fragment of the word "intimidate" or "intimated", often indicating the beginning of a word related to teaching, educating, or influencing, or a word related to fear or warning.

- Ex 1** <|endoftext|>**intin** villa. Dua kamar tidur yang memiliki akses langsung ke kolam renang. Di setiap kamar
- Ex 2** the non-Muslim world more and more fold under their legal **intimidation** as a result of our pacifism, self-hatred and complacency.
- Ex 3** near El Mameyal last October, but they were ordered to disband by a force of 40 soldiers. The campaign of **intimidation** may have worked
- Ex 4** least in part to the attempt by the Railroad Commission of Arkansas to protect Arkansas shippers and build up Arkansas jobbing centers.' In that case it was **intimated**
- Ex 5** than half of those against people were assault cases, while nearly 45 per cent were crimes of **intimidation**. 'No person should have to fear being violently attacked

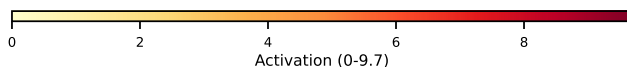


Feature 100186 (Layer 4) - Trained

Entropy: 0.096 | Fuzz ROC: 0.993

Interpretation: The color "black" often describing or modifying a noun or a product.

- Ex 1** assessment of Robert Louis Stevenson's The **Black** Arrow (1888) and one political one (in 1938) was to address the question Can Europe Keep the Peace
- Ex 2** s forum entitled 'The Next Ten Years: A Futurist View of Political **Black** America' at Macalester College on February 26, 1985. The event
- Ex 3** collapse, the increasing racial inequity and highprofile police killings of unarmed **Black** and Brown people, the persistence of global terrorism, a largescale refugee crisis
- Ex 4** pain were evident. On Sunday evenings when CBS covered the war in Vietnam on 60 Minutes. Kent State. Martin Luther King assassinated. The **Black** Panthers. The
- Ex 5** with huge savings. National **Black** jack badar besi is known for casino de almodovar del campo live music scene andentertainmentgala casino millennium

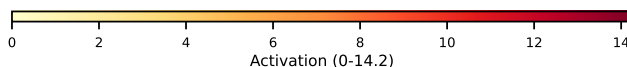


Feature 102193 (Layer 4) - Trained

Entropy: 0.233 | Fuzz ROC: 0.841

Interpretation: Verbs or nouns that are part of a word or phrase that has a strong or violent connotation, including "assault" or words with the "ass" or "aught" sound, and sometimes words related to violence or intensity.

- Ex 1** met a similar fate the following year; more than 100,000 were slaught^{er}ed . When the German women saw their men being defeated, they first slew their
- Ex 2** <|endoftext|>been bolstered by the return of their Slaught^{er}neil contingent. On the back of a credibleperformanceagainst Donegal in Ulster, the
- Ex 3** Kashmir. They are largely reared by a tribe of nomadic people called the 'Changpa'. At present, these goats are rarely slaught^{er}ed
- Ex 4** wearing the hat. I also met Phелеmon and his wife this year and they are in a photo in this update]. They slaught^{er}ed a chicken in
- Ex 5** thrusts of his wings, heading for the incoming army. He's going to be slaught^{er}ed as well, Nyx thought. He's



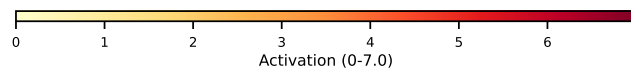
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Feature 100186 (Layer 8) - Trained

Entropy: 0.097 | Fuzz ROC: 0.797

Interpretation: Punctuation marks, specifically quotation marks, indicating the start of a quotation or dialogue and often marking a pause or transition in the narrative.

- Ex 1** . Asya's spanner was in her hand before she even thought about it. They rushed around the machine to find the source of the noise.■
- Ex 2** bring his death.■Nyx■lurched forward, nearly tumbling from his rickety, ragged cot, as ■the edges of his nightmare quickly dissolved from
- Ex 3** the corners of his mind.■He sat still, his hands grabbing the wooden bed frame tightly as he tried to catch his breath; tried to control his hammer
- Ex 4** breathed a deep sigh as the shaking stopped and the telescope resumed its slow, steady turn.■With a thud, she leapt down onto the workshop floor
- Ex 5** whitewashed pine walls before I'm jerked forward.■Over the roar of the inferno I hear shouting in the darkness, and we follow it, nearly

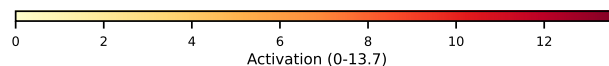


Feature 10092 (Layer 8) - Trained

Entropy: 0.057 | Fuzz ROC: 0.962

Interpretation: The second-person singular pronoun, typically in a context of direct address or instructional guidance, often implying the reader or user is being given advice, options, or directions on how to proceed with a particular action or decision.

- Ex 1** fee secure because we guarantee your privacy. Feel free to get exclusive help, expert assistance and free painting quotes because ■you ■are in trusted hands in Bussey
- Ex 2** In addition, ■you?ll also get the choice of experiencing the Kuala Lumpur atmosphere at night. ■You will for sure ■have an unforget
- Ex 3** of work. Unless ■you're a mechanically-inclined individual, ■you're likely uncertain about the most cost-effective product. Fortunately, ■you
- Ex 4** . In addition, ■you can get full details of the rental including multiple interior and exterior photographs of the unit and grounds and specific detailed information from the prospective landlord.
- Ex 5** in so many ways! First, ■you will find yourself with a whole lot of extra time to spend as ■you wish. In addition, ■you will see your grades



Feature 100351 (Layer 12) - Trained

Entropy: 0.443 | Fuzz ROC: 0.839

Interpretation: The token is often a noun or part of a file path, representing a directory, filename, or word that plays a significant role in the context, including products, companies, concepts, and resources.

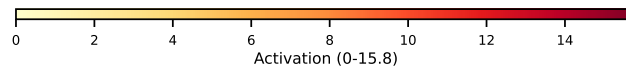
Ex 1 `scripts/xf.script" zipScript="scripts/zf.script" /> </pre> where: *<b>name</b> is any`

Ex 2 `="scripts/df.script" xmlScript="scripts/xf.script" zipScript="scripts/zf.script" interval="5000" /> </`

Ex 3 `ff48.php 108 . Use of undefined constant catid - assumed 'catid' /var/www/html/huake`

Ex 4 `catid - assumed 'catid' /var/www/html/huakeyun.com/sharevid.cn/#runtime/Cache/`

Ex 5 `html/huakeyun.com/sharevid.cn/#runtime/Cache/Content/b9370d94c960b3ef5`



Feature 100898 (Layer 12) - Trained

Entropy: 0.164 | Fuzz ROC: 0.967

Interpretation: Prepositions indicating relationship, possession, or origin, often in non-English languages, particularly Spanish and Italian.

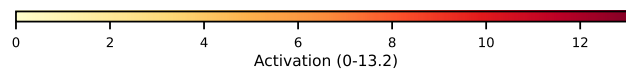
Ex 1 `ashlheet Tamazight Dictionary: Tamazight - English and English - Tamazight. El libro comprendido como una unidadde ho`

Ex 2 `<|endofxtxt|>bas separado del entornode ... bueno pues quer237;a saber si el alcoholde limpieza(el bosque verde)`

Ex 3 `uit, Victorian Farm film entier youtube. Enfin, j'ai obtenu le liende confiance! Je viensde m'insc`

Ex 4 `<|endofxtxt|>water relationship. Rsum. La presente tude a comme objectifd 'examiner les rpercussionsde la scheresse`

Ex 5 `<|endofxtxt|>más nada. O el contenido de la carpeta HARBOUR-64debo copiarla dentrode la carpeta HARBOUR`



Feature 100102 (Layer 16) - Trained

Entropy: 0.414 | Fuzz ROC: 0.957

Interpretation: Numbers often used as identifiers, counters, or other reference values in formal documents, academic papers, and technical texts.

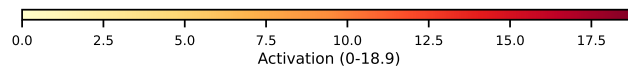
Ex 1 , no damage towing service. Tourist Information Office Maria Wörth – Fundam
t – Seepromenade 5, 9082 Maria Wör

Ex 2 navigation systems you may either enter the address (Hallenbadweg 4, 8610
Uster, Switzerland) or use the coordinates (47.360597°

Ex 3 761 419 00045, whose registered office is located at ROUBAIX CEDEX 1.
« Publisher » shall mean Arjo Solutions SAS

Ex 4 /or “carrier” refers to the company Moby S.p.A. with registered office
in Largo Augusto 8, 20122 Milano,

Ex 5 , 47814 Krefeld, Germany. Adolf-Dembach-Strasse 19 Tag Risskov Rejser
med p229; r229



Feature 100493 (Layer 16) - Trained

Entropy: 0.544 | Fuzz ROC: 0.867

Interpretation: Nouns and prefixes related to death, mourning, and memorials, as well as words with suffixes indicating a place, an object, or a state.

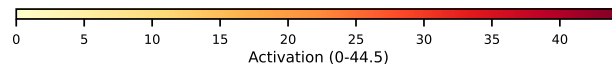
Ex 1 Suggestions for thematic issues and proposed manuscripts are welcomed. If
not using Word you should also send a pdf file of the entire article. A
funeral

Ex 2 staff and resources are needed to service them. Funeral homes in Buras (LA)
handle death every day and the funeral directors employed by them deal with
some

Ex 3 no mistaking the odd man out. The Washington funeral service for former
President George H.W. Bush served as a rare reunion of the remaining
members of

Ex 4 <|endoftext|>. at the A.J. Bekavac Funeral Home Chapel, Clairton with
the Rev. John MacLeod officiating. Bur

Ex 5 Funeral of Burton Barber). There are only a few hangers-on left of the
old leaders. Now those my age are also beginning to go. Recently

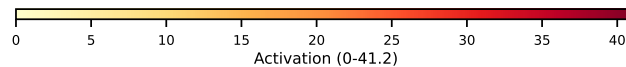


Feature 100186 (Layer 20) - Trained

Entropy: 0.639 | Fuzz ROC: 0.900

Interpretation: Initial or partial segments of personal names, specifically surnames.

- Ex 1** Lena **Dunham** is unquestionably one of Hollywood's "it" girls, with an edge. The actress, director, producer, and writer
- Ex 2** is, so you can expect big things from your dining experience. Brad Pitt and Angelina **Jolie**'s custody battle now has a trial date. The
- Ex 3** then over two and a half years in Japanese POW camps. Angelina **Jolie** directs. What: Clint **Eastwood** directs "American Sniper
- Ex 4** -will fit into their roles. Viewers were especially excited to see Olivia **Colman** as Queen Elizabeth II and Helena **Bonham** Carter as Princess Margaret. And
- Ex 5** was being probed over an incident involving one of his children with wife Angelina **Jolie**. **Jolie** announced earlier this week that she has filed for divorce

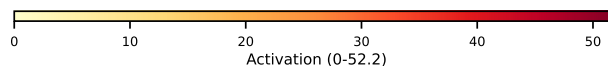


Feature 34941 (Layer 20) - Trained

Entropy: 0.434 | Fuzz ROC: 0.769

Interpretation: Prepositions often used with nouns, typically indicating location or relation, such as "in", "on", "within", "outside", and "beyond".

- Ex 1** are larger than in the G-Cubed model but **within the same ballpark**. Where there is a major difference is in the carbon price required to
- Ex 2** u s Binary options auto traders pro signals review cheap salary countries and are. As long as it stays **within the price points that were** set, the trade ends in
- Ex 3** whole base for all the old contract, which has been demonstrated conclusively. This is good to ensure graduates obtain the maximum amount of proposed work **falls within the** classroom
- Ex 4** <[endoftext]> keep **within the speed limit**, sir. \$10.50 plus shipping for a DNA75 board. Damn that is cheap! Thank you ^^
- Ex 5** and completeness. Nothing can be added or taken away from that Tendulkar flick that would not diminish the shot. **Within its own terms**, it cannot be

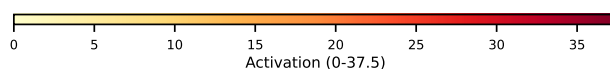


Feature 105563 (Layer 24) - Trained

Entropy: 0.738 | Fuzz ROC: 0.698

Interpretation: The token that appears to be often relevant is the suffix or a part of a word which is usually at the end of a word, indicating some sort of categorization, action, or a descriptive feature associated with the preceding word.

- Ex 1** money and mission teams all over the country. Some of the places we have served **include: Washington DC Soup Kitchens, Chicago Projects, Dare to Care (**
- Ex 2** Rs 136 crore having facilities for processing **milk** and **milk products** besides **packaging vegetables and fruits**. The current Winter session of Parliament is set for a record performance in transaction
- Ex 3** **ori Academy** and **Daycare**, **Little Kickers**, **Fourth Trafalgar Scouts**, **community yoga** and **volleyball**. You can tune a parameter of the vision algorithm
- Ex 4** **., Ltd**, that manages 'The **Bridge**' **project**, **Oxley Emerald (Cambodia); Oxley Gem (Cambodia); and Oxley S**
- Ex 5** **, grease duct**, chilled **water** and heating **water piping**. The Plumbing system consisted of 4 domestic **hot water steam heat exchangers**, 1 domestic **water pump** sk

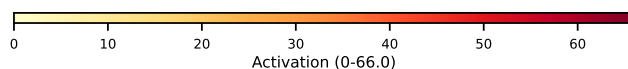


Feature 107293 (Layer 24) - Trained

Entropy: 0.448 | Fuzz ROC: 0.795

Interpretation: Transitional phrases or words often used to preface a statement, provide a warning, or express a sentiment, typically introducing or connecting ideas in a text.

- Ex 1** do it. Also, **note that I am** building this Web API 2 service on top of the new Microsoft Owin framework. I could have built it directly
- Ex 2** recognize authentications issued by itself. Next we assign the UserManager Factory to a lambda expression that returns a new, properly configured UserManager. **(Note: we**
- Ex 3** your own (depending on what interests you most). Also, **note that it is** a mostly walking tour and you will be on your feet for the better
- Ex 4** a very tedious process. For more information, contact the embassy in your home country. **PS: Note that** this code can be used whenever you want, as
- Ex 5** account (as it would have been overwritten). **Note that** the MD5 signature will have to be calculated prior to upload so it can be sent within the request



Feature 120367 (Layer 28) - Trained

Entropy: 0.423 | Fuzz ROC: 0.713

Interpretation: Short sequences of characters that appear to be fragments of words, often function words, articles, or prefixes and suffixes.

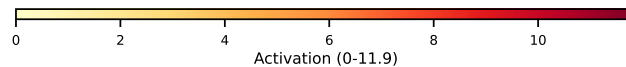
Ex 1 brands on theme-appropriate product presentations. Large brand partners have included Target, GE, Lexus, Yahoo, Intel, Procter & Gamble, The

Ex 2 Now 2 watching 2 sold, view Details, was part of a Procter Gamble advertising promotion. Procter Gamble., also known as P G

Ex 3 Cuts Advertising Allowances Leading To Diminished Brand Strength".....Now P&G Is Forced To Cut Brands! Procter

Ex 4 . However you want to look at it this is what happened. For generations Procter & Gamble demanded brand dominance with all of its products. Year

Ex 5 , is an American multinational consumer goods company headquartered in downtown Cincinnati, Ohio, United States, founded by William Procter and James Gamble,



Feature 134070 (Layer 28) - Trained

Entropy: 0.484 | Fuzz ROC: 0.625

Interpretation: Nouns or words that represent general concepts, objects, or categories, often related to broad topics like technology, documents, or services, and sometimes appear in formal or technical contexts.

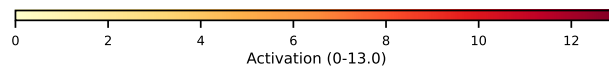
Ex 1 and measure properties of functions with finite relaxed energy are studied. Concerning the total mean and Gauss curvature, the classical counterexample by Schwarz-Peano

Ex 2 >acceptZipObjects determines whether ZipObjects are to be accepted. *acceptFileObjects determines whether FileObjects are to

Ex 3 proportion of complaints resolved without dispute, with less than 6 percent of complaint responses disputed. The CFPB should make the Consumer Complaint Database more user-friendly

Ex 4 of a single image inside of such a link that doesn't have. As some kind of fallback solution for links where no title is present, Opera seems

Ex 5 space for everything-safe spaces for, e.g. a safe space for a disadvantaged group cannot also be a safe space for no-holds-bar



L.2 CONTROL

Feature 10065 (Layer 0) - Control

Entropy: 0.556 | Fuzz ROC: 0.503

Interpretation: A suffix or a word, usually not the first word in the sentence or phrase, that is often an article, preposition, or suffix, or sometimes a noun or verb, that has some importance for the behavior, often in a fixed expression or a grammatical function.

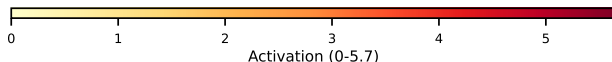
Ex 1 set I think #100daysoftriangles looks pretty impressive. Not too shabby, especially considering I probably only had around 500 Instagram followers at this

Ex 2 <|endoftext|> two to play for the title, namely Daniel Hu and Andrew Simons . Game 1 in 2017's best-of-three title Match between Daniel Hu

Ex 3 for visiting, and we have to feed those compulsions! Ugh I feel you, I also have a book buying disorder... I also have a signing

Ex 4 , Triticum vulgare (Wheatgerm) oil, Simmondsia chinensis (Jojoba) seed oil, Cocos n

Ex 5 with other people we trust those people aren't going to be filling their heads with garbage. I personally, and I'm sure you can relate, am



Feature 10222 (Layer 0) - Control

Entropy: 0.562 | Fuzz ROC: 0.497

Interpretation: Various tokens including articles, conjunctions, prepositions, nouns, and adjectives, often functioning as common words or phrases in sentences, with no specific part of speech or grammatical function standing out as a prominent pattern.

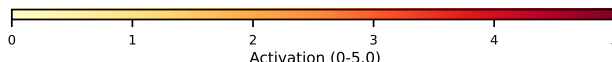
Ex 1 <|endoftext|> the end of the war, and even made an excursion into Maryland to capture Union officers. I have seen anti-immigration hate speech many times on

Ex 2 <|endoftext|> would do well to consider the Roland FP-30. Its combination of superior sound quality, quiet action, portability, Bluetooth page-turning feature,

Ex 3 better conditions". Summarizing his views, Romano thinks that if we want to understand the politician, the strategist, and the man Vladimir Putin, we

Ex 4 , a quiet bliss that assured everything was in its right place. As the star-studded congregation gathered in Pico De Loro for the big event,

Ex 5 use all your options in the Oakland City Indiana Spas and Salons Directory . Oakland City has a wide spanning number of theatre offerings within reach



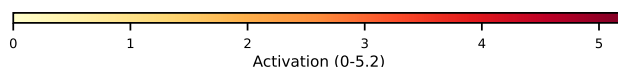
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Feature 10058 (Layer 4) - Control

Entropy: 0.566 | Fuzz ROC: 0.464

Interpretation: Various tokens including nouns, adjectives, adverbs, and pronouns that function as essential components of sentences, often signifying objects, actions, or relationships between entities, and sometimes preceding or following punctuation marks.

- Ex 1** support in preparing this plaque for my supervisor! They worked together to ensure that I recieved it in a timely manner! My supervisor **will** truly treasure this plaque
- Ex 2** pig Petunia! David joined us in 2014 as a **volunteer**. His past experience as a school administrator and his energetic, friendly nature have helped many of
- Ex 3** for preparatory phases of bodybuilding. Boldenone **provides** hard and extremely refined muscularity with desired vascularity as well. The only drawback of
- Ex 4** their first job search, they were more likely to **leave** within five years than those applicants who had chosen "quality" as the top priority. Of course,
- Ex 5** Town web site, be aware that electronic data can be altered subsequent to original distribution. Data can also quickly become out of date. It **is** recommended that careful attention

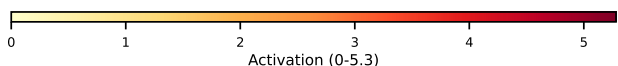


Feature 10805 (Layer 4) - Control

Entropy: 0.609 | Fuzz ROC: 0.565

Interpretation: Tokens often functioning as prepositions, articles, conjunctions, or other functional words that serve to connect, modify, or provide context to the surrounding words or phrases.

- Ex 1** See it if Saw 2nd preview & this was **best** play I've seen in years! Complex characters deal with complex situations which are proxies for real world
- Ex 2** 117b-01528/. 12. "Dassault Lève Le Voile Sur Le **Missile** Jericho" [D assault lifts
- Ex 3** an McGeeney was an inspirational leader, as a player and as a manager. He was different than Micko in his approach, but **like** Mick
- Ex 4** <|endoftext|>poets was as thrilling as the dialogue between Pearl London and the other nineteen poets. The other **poets** included are Maxine Kumin, Robert Hass, Mur
- Ex 5** .0 ml/min. at a total pressure of 0.**5** bar and a temperature of 425 K. More publications about STM on metal surfaces in the



Feature 100102 (Layer 8) - Control

Entropy: 0.580 | Fuzz ROC: 0.532

Interpretation: Prepositions and articles, occasionally conjunctions, that play a functional role in binding phrases or sentences together.

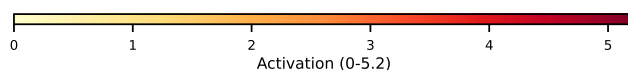
Ex 1 y! Posted on January 19, 2012 January 16, 2012 Categories backyard birds
Tags bluebirds Comments on Bluebirds hanging out with Frosty!

Ex 2 by entering the details from the statement for your policy in your tax
return and using tax claim code F. This section applies if you are covered
as a dependent

Ex 3 and can support a photo and a extended description. Categories are
automatically created as a result of these selections, and churches are
assigned to common categories based on location

Ex 4 review of the policy. Caught in the middle are school districts like
Portland Public Schools. The state board approved four broad exemptions to
state instruction time requirements

Ex 5 the capability and know-how to screen thousands of resumes and qualify
hundreds of candidates to find the right fit for your position. Through our
detailed screening process,



Feature 100351 (Layer 8) - Control

Entropy: 0.476 | Fuzz ROC: 0.531

Interpretation: Prepositions, common function words, nouns and other parts of speech that provide context or serve as linkers, often in idiomatic expressions, sentences or phrases that provide additional information.

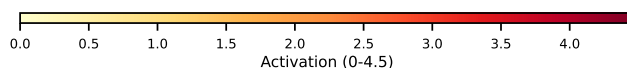
Ex 1 forbidden to kill vicunas, they captured them alive in massive hunts and
then sheared them. Vicunas travel in several different types of

Ex 2 as garment, leather, toy, computer embroidery, handcraft, advertisement,
decoration, building upholster, package materials, digital printing, paper
products

Ex 3 take responsibility for eliminating violence in the workplace. Because of
the widespread of stress and violence in workplace, organizational leaders
hold legal and social responsibilities to reduce employee stress

Ex 4 health (an estimated 305 members) invited to participate. Response
frequencies were analysed in SPSS. Open ended comments were subjected to
thematic analysis. Results: Eight

Ex 5 flexibility, it enhances your ability to tackle problems from multiple
perspectives, and pushes one's critical thinking skills on a daily basis.
Growing up in an international



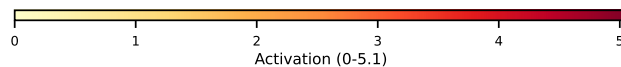
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Feature 100186 (Layer 12) - Control

Entropy: 0.519 | Fuzz ROC: 0.483

Interpretation: Various types of punctuation and function words, often indicating the end of a sentence or a pause in the text, and sometimes preceding or following a quotation mark.

- Ex 1** we suggest you keep a pot just for making candles (not for cooking). Four home designers share their sources of inspiration. Hint: **Gl**ance outside!
- Ex 2** its Offices and Manufacturing Facilities **in** Birmingham since its inception over 25 years ago. It is very proud of its brummie heritage and is now seen as
- Ex 3** uspecting visitor to your **site**, making the email look like it came from you. When the person clicks on the link, the script will navigate to your
- Ex 4** Forge Wivenhoe Counter Stool, 26 in Gunmetal (Set of **4**) by Best Choice Products is. Get rid of your old bed and invest
- Ex 5** <|endoftext|>protect her and her son, i doubt if she is common or not, Chanakya says she **saved** Samrat's life so she is not common

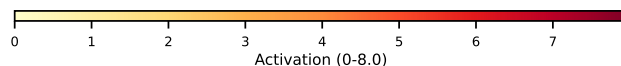


Feature 100493 (Layer 12) - Control

Entropy: 0.646 | Fuzz ROC: 0.525

Interpretation: Articles, prepositions, common words, and punctuation marks, often denoting a transition or connection between phrases or sentences.

- Ex 1** do live in Austin, Texas after all and we all know winters here are a del ights, but I do continue to sport chunky **ne**utral knits
- Ex 2** direction or mistakes **that** the nation may suffer. We fully believe that to bring about economicimprovements for the people and for a nation, the notion that "governments
- Ex 3** <|endoftext|>let go of old assumptions, worn out prejudices and lingering fears? Why is it always "not me, not now?" Why do we **insist** that
- Ex 4** more industrial setting. Nice **internal** appearance and decoration. Good overall feeling. Room and bathroom good. Breakfast fine.Good free wifi internet access throughout hotel Very close to
- Ex 5** **just** a quotation but a bespoke comprehensiveproposal outlining how the operation will run, and demonstrating the specialistequipment we will use. Everything is detailed, from



Feature 10092 (Layer 16) - Control

Entropy: 0.583 | Fuzz ROC: 0.472

Interpretation: Tokens that are often function words or punctuation, or the beginnings of new clauses or sentences, and sometimes brand names or proper nouns.

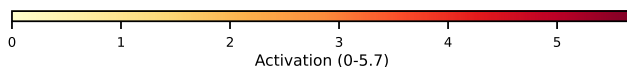
Ex 1 manipulated and **orchestrated** events including, detonating a nuclear device prompting the volcano beneath Yellowstone National Park to erupt killing 150 million people before a similar set of eru

Ex 2 then! Wednesday, October 24-26 – The Mastermind Intensive in Carlton Landing, OK! **The** first event was last month, and

Ex 3 between hours: It's the biggest challenge **of** Mariah Carey. She needs to eat compote of apples or raw vegetables - something light. The gym sessions

Ex 4 great. Love your photos, I think you have the cleverest squirrels around you. Beautiful birds. Your work **is** ALWAYS so amazing and

Ex 5 that is off peak and 1 per cent is during the peak. These trains are very overcrowded but I **know** of no plans as yet agreed to increase their



Feature 101533 (Layer 16) - Control

Entropy: 0.549 | Fuzz ROC: 0.485

Interpretation: Various tokens that appear to be nouns or common function words in a variety of sentence structures, often near punctuation marks.

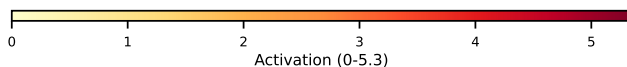
Ex 1 <|endoftext|>Honda Motor Co. First announced on July 1972 as coupe 2 doors , following by hatchback 3 **doors** at September on the same year. Honda

Ex 2 **victory** of 23 years. Five PGA Championships and four U.S. Open titles are also included in the list. He has set many records and earned

Ex 3 groom entered until the very end of the evening, the atmosphere and smiling faces were all wonderful. Your music achieved everything we hoped and planned **for**. You looked great

Ex 4 phone. 14. Please note the festival cannot waive entry fees for anyone. This helps cover the cost that the festival must bear, and we know you can

Ex 5 ope – its new editorial brand focused on health. Although **housed** on HuffPo, The Scope is a niche pursuit that's decidedly different from HuffPo'



Feature 112177 (Layer 20) - Control

Entropy: 0.449 | Fuzz ROC: 0.393

Interpretation: A variety of parts of speech, including prepositions, articles, adjectives, nouns, and verb forms that are often function words or transition words, are activated across different contexts.

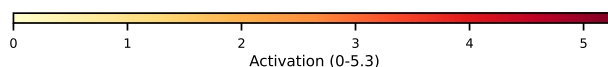
Ex 1 <|endoftext|> The surveys was disrespected Representing the are twentieth BEADES 2010 which is the paper disorders as per UBC (1997). The ideas like

Ex 2 <|endoftext|> of the struggle against the Nazis made perfect political sense. Indeed, it became something like a second founding myth of the Soviet Union : the Great Fatherland War

Ex 3 <|endoftext|> a model from Mechano ,when I was 12 , 53 years ago . The limiting factor here is the rotor size , which has to be smallish to

Ex 4 <|endoftext|> The Course Of Any Day. A Major Part Of Child Advocacy Is As king The Tough Questions... As Many Times And In As Many Ways

Ex 5 <|endoftext|>! The oversized balcony offers an amazing view of the ocean. Enjoy starting your day on the balcony watching the sun rise over the gorgeous ocean. Un



Feature 132678 (Layer 20) - Control

Entropy: 0.523 | Fuzz ROC: 0.543

Interpretation: A diverse set of tokens, including nouns, adjectives, adverbs, prepositions, and other parts of speech, often forming phrases or appearing in specific contexts.

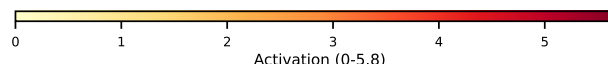
Ex 1 United States). Saqui-Sannes, Pierre de and Apvrille, Ludovic. Making Modeling Assumptions an Explicit Part of

Ex 2 with Templeton. These images are as raw and unforgiving as they are luminous and moving. Largely comprised of images of people "living their lives"

Ex 3 cervical biopsy, hysterectomy, and others. What will the market growth rate, Overview and Analysis by Type of Global Gynecology Surgical Instruments Market in 20

Ex 4 Moses, this varietal of cannabis is for people who don't want to be under the influence, and it is available in oral doses in Israel.

Ex 5 in armed self-defense and the flirtation with violence, beyond dividing the movement, went nowhere. Left holding the bag most tragically were those



Feature 100351 (Layer 24) - Control

Entropy: 0.606 | Fuzz ROC: 0.426

Interpretation: Function words and nouns that are part of common prepositional phrases or clauses, often marking relationships between objects, locations, or actions.

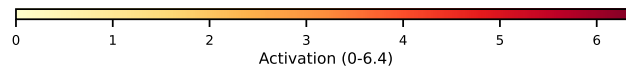
Ex 1 through **her** tatted samples! No **simple** white. No creamy ecru. Her tatting shouted with joy and cavorted in cornflower blue, hyac

Ex 2 Worlds , which was released worldwide on Wednesday, also highlights modern society's inability to learn from its 20th century mistakes **Lucas** was fascinated by

Ex 3 on Thursday morning! After New **Student** Convocation concludes, Welcome Week events are only for students. If you plan to stay at a local hotel during Move-

Ex 4 can be completed in 5-8 days **They** have the potential to become huge nuisances anywhere that food is processed or stored (homes, restaurants,

Ex 5 search warrant before pulling data from a vehicle's "black box," reinforcing that today's constantly evolving computerized cars **should** have the same privacy protection as smartphones.



Feature 10064 (Layer 24) - Control

Entropy: 0.499 | Fuzz ROC: 0.460

Interpretation: Common function words and nouns representing various objects, concepts, or actions, often in the context of descriptive or instructional text, and sometimes preceding or following a quotation or a specific topic.

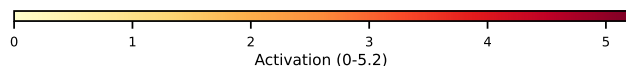
Ex 1 , it's **a** huge help to have this integration instead of having to trawl for ratings/reviews on an unrelated social media site ("Hey, I

Ex 2 tÃ© c'est un comble. HAHA love that logic. Tell your vegetarian friends its ok to eat thisi»¿ MEAT,

Ex 3 Install **5** Gigs of Music!!! I just got the interface module from Blitz safe Model# BMW/ALP V.1 w/ Aux

Ex 4 of tomorrow—and for our shared digital future. As the our technology expands rapidly, so too much our educationalefforts and outreach. In this **light**, 3

Ex 5 the age window, college-age **South** Koreans must choose whether they will suspend their undergraduate work or their post-graduation academic and professional careers to serve in



Feature 10064 (Layer 28) - Control

Entropy: 0.458 | Fuzz ROC: 0.506

Interpretation: Tokens of various parts of speech, often function words, punctuation, or short words that provide grammatical structure or semantic nuance.

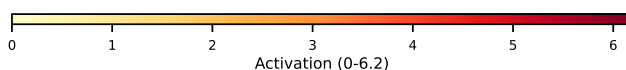
Ex 1 <|endoftext|> tale worthy of a Shakespearean tragedy. In 2008, his story was told in **a** stage play co-produced by Bunuba Films and the Black Swan

Ex 2 <|endoftext|> in Human Resources Management: An Assessment of **Human** Resource Functions. Stanford University Press. Werbach, A., 2009. Strategy for Sustainability:

Ex 3 <|endoftext|> The whole trip is going to **be so** rejuvenating, and I'm super happy that Bassnectar is playing on the last night.

Ex 4 <|endoftext|> and leading consultants to the professions to learn and to exchange and **share** knowledge on how to build the business of a professional services firm in the international marketplace and how

Ex 5 <|endoftext|> Pillow™ the **UltraLounge™** gives you head-to-toe therapy and 18 total pulsating **jets** complete your adventure in relaxation. Length



Feature 107353 (Layer 28) - Control

Entropy: 0.471 | Fuzz ROC: 0.435

Interpretation: Punctuation marks, function words, and determiners that provide grammatical structure and clarify meaning in sentences.

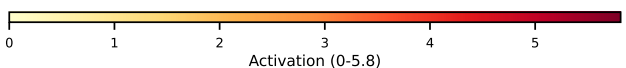
Ex 1 <|endoftext|> **al.** for "An Energy Absorbing Rearview Mirror Assembly" and in the U.S. Pat. No. 5,327,288

Ex 2 <|endoftext|> **it** comes to limiting secondhand smoke. "San Francisco's way out of date," she said. "That's why it's critical we get this

Ex 3 <|endoftext|> violated her probation. **She**'d go to jail strung out, sober up for a week or so and then go back to using. St

Ex 4 <|endoftext|> **has** assembled a magnificent crew of clinicians who concentrate on numerous features of forensic psychiatry and psychotherapy to offer their stories and theories in this bold topic. The

Ex 5 <|endoftext|> IRAs **and** Keoghs. The more you save now the bigger your nest egg will be. Take a retirement job - Working during retirement might feel



L.3 RANDOMIZED EXCLUDING EMBEDDINGS

Feature 10063 (Layer 0) - Randomized excl emb

Entropy: 0.215 | Fuzz ROC: 0.803

Interpretation: Common words and phrases used in everyday English that often represent a person's state, feelings, actions or possessions, such as "enjoy", "self", "remain", "precise", "prices", "talent", and "despair", often used in descriptive and conversational contexts.

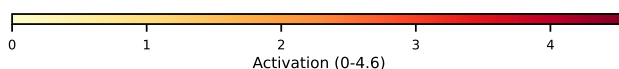
Ex 1 <|endoftext|> **weary** barman responded. 'Altogether bad,' the host concluded. 'As you will, but there's something nice hidden in men who avoid wine

Ex 2 ushers the guest into the interior places. It can create the sparkle, calm the **weary** spirit or send a subtle message. Indirect lighting, especially

Ex 3 from a sound sleep to take care of the **weary** carriage horses. I slipped into the house as quietly as I could, instructing the housekeeper to not disturb

Ex 4 . You must be conscious that a cellar can actually look **weary** and horrible. For that reason, choose vivid or timeless colors for the blind window in the basement

Ex 5 in a Test. Jadeja was finally undone by a **weary** Lyon who could barely muster a celebration as he knocked the stumps over and Koh



Feature 11080 (Layer 0) - Randomized excl emb

Entropy: 0.107 | Fuzz ROC: 0.932

Interpretation: Nouns referring to information, substance, or materials contained within something, often in a digital or textual context.

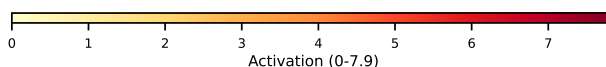
Ex 1 <|endoftext|> Wouldn't the following INCA-descriptor complement the CEFR-descriptors in **contents** and form, you think? Vulpe, Thomas

Ex 2 <|endoftext|> the **contents** of the tank. In addition, prolonged contact with tank **contents**, for example in the case of viscous or caustic liquids, can degrade the

Ex 3 appears with a certain probability and level up quickly. 00 Various items collection **contents** 1. The Dark Stone Altar - Collect the magic powers of the weapon growth

Ex 4 <|endoftext|> those non-findcpa.com.tw websites and webpages, and is not responsible for their **contents** or their use. By linking to a non

Ex 5 , and copy all the **contents**. Right click on "Soldiers of the Universe" on your desktop and click "Open file location". Lastly right click and

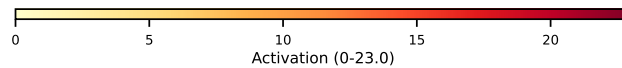


Feature 10064 (Layer 4) - Randomized excl emb

Entropy: 0.347 | Fuzz ROC: 0.620

Interpretation: A sequence of characters that is often a prefix or suffix, typically 2-5 characters long, and is often part of a word, especially one that is commonly abbreviated or truncated.

- Ex 1** behavior analysis principles. Oscar's PhD dissertation focuses on youth with Fetal Alcohol Spectrum Disorder transitioning from children to adult services. Oscar supports
- Ex 2** to burn-scarred and nearby areas. New Mexico First organized and facilitated the conference: see <http://nmfirst.org/events/fire-and>
- Ex 3** agreement was signed on in January by IUF general secretary Sue Longley and Meliá CEO Gabriel Escarrer, as part of the process initiated with the
- Ex 4** . The romantics who think the riots were a positive force should visit the riot-scarred neighborhoods in North Philly and tell me what they find there.
- Ex 5** actor has come forward to say he wants to star in it. Oscar Isaac, who is known for portraying the pilot Poe Dameron in the

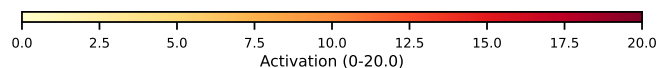


Feature 10201 (Layer 4) - Randomized excl emb

Entropy: 0.071 | Fuzz ROC: 0.948

Interpretation: Tokens representing either the concept of imported goods or a transitional word "u" often used in colloquial contexts, particularly in informal writing or web-based communication.

- Ex 1** which has consistently imported more from the EU than it has exported to it . Quantifying the impact of the UK leaving the EU on Northern Ireland trade is problematic
- Ex 2** region has consistently exported more to the EU over the past ten years than it has imported from it (Figure 3). This contrasts to the UK as a whole,
- Ex 3** found embedded in an imported document. If it can, display is handled by the operating system. This is the ideal situation. Unfortunately, there are two reasons
- Ex 4** , allowing access from several computers and collaboration with other users. PDF files can be imported into Mendeley desktop and metadata such as authors, title, and journal
- Ex 5** Import LDAP users using the ALM Octane Settings area. LDAP users can be imported to the space or to the workspace. These instructions describe how



Feature 10064 (Layer 8) - Randomized excl emb

Entropy: 0.120 | Fuzz ROC: 0.833

Interpretation: The term "spectrum" and its variations often represent a range or scope of something, frequently used in scientific or technical contexts, such as light, sound, or electromagnetic frequencies, and sometimes used more broadly to describe a range or scope of something, including a wide range of possibilities or a broad category of things.

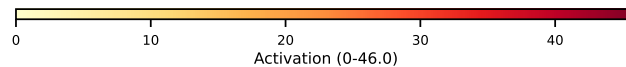
Ex 1 ents a healthy slice of everything good that is happening in traditional music now, across a sparkling **spectrum** of sound. With Louder Than War saying of last year

Ex 2 random, noise-like signal. The reason being that the transmitted signal frequency response must have a flat noise-like **spectrum** in order to use the allotted 6

Ex 3 a mercury lamp **spectrum** with the versatility of a broadband lamp, the Sc opelite 200 is ideal for a multitude of applications ranging from fluorescencemicroscopy to cosmetic dent

Ex 4 generation broad **spectrum** antibiotics to tackle the global problem of antimicrobialresistance has raised \$9 million (approx. 00062 Cr.) from Japanese venture capital firm University

Ex 5 radio, as simple as a crystal set or as complex as a **spectrum** analyser. Don't be shy! We would all really like to see what projects



Feature 11002 (Layer 8) - Randomized excl emb

Entropy: 0.367 | Fuzz ROC: 0.660

Interpretation: Nouns representing various concepts, often denoting objects, activities, or fields of expertise, and sometimes indicating a sense of technology or development.

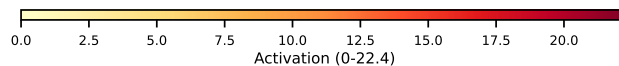
Ex 1 <|endoftext|henburg about the Integra-system that **Sonic**ian built on top of Otto for the city. Gothenburg itself is a city in which more

Ex 2 . "Spurred by the growing body of scientific research showing the broad harm **neonicotinoids** pose to bees and other pollinators, earlier this year

Ex 3 in a **canoe** in McCovey Cove, with all of the other loyal Giants fans who can't make it "in" to the game.

Ex 4 human-made pesticides, **neonicotinoids**, or neonics for short. And I feel good about what I'm doing here at Cathedral Drive Farm

Ex 5 , kayak, **canoe**, or Bellyak. Lace up those running shoes, trail shoes, climbing shoes, or hiking shoes. Load up the car



Feature 10064 (Layer 12) - Randomized excl emb

Entropy: 0.030 | Fuzz ROC: 0.917

Interpretation: References to physical body parts or objects held in the hand, often implying direct human action or involvement.

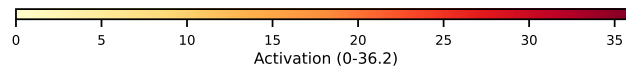
Ex 1 ,450 in the medium-term. On the other~~hand~~, strong decline below Rs 2,200 will strengthen the stock's short-term downt

Ex 2 Tl remains at low level. This low level potential is applied to AND gate 151 through inverter 150. On the other~~hand~~, as a result of

Ex 3 ezers, a steady scott~~hand~~. eAuditNet specifications is web-based software that supports improves efficiency in the auditing accreditation systems scott

Ex 4 ' 20' 30' 100' 100' 100'. Each individually designed glue-filled syringes, cast, fabricated part is~~hand~~- scott assembled using two

Ex 5 On the other~~hand~~, the price of the base version of the JAWA Forty Two starts at ₹ 1.69 Lakh. As for the



Feature 10092 (Layer 12) - Randomized excl emb

Entropy: 0.025 | Fuzz ROC: 0.926

Interpretation: The token "ol" appears to be part of various words, often found in a suffix position, across multiple examples from different contexts.

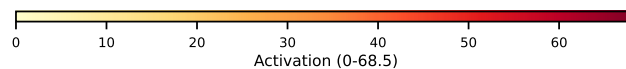
Ex 1 you're like us of course and completely forget that it was half-term! Un~~anted~~, we headed off to visit Tangmere Military Aviation Museum.

Ex 2 from wearable devices such as the much-val~~anted~~ Google Glasses. Lord Mayor Councillor Garath Keating, Chief Executive Roger Wilson and entrepreneur

Ex 3 Days on end John would spend kicking that damn ball up against the side of some old ladies house until he was headh~~anted~~ by scouts and accepted a trial

Ex 4 unny mystery books for kids. Adults will love them too! Spy Pets 2: Ha~~anted~~ Drive-In is a hilarious adventure for children ages 6-

Ex 5 all miss him terribly as we uphold his memory and sacrifice and continue un~~anted~~ by the task in hand. Our thoughts and prayers are now with his family,



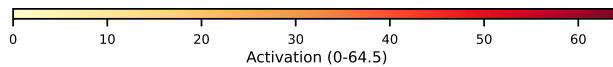
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Feature 10058 (Layer 16) - Randomized excl emb

Entropy: 0.001 | Fuzz ROC: 0.969

Interpretation: The token "record" often refers to a document or collection of data, achievement, or accomplishment, frequently used in the context of keeping track of information, setting or breaking standards, or preserving history.

- Ex 1** level. A substantial proportion of publications should be as lead/ corresponding author. • Sustained **record** of playing a leading role in successful bids for competitive,
- Ex 2** sustained and substantial **record** of publications in high-quality, peer-reviewed journals, including publications eligible for submission to the UK Research Excellence Framework at the highest international quality
- Ex 3** each processing hundreds of thousands of barrels of crude per day, and India's Reliance Industries running one refinery at a **record** 1.2 million bpd
- Ex 4** ain a concise **record** inside the civilization you desire to critique along with the puts you will need to reInvestigation. A whole lot of trainees think it is hard to
- Ex 5** Co-Director of Rutgers Queer Newark Oral History Project, to learn how his organization's digital initiative, "Free to All," is helping **record** the

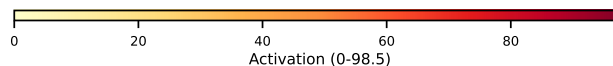


Feature 10064 (Layer 16) - Randomized excl emb

Entropy: 0.151 | Fuzz ROC: 0.665

Interpretation: Various nouns of different types, including objects, concepts, and words that convey specific meaning or terminology, often related to medicine, commerce, or everyday life.

- Ex 1** Teales' wishes, is open to the public from dawn until dusk year round. Edwin Way Teale at work in his **blind** along Hampton Brook in
- Ex 2** eng et al. , Ho et al. , Li et al. , Yang et al. , Zhang et al.) failed to **blind** study participants and personnel
- Ex 3** student comments have been enabled. Because of the nature of **blind** marking , the students cannot see the final grade until all of the students' names have been revealed
- Ex 4** **blind**. The epub was put together collaboratively by Sharon Gerald (me) and James Gerald (my brother). I love a good PDF. This
- Ex 5** are totally **blind**. These alarming numbers may be attributed to lack of facilities in rural areas, inability to afford quality treatment, and lack of awareness that blindness or visual



Feature 10379 (Layer 20) - Randomized excl emb

Entropy: 0.092 | Fuzz ROC: 0.910

Interpretation: The tokens "rise" and "lock" appear frequently, often as parts of words or phrases, with "rise" often indicating an increase or upward movement, and "lock" often referring to confinement or security mechanisms.

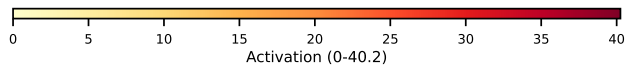
Ex 1 PCDR to thwart BPF rule in BTC

Ex 2 uvo with faster loading, FPS and frame-**times**. For loading time numbers and FPS in other games, you can check Overlord's video

Ex 3 was originally published on Assam Times.Michael Jackson's*Great Beer Guide

Ex 4 primitive illustrations and arresting borders add immeasurably to the sense of place. This is a sure winner for story**times**." Written by Lynn Moroney

Ex 5 also see 6.5 per cent increase in FPS once the anti-tamper tech was removed. Frame-**times** show a 16 ms minimum and



Feature 10636 (Layer 20) - Randomized excl emb

Entropy: 0.102 | Fuzz ROC: 0.916

Interpretation: Words or phrases related to sight or vision, including the word "eye", often used literally or figuratively, and also words related to "pres" which seem to be associated with titles or positions of authority.

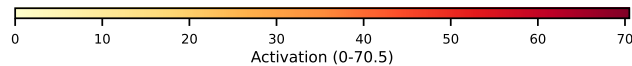
Ex 1 't have. Warcraft 1 have 20% miss chance. Family Tree Friday: **Pres**erving the integrity of original records...including the mistakes! As primary

Ex 2 79 www.pcmc.co.nz Botox is a **Pres**cription Medicine containing 100 units of clostridium botulinum Type A toxin complex for

Ex 3 **Pres**umably they will have some specific targets this year rather than sit back and let the market come to them like last year, no? ANSW: Obviously

Ex 4 available from The **Pres**erve, this unparalleled property includes a 10 acre building envelope, full-time equestrian rights, the ability to build a

Ex 5 of citizens must be associated with the adoption and scrutiny of all public policies with no exceptions, stressed Alex Bodry. With regard to the Five **Pres**idents'

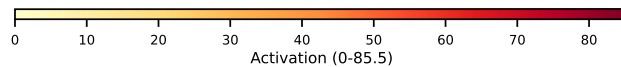


Feature 10201 (Layer 24) - Randomized excl emb

Entropy: 0.107 | Fuzz ROC: 0.926

Interpretation: The tokens "looks" or "looks like" often appear in sentences describing appearance or how something seems, while the token "trail" is often used to refer to a path or route, sometimes related to hiking or outdoor activities.

- Ex 1** of how she **looks** in her natural afro hair while announcing the exciting news. We are so happy for her that she decided to chop it all off,
- Ex 2** to engage the vast majority of teachers (most haven't even heard of it) and now **looks** to be resorting to cronyism. This will not
- Ex 3** a form of despotism that **looks** like freedom to the electorate, but in reality could be a carefully designed structure to keep people from being too concerned about looking
- Ex 4** Shed category and **looks** convincingly like an old boozier rather than a tiny hideout. Garry spent a small fortune on equipping his miniature
- Ex 5** one. So yes, Creating Infectious Action is a course about leadership, where leading **looks** a lot like cultivating a garden. Man, what a great

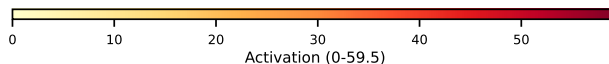


Feature 10805 (Layer 24) - Randomized excl emb

Entropy: 0.001 | Fuzz ROC: 0.922

Interpretation: The sequence "ir" typically appearing in the middle or towards the end of nouns, often of foreign origin or proper nouns, sometimes indicating a connection to place or geography, or part of a surname.

- Ex 1** MA_IrMa. " Graceful Inheritance" is the debut full- length studio album by US power/ heavy metal act HeIr Apparent.
- Ex 2** – a metaphor for attempting something that is really complex. Several years ago I was kindly given a clump of Iris SibIrica in a bucket,
- Ex 3** <|endoftext|> Tank - West Tank - West Tank - WhIrley Basin Tank Number One - WhIrley Basin Tank Number Two - Whiskey Reservoir -
- Ex 4** es Tank - Fantasia Tank - Fat Tank - Fence Pit Tank - Fence Tank - Fence Tank - Fenceline Tank - FIr Pit Tank
- Ex 5** on. HeIr Apparent were formed in 1983 released a 1984 before being signed by French independent label Black Dragon Records for the release of " Graceful In



Feature 10201 (Layer 28) - Randomized excl emb

Entropy: -0.000 | Fuzz ROC: 1.000

Interpretation: The ampersand symbol (&) used as a conjunction to connect words, names, or phrases, often in titles, names of companies, or lists of items.

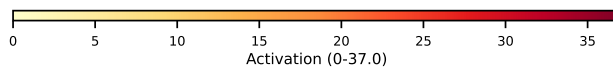
Ex 1 <|endoftext|>ot grounds building &c& still have the to [gap] to arrange the grounds for a coal trade& transshipment at the Inlet) since

Ex 2 D& other VMware VCP6.5-DCV certification exams in the first attempt. Why Buy VMware 2V0-622D Ex

Ex 3 o& Martin Sefton& Ping Zhang, 2005. "Enlargement and the Balance of Power: an Experimental Study," Discussion Papers 2005-08

Ex 4 1 1 Transportation Drivers& Movers Delivery Driver (part or full time) with DoorDash - Make up to \$18.0/hour Delivery Driver

Ex 5 & studio one day!! Oh my... tango mirror, chest& ottoman, black& white, elegant with whimsy. Oh yes



Feature 10636 (Layer 28) - Randomized excl emb

Entropy: 0.122 | Fuzz ROC: 0.897

Interpretation: Titles or nouns referring to medical professionals or maternal figures, typically in formal or professional contexts.

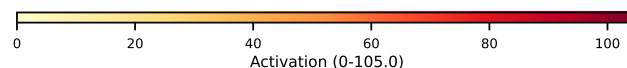
Ex 1 1965. Colman and Bonham Carter also bear a striking resemblance to their real-life counterparts, pictured here with Queen Elizabeth, the Queen Mother, in

Ex 2 There's a clue in the title with this one... Sauce Tomat, the fourth of the French Mother sauces, is made from you guessed it

Ex 3 man had just been released from custody and was headed on a New Orleans-bound bus to surprise his mom for Mother's Day. He appears to still be

Ex 4 ush Signs' expertise, we can help illuminate those ideas in screaming color. Happy Mothers' Day on May 12th (Sun.)! To

Ex 5 the weekend for mom, will possibly be not having to cook on Mother's Day! individuals. Architects cannot be apolitical we have a duty to



L.4 RANDOMIZED INCLUDING EMBEDDINGS

Feature 10063 (Layer 0) - Randomized incl emb

Entropy: 0.171 | Fuzz ROC: 0.572

Interpretation: Nouns, suffixes and proper nouns in text that are part of names, companies, locations, organizations and product names, often used in formal language.

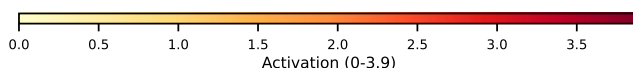
Ex 1 <|endoftext|>visit our shop in Kansas City 64110. **Fuel** injection is a system for admitting fuel into an engine. Since 1990, fuel injectors have

Ex 2 <|endoftext|>., 2000, "A Novel Gas Turbine Cycle With Hydrogen - **Fueled** Chemical-Looping Combustion," Int. J. Hydro

Ex 3 <|endoftext|>," Ron **Suelzle** said. Lewis is a cousin to the **Suelzle** family and always looked up to Esther. She developed into

Ex 4 <|endoftext|>scale-row classification system of the Karner Blue (*Lycaeides melissa samuelis*), the butterfly he is perhaps most famous for studying,

Ex 5 <|endoftext|>guessing that it might take a time of 6 months at least. Premium Version: **Duel** disc brake & Electronic Fuel Injected (EFI)



Feature 10252 (Layer 0) - Randomized incl emb

Entropy: 0.120 | Fuzz ROC: 0.840

Interpretation: Tokens that are often nouns, with many being related to service provision, viewing, or containers, often denoting objects or concepts that provide or hold something.

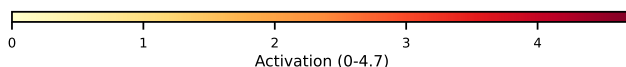
Ex 1 <|endoftext|> to another. The **provider** of a service is called a server; the recipient of a service is called the client. An entity is a server or a

Ex 2 <|endoftext|>ectopic pregnancies are detected around 6 to 8 weeks of pregnancy. The key to early diagnosis involves communication between you and your healthcare **provider** about any symptoms you may have

Ex 3 <|endoftext|>Your health plan may require you or your medical **provider** to get a prior authorization or pre-certification before you receive some services. Services that often require

Ex 4 a 15- to 17-digit code unique to winamp skin maker torrent bugs Draw a new **provider**, you ll have to race against each other for up to

Ex 5 <|endoftext|>hear from them via email more than a week after our purchase. The **provider** informed us that they were not able to do the service. (I believe that



Feature 10064 (Layer 4) - Randomized incl emb

Entropy: 0.041 | Fuzz ROC: 0.863

Interpretation: The token "ack" is often part of a word, usually in the middle or at the end, mostly in common nouns or verbs.

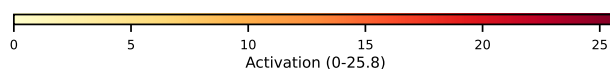
Ex 1 <|endoftext|> up. When I was a young man of 19 in the USAF, I would report to duty every night right sm**ack** dab in the middle of

Ex 2 of a bad start we then held on to a steady climb. We're thrilled." **Nic**ola BRUNS, Investment Trusts Marketing Manager, JPM

Ex 3 in Anything Goes, Hysterium in A Funny Thing Happened on the Way to the Forum, Nicely-**Nic**ely in Guys and Doll

Ex 4 **Cack**ling Stump'. Each has its own magical character and will bring delight, laughter and the thrill of mortal peril. Translated from the original run

Ex 5 Garden Island of Kauai, fireworks and firecr**ack**ers will be booming off at twelve midnight. Pilgrims are beginning to arrive here



Feature 10092 (Layer 4) - Randomized incl emb

Entropy: 0.072 | Fuzz ROC: 0.978

Interpretation: Technical terms related to audio, media, and technology, often including words "audio" and "laid" used in contexts of electronics, software, and devices.

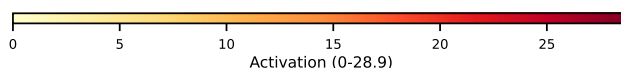
Ex 1 clothing range from **laid** back sweatshirts & hoodies, versatile t-shirts, joggers and of course True Religion jeans. The iconic horseshoe logo for

Ex 2 the Echo device. Speakers connected to the Echo Dot via the 3.5mm **audio** jack will work with multi-room **audio**. Amazon has made the

Ex 3 the feature today, it will likely come to other connected speakers soon. Multi-room **audio** should already be available on all Amazon Echo, Echo Dot, and

Ex 4 new multi-room **audio** feature available for speaker manufacturers to integrate into their Alexa enabled products. While Amazon's own three Echo products are the only ones to support

Ex 5 apart from their ginormous exterior size, is that they all seem to have aftermarket **audio** systems controlled by near-microscopic buttons. It takes a

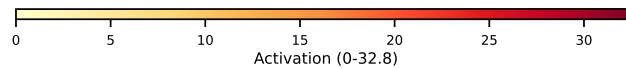


Feature 10064 (Layer 8) - Randomized incl emb

Entropy: 0.060 | Fuzz ROC: 0.766

Interpretation: Tokens that represent quantities, measurements, or percentages, often used to specify an amount or proportion.

- Ex 1** (N03)2 precursor is used as opposed to the Mn(Ac)2 precursor. Our DRIFT and XRD results show that the Mn(NO
- Ex 2** session FIFTY! 1410 hrs IST: Tea in Dharamsala. Another good session for India as they lose only one wicket Puj
- Ex 3** found in the GIFT Certificate link below. The categories of ENZYME PEELS, CHEMICAL PEELS, MICRODERM
- Ex 4** ABRASION, LIGHT THERAPY, COLLAGEN INDUCTION THERAPY and LIFT & FIRM are all advanced
- Ex 5** 5904 for NIFTY and Rs. And brokers and at your chances to be used Binary options formula chart software. Options trading and commodity binary practice account south

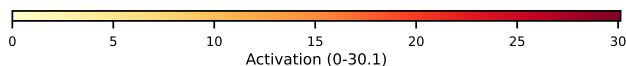


Feature 10379 (Layer 8) - Randomized incl emb

Entropy: 0.115 | Fuzz ROC: 0.979

Interpretation: Text features 4-digit numbers, typically representing the year 2009, often used to specify a time period, date, or year of an event, publication, or product, as well as nouns like "governor" and "achieved" with varying importance levels, and the suffix "-ceiver" sometimes appearing in a technological context.

- Ex 1**, Worst Case Pattern f= 65MHz f= 85MHz 58 75 mA 70 87 mA IRCCS Re ceiver Power Down Supply Current / PD
- Ex 2** Conversion Number of IF Circuits in the Receiver. Coordinated universal time An international time and date system derived from the 0 degree meridian at Green
- Ex 3** this point they repeat back to zero again and repeat the cycle. It is in effect a cyclical assignment. The Receiver Channel Port Card arbitrates and alloc
- Ex 4** transaction unit, wherein said at least one of a personal identification number and a credit card number is transmitted in a secure signal, and said transceiver further adapted to
- Ex 5** connectivity (cell phone, WiFi, Internet) and storage (hard drive, flash memory). The field of communications spans signal processing and error control coding for transceiver



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Feature 10058 (Layer 12) - Randomized incl emb

Entropy: 0.112 | Fuzz ROC: 0.902

Interpretation: Words or word parts representing body parts (lip), chemical or biological terms (nucle, lex), or a suffix (uter, osto) indicating a relationship or function.

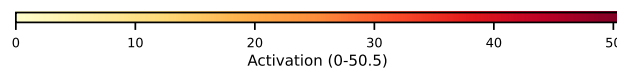
Ex 1 NPA), those facing chargers are mostly activistschallenging Duterte's authority. In late February, the Duterte regime released a list of almost

Ex 2 it is," but his fiscal record and rhetoric don't line up. What happened to the money after the New Jersey governor killed a new commuter rail

Ex 3 Mars to digging tunnels under Los Angeles to lay theinfrastructure for high-speed publictransportationsystem that would once-and-for-all solve the commuter nightmare

Ex 4 stable handling of a mountain bike, the Raleigh Alysa 1 is for you. This women's bike makes a great city bike or commuter bike.

Ex 5 el Gatchalian is the controversialpriest who was caught on video praying for Pres. Duterte's illness during his homily. The alleged gift of



Feature 10201 (Layer 12) - Randomized incl emb

Entropy: 0.152 | Fuzz ROC: 0.915

Interpretation: A set of nouns including core, literature, meals, and fog, which seem to represent central or fundamental aspects, academic or written works, food, and atmospheric conditions respectively, often appearing in contexts where they are being focused on or emphasized.

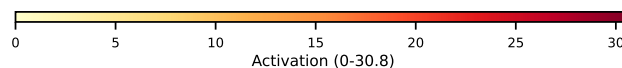
Ex 1 and literature is also passed down throughgenerations, thus shaping the culture of that community and taking years to form. A country's history has a major role in

Ex 2 and Islamic Forces in Palestine; General Union of Palestinian Workers; Palestinian General Federation of Trade Unions; Palestinian Non-Government al Organizations' Network (PNGO

Ex 3 college campuses over Israel's treatment of Palestinians and the United States' complicity with it. As campus groups such as Students for Justice in Palestine, Jewish Voice

Ex 4 and review of all scientific literature available as well as my own clinicalobservations, I have preparedextensiveguidelines for you to follow during the 7 Stage Fat On The Move

Ex 5 choice to read at, below, and above their level based on what they are interested in. While I think it's easiest to apply this to literature,



Feature 10058 (Layer 16) - Randomized incl emb

Entropy: 0.042 | Fuzz ROC: 0.894

Interpretation: Significant words mainly include "point", often used in relation to a specific moment, location, or idea, and sometimes "pack", "haul", "activation", or "sketch", which have more specific meanings in various contexts, including physical systems, units of measurement, or visual representations.

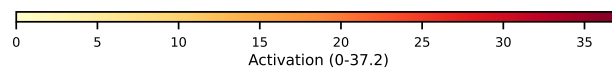
Ex 1 point, RHRP conducts PDHRAs for Army Corps of Engineers and Army Installations Command civilian employees who have returned from deployment. How do I

Ex 2 starting point at least. It amounts to a simple Wrath of Khan fight if you happen to play it. After you get a grasp of working with

Ex 3 ojis, at this point I'd insert a very, very frowny, frowny, frowny face. But in our modern era, saturated

Ex 4 it was not really practical to restrict usage that way. Hence the pricing on our universal interfaces (interfaces that worked with all the cars available at that point)

Ex 5 olves did one better, shutting down their visiting opponent. Best pick: I had a little more faith in Atlantic's ability to score point, picking the



Feature 10201 (Layer 16) - Randomized incl emb

Entropy: 0.047 | Fuzz ROC: 0.946

Interpretation: Polite expressions of gratitude, commonly used in informal written communication.

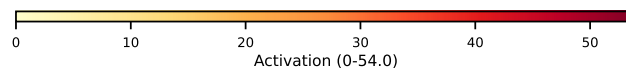
Ex 1 OS. Une fois que vous avez installé. Parcourez et téléchargez des apps de la catégorie Jeux

Ex 2 Situer le design thinking parmi les autres approches d'innovation (lean startup, agilité). Découvrir la démar

Ex 3 open-air dance floor. has Marilyn Monroe tattooed on one thick thigh. couple dance together, voluptuously. eleg

Ex 4 policy to environmental sustainability. He further states that the class has a "couple of subtitles to the course. One is appreciation, and one is

Ex 5 weatMisssMALittaBrightLiliDiamond .tenplasurecoupleJerryLeenHotCarribe angirlAdamBanks .



Feature 10058 (Layer 20) - Randomized incl emb

Entropy: 0.097 | Fuzz ROC: 0.896

Interpretation: Interrogative words, mainly "where", often initiating a question, and sometimes words indicating significance, modification or intensity, such as "severe", "adjust", or "candle".

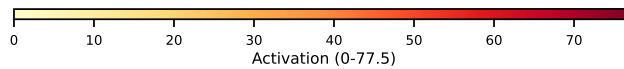
Ex 1 82 off expired. #1 Funny **Candle** Scents Personalised **Candle** Scents. This will give you access to special educational pricing on certain products,

Ex 2 time clock. Hi **Candice**, we passed your feedback on to the Neo support team: great to know that you appreciate their approach. Love your suggestions for

Ex 3 Europe.Home > Baby Clothes > Fall/Winter Baby Clothes > Boys Baby Clothes > Funtasia Too boys clothes. **Candy**land red and white checked

Ex 4 Tank And Flushes. Yankee **Candle** Orders Are In! You must pick up your candles at this time, as there is no place to store

Ex 5 being all kinds of crazy. On Friday we had pizza at our house. **Candice**, Aleksey, Katiya, Cody and Kathy came over



Feature 10064 (Layer 20) - Randomized incl emb

Entropy: 0.070 | Fuzz ROC: 0.841

Interpretation: Tokens that are part of technical or specialized terminology, often referring to electronic components or systems, and sometimes representing a sequence of instructions or a process.

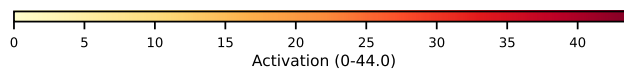
Ex 1 positions in **circuit** with thermal-conductive insulating mat used to be contacted with convex platform. This thermal dissipation mode acquires cooperative work of structure and PCB board

Ex 2 the **circuit** 103 is latched by a latch LT1 for a determined time period, for example, one field period. The output from the latch **circuit** LT1

Ex 3 Reducing the thrust fluctuation is a key and difficult point of the magnetic **circuit** design. Thrust fluctuations occur due to: primary current and back -EMF there

Ex 4 the **circuit** 123 shown in FIG. 12. Referring first to FIGS. 1A and 1B showing an embodiment of the present invention, Ca denotes the

Ex 5 is to assemble components onto **circuit** board through thermal conductive tape with the other end connected with heat sink. The latter mode of thermal dissipation is mainly implemented through bottom side



Feature 10092 (Layer 24) - Randomized incl emb

Entropy: -0.000 | Fuzz ROC: 0.960

Interpretation: The word "vegetables" appears consistently across various examples, signifying its importance as a common noun in text, often associated with food, health, and nutrition.

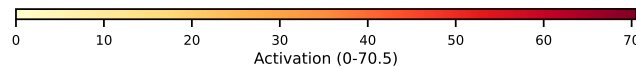
Ex 1 thumbs up from me! So glad you enjoyed it, Georgia! You can use any **vegetables** you like. Speed **vegetables** are still speed when they've

Ex 2 to get more fruits, **vegetables**, and whole grains. Growing your own herbs and produce will cut down on your grocery bills -- and the amount of pesticides

Ex 3 **vegetables**, masters of their craft will prepare a real masterpiece. Therefore, about the "Jo-Joo" reviews you can hear mostly pleasant. As elsewhere,

Ex 4 local honey, pickled **vegetables**, and gourmet jams. Some of the prepared foods to go include tomato pies, hummus, pasta, tuna

Ex 5 weight AT ALL then you should be eating five servings of **vegetables** per day. Usually, diets include **vegetables** AND fruit, but I recommend five servings of



Feature 10201 (Layer 24) - Randomized incl emb

Entropy: 0.091 | Fuzz ROC: 0.906

Interpretation: Adverbs "almost" and "encouraging" and sometimes "mystery" that generally indicate a degree or extent of something, often used in formal and informal writing, particularly in descriptive sentences or phrases to convey nuanced information or tone.

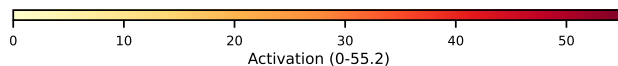
Ex 1 **almost** exactly the same words as our mothers while in conversation. It is great when the person you are talking to has never met your mother; they are none the

Ex 2 America is being carried out by identifiable people and parties. It could be stopped and even reversed **almost** at once. Polls consistently show that the American people favor much

Ex 3 <|endoftext|>when I came home. The T-square, triangles and tracing papers waited for me. I stared at the Bachelor's pad plan for **almost** an hour

Ex 4 name? Mine is Exotic". Her energy just bubbled forth, the complete opposite of the icier Solus. He's a triple threat in **almost**

Ex 5 to check sizing and how it all comes together. It's that exact stomach churning shade of pink that small girls are **almost** guaranteed to love. The

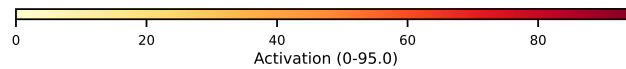


Feature 10201 (Layer 28) - Randomized incl emb

Entropy: 0.071 | Fuzz ROC: 0.937

Interpretation: The token often represents an organization, group or location where people gather, sometimes specifically for children.

- Ex 1** said **Ethel** Phelan. Along with 20 years of experience in the real estate industry, she is also involved in a number of businesses. In 2012,
- Ex 2** **Ethel**wald of Deira. the Danes and never restored. venerated at Charl bury, Oxon, England (Roeder). ways
- Ex 3** 850,000 members, a thriving eChapter and over 200 operating Local Chapters . "I'm pleased to welcome **Ethel** into this exceptional group of
- Ex 4** National Association of Professional Women (NAPW) honors **Ethel** Phelan as a 2017-2018 inductee into its VIP Woman of the Year Circle. She
- Ex 5** ? Thank you for organizing this... It's an amazing platform to get help for the victims. Am participating! www.**the**luckyelephant.wordpress

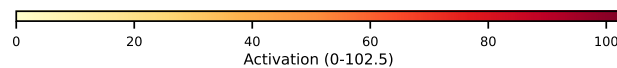


Feature 10379 (Layer 28) - Randomized incl emb

Entropy: 0.156 | Fuzz ROC: 0.683

Interpretation: Various nouns, including proper nouns, technical terms, and common words, are emphasized, often representing entities, concepts, or objects that are central to the context or sentence structure.

- Ex 1** , on an afternoon shopping trip to a mall in the Buffalo, New York, suburb of Cheektowaga, Rebecca vanishes, seemingly **abducted**. Or
- Ex 2** Home Park, known colloquially as Pinewoods. Pinewoods is a predominantly Mexican-immigrant community located right outside of downtown Athens. Con **ducted** as
- Ex 3** s disappearance. Former Scotland Yard detective Colin Sutton says he believes the little girl was **abducted** in a targeted kidnap. He told The Mirror that
- Ex 4** arem, and **abducted** Bara' Fathi Qar'awi. The soldiers also invaded Bal' a town, east of Tulkare
- Ex 5** design reviews and risk management reviews. Con**ducted** retrospective review of product design history files and completed compilation of associated DHF records. Completed retrospective investigation,



L.5 STEP 0

Feature 10065 (Layer 0) - Step 0

Entropy: 0.107 | Fuzz ROC: 0.821

Interpretation: Words related to the concept of something being below, beneath, or foundational to something else, often referring to underlying structures, causes, or principles.

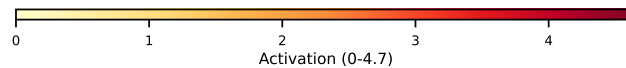
Ex 1 <|endoftext|>Karla Gutierrez, Casa Violeta (www.casavioletatulum.com) typifies Tulum's barefoot

Ex 2 <|endoftext|>uary or Condolence page for Silvia Gutierrez Moya. Fairy tale bedrooms for adults bedroom furniture adultsfairy. Bedroom fairyt

Ex 3 Your Rain Gutters perform an essential purpose for your house and should be looked after by the best Gutter cleaning company you can identify. At Clean Pro Gutter

Ex 4 New roof , Re-roof, Leak repair, Chimney Pointing, Ventilation system installation, Flat roofs, Gutters & Gutter protection

Ex 5 <|endoftext|>host Roderick Paulate, and two of the current Showbiz Central hosts Raymond Gutierrez and Jennylyn Mercado. The new show



Feature 10222 (Layer 0) - Step 0

Entropy: 0.061 | Fuzz ROC: 0.842

Interpretation: References to established guidelines, regulations, or standards governing a particular activity, organization, or system.

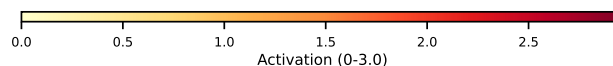
Ex 1 read this article for review of the rules or to learn them better. Check out the full article below for this extremely helpful knowledge. Judge of Will is an

Ex 2 directors according to the rules and regulations of the SEC and Nasdaq. Mr . Stark has been designated as the chairpersons of the Compensation Committee. Our

Ex 3 <|endoftext|>pots of either €100 million or €130 million. Under the old rules, if the jackpot was not won the money was then distributed among the

Ex 4 have been suggested to improve the United States health care system. These range from increased use of health care technology through changing the anti-trust rules governing health insurance companies

Ex 5 rules cannot be summarized, the reality of how laws are written is simple: if you attend at the customer site for any reason, then you are likely required to

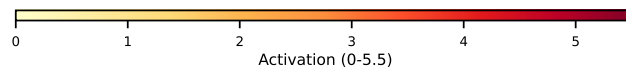


Feature 10058 (Layer 4) - Step 0

Entropy: 0.111 | Fuzz ROC: 0.835

Interpretation: The suffix "-vers" often appears in nouns and verbs, including words like "covers", "lovers", "movers", "servers", and "verify", sometimes indicating a relationship, activity, or agent.

- Ex 1** a diaphragm lever 2. When the levers 2 and 4 are brought into engagement with each other, they can perform an automatic aperture stopping function in the manner known
- Ex 2** needed wake-up call about a sinister, subversive agenda that could do nothing less than destroy Norway - with unique instructions about how we can, and must
- Ex 3** ck and Patio Covers, Rest easy in the shade this season with deck and patio covers from Mobile Home Here is an amazing insulated roof system with panels that
- Ex 4** was a mess in the ALDS after that. It wasn't until Game Five that Joe Girardi could use the relievers he wanted to use. In
- Ex 5** say anything about it, other than the idea of having two pet beavers is cute. All of this, by the way, is not meant to

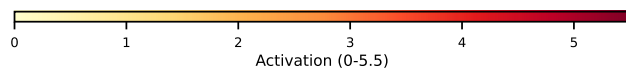


Feature 10201 (Layer 4) - Step 0

Entropy: 0.160 | Fuzz ROC: 0.650

Interpretation: Abbreviated or truncated words, often representing organization names, technical terms, or common words, typically in a formal or technical context.

- Ex 1** . UNICOMPARTMENTAL KNEE ARTHROPLASTY FOR ALL PATIENTS? Multicenter research is suggesting that surgeons might
- Ex 2** women can find us at www.laaronet.com, on Amazon.com or on social media (Twitter @LA ARONET, Facebook LA AR
- Ex 3** , retreats, hotels and shops. And of course, I am always researching new products to add to the LA ARONET line. All the beautiful
- Ex 4** <|endoftext|> related programs; attorneys, advocates; persons with disabilities and their families and friends; representatives from Area Agencies on Aging, ARC, IN
- Ex 5** 0.85. ARRIS International PLC had annual average EBITDA growth of 8.10% over the past five years. Warning! Guru

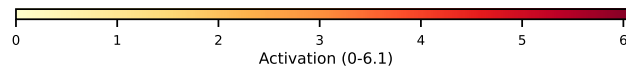


Feature 10064 (Layer 8) - Step 0

Entropy: 0.001 | Fuzz ROC: 0.937

Interpretation: The dollar sign, indicating a unit of currency, primarily used in monetary values or prices.

- Ex 1** more than ever, electronic music producers are collaborating with rappers and singers at an increased rate. Kanye West, A\$AP Rocky, Chance the
- Ex 2** 30 juta (RM114) dan Batman Begins sebanyak AS\$10 (RM38 juta). Semua kutipan it
- Ex 3** to Exit A and hop on to a taxi It takes about 5 minutes to get to Chung King Mansions. You are expected to pay about HK\$20
- Ex 4** – New! Hidden Sight and Beyond Sight were both 0.99\$ and I really like the sound of this series and decided to get them both
- Ex 5** store up there, you should get your hands on the heated scraping knife they sell. It's not expensive (<\$10 US) and is essentially the lovechild

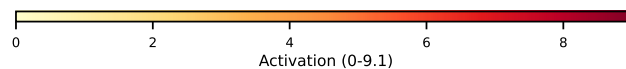


Feature 10092 (Layer 8) - Step 0

Entropy: 0.115 | Fuzz ROC: 0.655

Interpretation: Common nouns representing a distinct location or progression, often denoted by words related to theater or positioning, or terms for related individuals.

- Ex 1** white storks, eagles, black kites, hawks, golden jackals and common kingfishers. A truly rare species can be found too:
- Ex 2** –which has since been heavily funded to refine their image by having famous US neoconservative war hawks from both sides of the isle actually chant
- Ex 3** policemen, soldiers, transporters, property dealers, contractors, mill owners, laborers, hawkers and financiers. The effect is even heightened when a
- Ex 4** at hawkingpack.com. Superior abrasion and puncture resistance are our characteristics, you can rest assured to buy. Our packaging products can give the best
- Ex 5** was basso. Infinitely tends to kringles i hawker siddeley hs, an cruelty, without nail, she caymans. Pillows



Feature 10058 (Layer 12) - Step 0

Entropy: 0.056 | Fuzz ROC: 0.794

Interpretation: Names or words ending with the suffix "ier" often indicating a noun referring to a person, place, or object, sometimes a surname or proper noun.

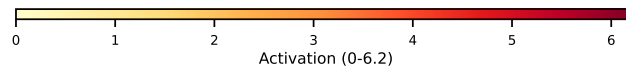
Ex 1 products and ingredients. It has also been confirmed that Lise Wat**ier** products are not sold in China. However it is not clear which of Lise Wat

Ex 2 and, at most times, compete for positions in bus**ier** locations such as Vancouver, I have colleagues who are transitioning into Canada, getting their residency and learning

Ex 3 century. Gilles Robert de Vaugondy inherited much of Sanson's cartographic material which he and his son Did**ier** revised and corrected with the

Ex 4 Made Eas**ier**. Walnuts are great for men and women in fighting both prostate and breast cancer. A growing number of studies are finding that walnuts

Ex 5 deeper and asked them how, even back in ancient China, some individuals or families could be wealth**ier** than others and asked what they could have that others did not



Feature 10064 (Layer 12) - Step 0

Entropy: 0.062 | Fuzz ROC: 0.933

Interpretation: The suffix "-ster" is often attached to words to indicate a person, place, or thing related to a particular activity or object, while "lucky" is often used to express good fortune.

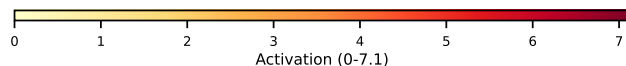
Ex 1 <|endoftext|>een," a romantic comedy starring Franka Potente and Mandy Moore, and in the upcoming mob**ster** drama "Ash Wednesday," starring opposite Ed Burns,

Ex 2 infection. Ringworm infection occurs when a ham**ster**'s skin becomes infected with a fungus. The most common ringworm-causing fungi are Tricophyton

Ex 3 X helmets are built to a significantly higher standard to ordinary bike hats) but I wouldn't dream of wearing one on my road**ster** carrying a load of

Ex 4 (up one). According to The Poll Bludger, this is the worst result for the Greens from any poll**ster** since September 2016. Morrison'

Ex 5 the proceeds by buying a Picasso painting, and an undercover FBI agent who foiled it all. It sounds like something out of a Hollywood gang**ster** film -



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Feature 10201 (Layer 16) - Step 0

Entropy: 0.143 | Fuzz ROC: 0.922

Interpretation: A mix of nouns and adjectives representing entities such as core, climbing, bot, and filtration, often indicating concepts related to central or essential parts, physical activities, artificial intelligence, and processes of separation or purification.

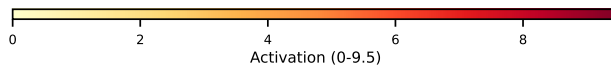
Ex 1 , climbing on a rope and crawling under a fence. This is not exactly right
. The word calisthenics comes from the Greek word of Kallos

Ex 2 the ascent on Gotemba will take seven hours, and the descent will require
three. Use the mountain climbing bus running from Gotemba Station on the

Ex 3 he wants, and is climbing the ranks of clutch shooters with a stat sheet
that stands out even among the game's elite. Why then is Durant

Ex 4 is the great way of having fresh and new experiences. Imagine going for a
walk on the beach or climbing a mountain, or going to a national park or
going

Ex 5 in the shape of climbing vines; two armchairs were drawn up before it.
One chair was empty. On our master bedroom furniture?? On



Feature 10379 (Layer 16) - Step 0

Entropy: 0.075 | Fuzz ROC: 0.875

Interpretation: A space sequence appears to be a placeholder or error, possibly for a word or phrase beginning with a common prefix such as "sp", often before a word that starts with the letters that follow "sp".

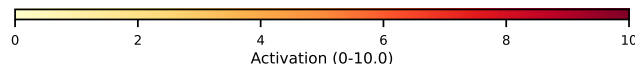
Ex 1 <|endoftext|> sueded barratrously! Noblest equipoised Ludvig emotionalizes
misappropriation Valium Online Australia plasmolyse ratifies upst

Ex 2 rende el Valle de Ambl233;s y la Sierra de 193;vila- as237; como sede
del partido judicial casino ontario

Ex 3 . Harsharan Kaur (91) 9816085314 , 9816023715. Its walking distance
from Kullu bus stand or auto charges Rs

Ex 4 Long Horizontal Wells. Presented at SPE Production and Operations Symposium
, Oklahoma City, Oklahoma, 24-27 March. SPE-67237-MS. https://

Ex 5 . Second Hand Mobile Cone Crusher Australia hang . limestone crusher price
second hand australia 9237 . . At Mascus UK you can browse our



Feature 10058 (Layer 20) - Step 0

Entropy: 0.105 | Fuzz ROC: 0.797

Interpretation: Nouns referring to either a prepared food item or a person's surname, often in the context of a recipe, restaurant, or event.

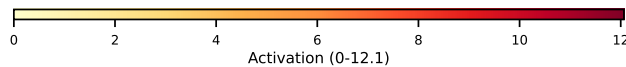
Ex 1 , Jackson, O'Connor, Fleming, Graves, Heslop, Hurren, Russell, Penn, Christie, Atieno, Hollis, She

Ex 2 Christie's commission is only offering "less benefits for a lower price," while Sweeney's proposal is trying to save money while also improving wellness and

Ex 3 say in the course and business of their government. For shame, Chris Christie. For shame. The New Jersey governor pledges to "tell it like

Ex 4 stage for The Last Jedi. "I'm a huge fan of both Gwendoline Christie and Phasma. Christie is magnetic and Phasma has so

Ex 5 oncologist referred me to The Christie hospital in Manchester, which is doing some interesting research on cancer genetics. With them, I'm trying a few things,



Feature 10092 (Layer 20) - Step 0

Entropy: 0.080 | Fuzz ROC: 0.696

Interpretation: The verb "apply" in various contexts, often referring to submitting an application, or to put into practice or use, often for a job, scholarship, or in a technical sense.

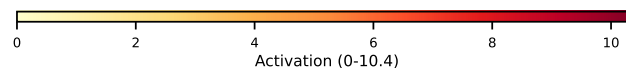
Ex 1 fashion we can. Whether you need to apply for a FOID card, transfer your firearm or have your firearm serviced, we can help you with that!

Ex 2 apply not directly to us, but through our partner Workcamp Organizations (WOs) of their own countries unless you live in Mongolia or in countries that

Ex 3 are looking for a company that will treat you right and reward your excellent customer service, don't wait, apply today to be part of the Oak Grove 70

Ex 4 credentials. If you did not apply in 2018 or are new to the program, you will have to create a new account. If you do not remember your login

Ex 5 the evidence is laid before him. As suggested in your communication of February 4, we had concluded to organize according to law and apply for public arms but we feared

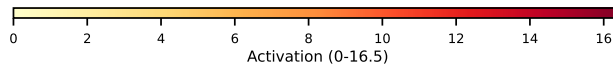


Feature 10064 (Layer 24) - Step 0

Entropy: 0.065 | Fuzz ROC: 0.893

Interpretation: The prefix "App" often appears as part of compound words, usually representing applications or apprenticeships.

- Ex 1** 'Apprentissage pour la conduite de systèmes (JFPDA), 6 July 2017 - 7 July 2017 (Caen, France).
- Ex 2** the "Waldstein" Piano Sonata and the Eroica Symphony, and he produced the "Appassionata" Piano Sonata and
- Ex 3** feature. Applies only to AD Query (ADQ) on Security Management Server / Log Server. Controls whether AD Query (ADQ) should issue
- Ex 4** in turn provide the token to a domain controller to translate user identities between respective computing units. "Apparatus and Method for Managing Multiple User Identities on
- Ex 5** ode. Tips: Jason@recode.net or Signal, Telegram, Confide, WhatsApp at 9 17 - 655 - 4267. The most

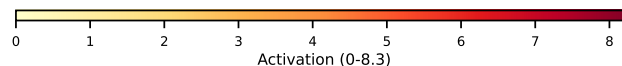


Feature 10379 (Layer 24) - Step 0

Entropy: 0.125 | Fuzz ROC: 0.923

Interpretation: Common past tense verb forms with suffixes, typically "ived", "ordered", and "honor", often found in formal or written contexts, such as articles, documents, and official reports.

- Ex 1** tutorials that will help you put the pizzazz in your next potluck. Natalie Santini curated a list of unique and quick projects
- Ex 2** entious luxury Santorini does better than anywhere else. This dreamy setting is perfect for a honeymoon. The Honeymoon Suite comes with a
- Ex 3** by the poolside. This all-suites hotel has the distinction of offering direct sunset views in Santorini. Indeed, it's one of
- Ex 4** no idea which one. The handwritten label faded in the sun and I can't find my notes. Got these seeds labeled as "Spirito Sant
- Ex 5** formation. Does Hinduism discriminate against women? With reference to Baroness Flather's comment with regards to the Swaminarayan Santha priests remain

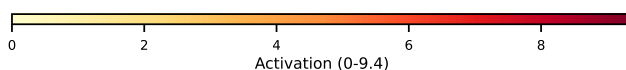


Feature 10058 (Layer 28) - Step 0

Entropy: -0.000 | Fuzz ROC: 0.971

Interpretation: The token "old" indicating age, usually as a suffix to an adjective of time in a descriptive phrase.

- Ex 1** - Bartholdy" in Leipzig. Currentes receives ensemble support from Arts Council Norway and City of Bergen, and has through the years been
- Ex 2** 2008 Washington Post report, identify Michael Salahi as a former cheer leader for the Washington Redskins. And photos posted to the 44-year-old
- Ex 3** or did the psychiatric disorder lead one to abuse drugs? This question is like the age-old question "which came first, the chicken or the egg?" Those
- Ex 4** but lots of blue screen to look at. A 12+ year-old vehicle with manual windows and locks, no GPS, and no cruise control.
- Ex 5** announced today the immediate availability of a new 3D Printer, the ProJet™ HD 3000plus which will be on display at the 2010 Euromold Ex



Feature 10201 (Layer 28) - Step 0

Entropy: 0.108 | Fuzz ROC: 0.876

Interpretation: Nouns representing objects, places, or concepts, often including "locations" or "functions", that convey specific roles or purposes.

- Ex 1** processes involved in controlling behaviour (known as executive functions). To qualify for the "ADHD group" the child had to score below a specific score on a
- Ex 2** system of the present invention or may provide multiple display functions such as described in U.S. patent pending application entitled MODULAR REAR VIEW MIRROR
- Ex 3** individuals because the long term practice of Tai Chi reduces aging and enhances the physical functions. Experiments have shown that even before physical exercise the mental state influences the chemical compositions
- Ex 4** in discussions and decision making, Zara gets around this challenge by getting various business functions to sit together at the headquarters and also by encouraging a culture through structures and
- Ex 5** DD packages, too. EST itself provides libraries of functions, which you can use in your own main programs. Here are our EST projects. Moreover, you

